From Learning & Training to Innovation with the Pharo Ecosystem

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BCI 2026

Inria

Berger Levrault

Inria
INVENTEURS DU MONDE NUMÉRIQUE

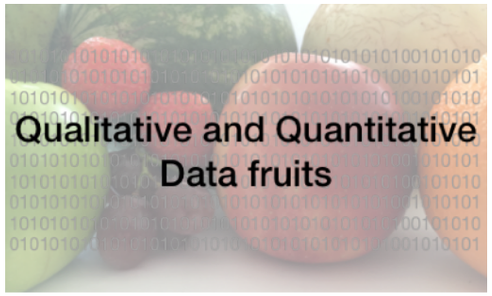
Lifeware

ENSTA Bretagne

THALES

ZWEIDENKER

Yesplan
Let's make it happen



esug
EUROPEAN SMALLTALK USER GROUP

GemTalk SYSTEMS

SPESENFUCHS

projector software

feenk

netstyle.ch

osoco
osoco software company

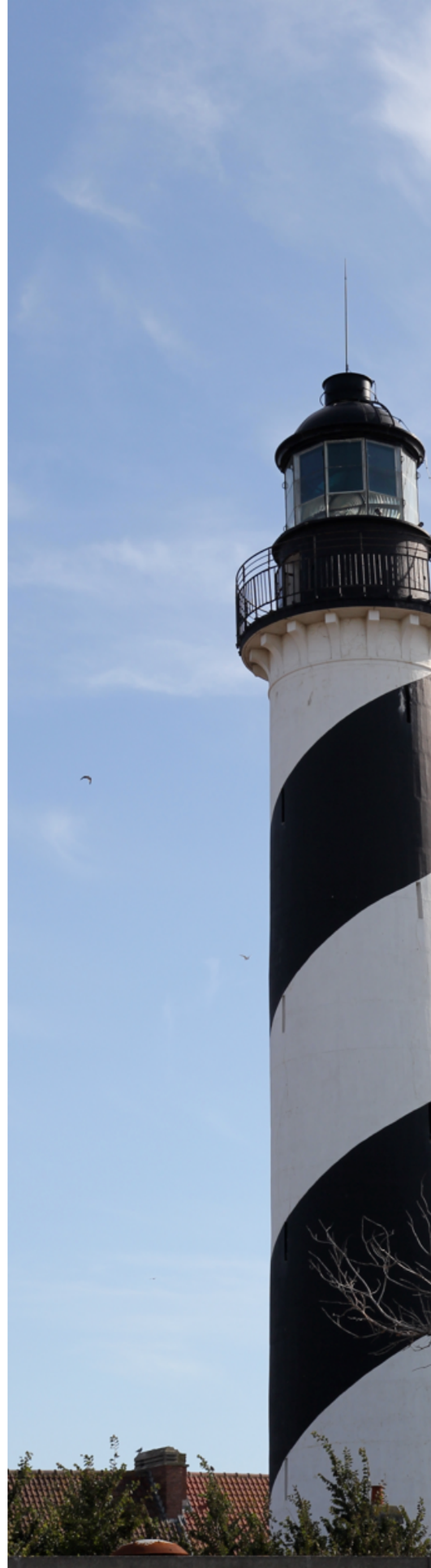
inspired!

SRA GROUP

telna

Université de Lille

FINWORKS
INFORMATION TECHNOLOGY SOLUTIONS



Expert in Software Evolution

Program analysis, Metamodels, visualization, testing, refactorings, metrics, rotten green tests….

https://ducasse.github.io

Expert in language design

Traits in Php7.0, Perl6O Influenced Traits in Scala

https://ducasse.github.io

Pharo?

- https://pharo.org
- https://consortium.pharo.org
- https://discord.gg/QewZMZa
- https://github.com/Pharo-project/Pharo

Pharo

- Language (syntax fits on half of a postcard)
- Pure model, reflective & simple OO model inspired by Smalltalk
- IDE with **powerful** tools
- Used in 40 Universities worldwide
- Industrial consortium and partners

Full syntax on 1/2 postcard!

method name
parameter

```
exampleWithNumber: x

	<syntaxOn: #postcard>
	"A ""complete"" Pharo syntax"
	| y |
	true & false not & (nil isNil)
	 ifFalse: [ self perform: #add: with: x ].
	y := thisContext stack size + super size.
	byteArray := #[2 2r100 8r20 16rFF].
	{ -42 . #($a #a #'I''m' 'a' 1.0 1.23e2 3.14s2 1) }
	 do: [ :each |
	   | var |
	   var := Transcript
	    show: each class name;
	    show: each printString ].

	^ x < y
```

pragma
comment
local variable
binary message
unary message
boolean literals
nil literal
block
keyword message
assignment
pseudo variables
instance variable
integer literals
byte array
array generated at runtime
literal array
symbols
floating point
character
string
scaled decimal
local block variable
block parameter
global variable
cascade
keyword message
return instruction

other method definition examples:
unary
+ binaryMessageArgument
keyword: arg
keyword: arg1 withTwo: arg2

https://www.pharo.org

PLACE STAMP HERE

Yes there is nothing else

```
exampleWithNumber: x

	<syntaxOn: #postcard>
	"A ""complete"" Pharo syntax"
	| y |
	true & false not & (nil isNil)
	 ifFalse: [ self perform: #add: with: x ].

	y := thisContext stack size + super size.

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	 do: [ :each |
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	  var := Transcript
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	^ x < y
```

pragma
comment
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global variable
cascade
keyword message
return instruction

other method definition examples:
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https://www.pharo.org

Still a real system :)

Pharo in Numbers

20 releases since 2008

Language Core

IDE + Tools + Frameworks

740 packages (tests included)

~ 27 000 tests

5 platforms (11 OSes)

~ #10 000 classes

~ #140 000 methods

https://github.com/pharo-project/Pharo (~430 forks, 15/123 international contributors)

A growing ecosystem

polymath

pharo-graphics

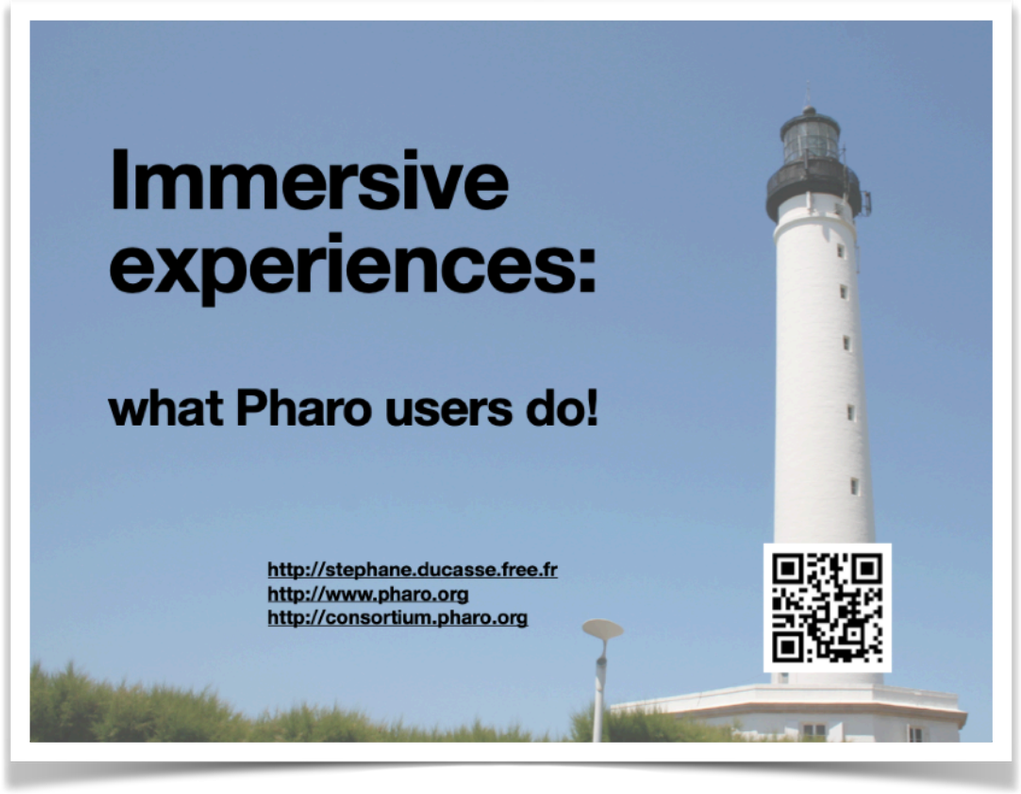
pharo-gis

pharo-container

pharo-ai

What companies are doing with Pharo

- https://archive.esug.org/ESUG2025/day4/401-duca-what-devs-are-doing-with-pharo.pdf
- https://archive.esug.org/ESUG2025/day4/408-duca-what-people-are-doing-with-pharo.pdf



Some examples

TEL NA.

SOLUTIONS PRODUCTS PLATFORM RESOURCES COMPANY

CONTACT US

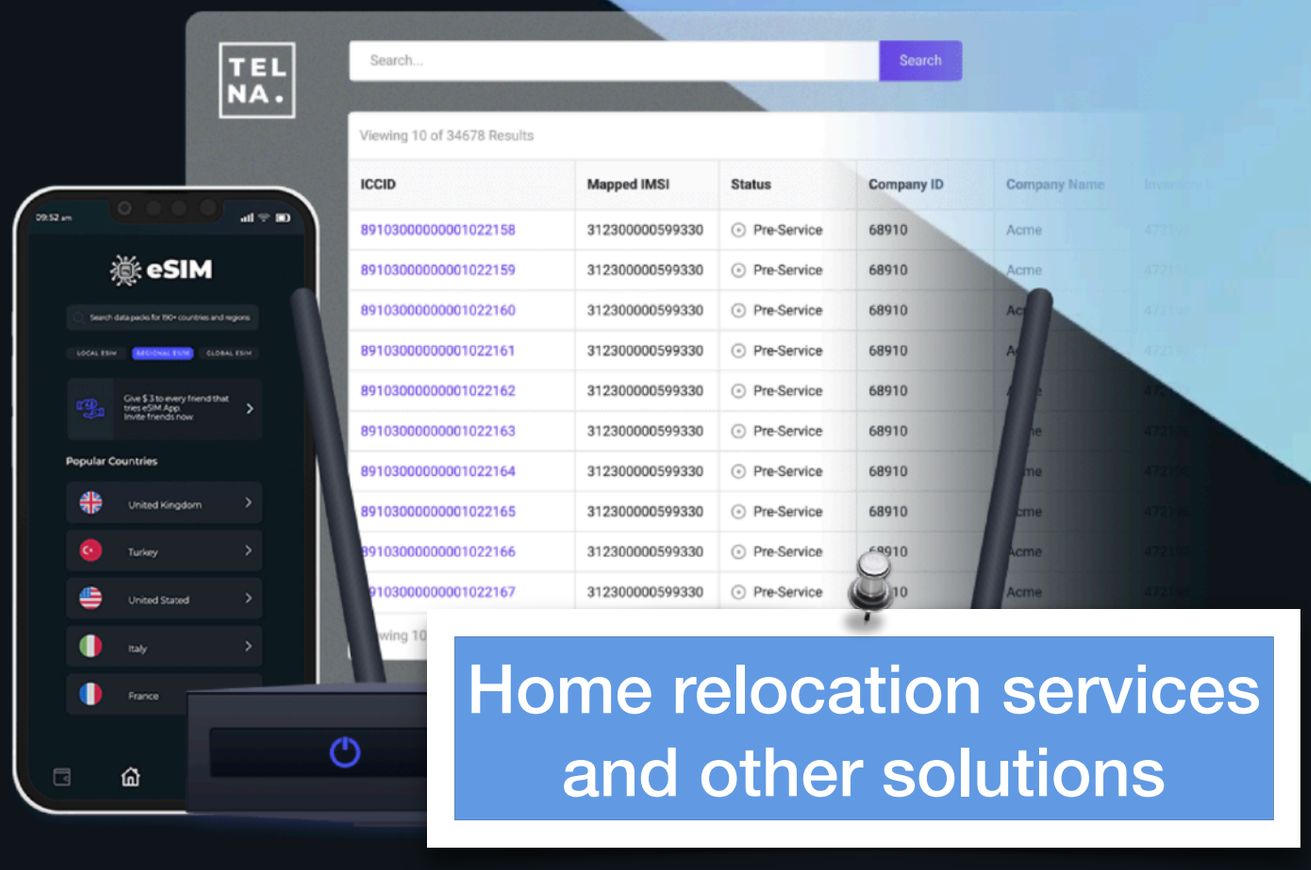
Global Cellular Connectivity platform built for platforms

Access endless possibilities through a single API with Telna, connecting you to cellular data coverage in over 200+ countries.

Designed for Connectivity Service Providers, our platform lets you prioritize your customers while we handle network infrastructure and relationships worldwide.

CONTACT SALES

aws EQUINIX KDDI Google Cloud



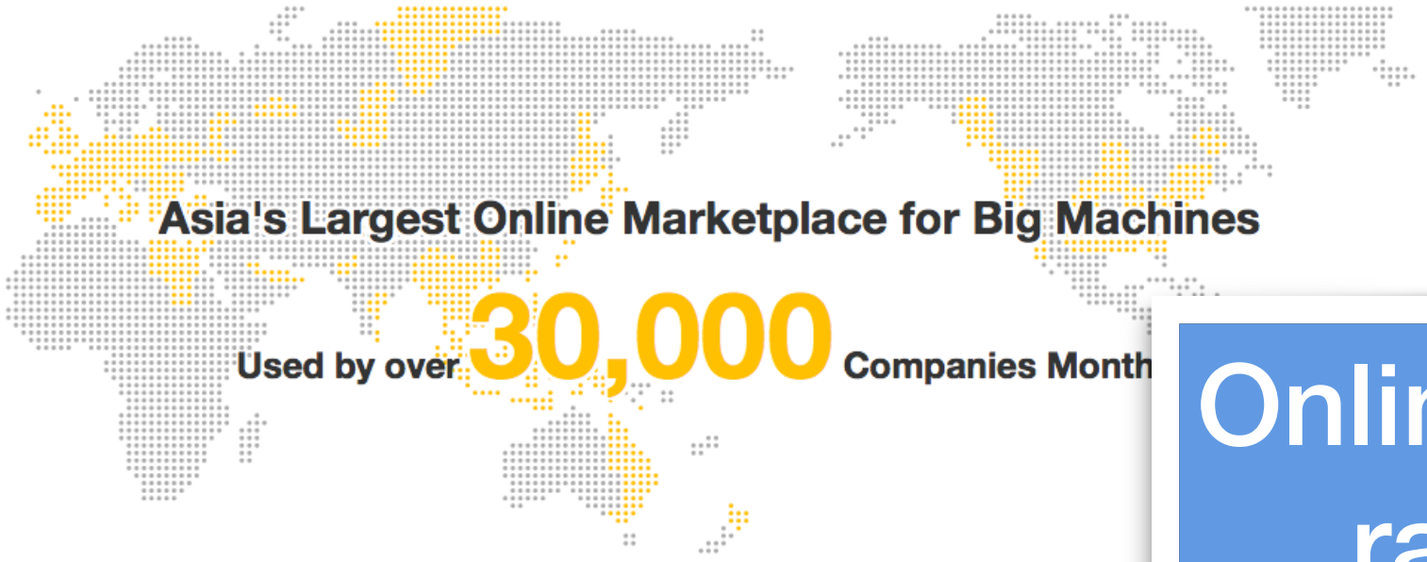
Asia's largest online marketplace for used construction equipment and machinery

How to Buy Login English

ALLSTOCKER

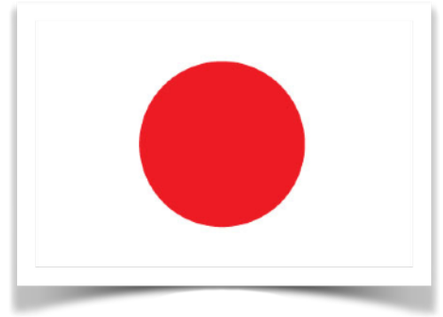
We find machines for you

+81-3-5847-8994



Asia's Largest Online Marketplace for Big Machines

Used by over 30,000 Companies Month



Online marketplace raised 4.7 M$

Find Machines

Search by Category

Construction Machinery

- Mini Excavators
- Excavators
- Wheel Loaders
- Dozers
- Cranes
- Rollers
- Graders
- Finishers
- Generators
- Welders
- Compressors
- Small Items
- Other Construction Machineries

Material Handling

- Forklifts

Farm Machinery

- Tractors
- Other Farm Machinery

Vehicles

- Dumps
- Trucks
- Other Vehicles

Others

- Attachments
- Parts
- Others

Search by Keyword

Enter a keyword (e.g. KOMATSU PC200)

Search

New Machines

Updated on: 2016/09/13 17:46:18 (JST)

See More ›

HOKUETSU
PDS175S
400,000 JPY

KUBOTA
U-55-6
3,500,000 JPY

OTHER
MGR2500
133,100 JPY

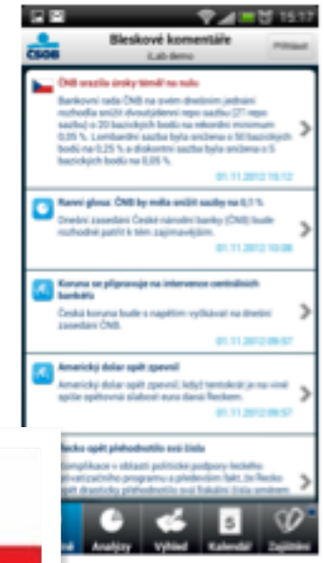
ISEKI
HL407G
198,000 JPY

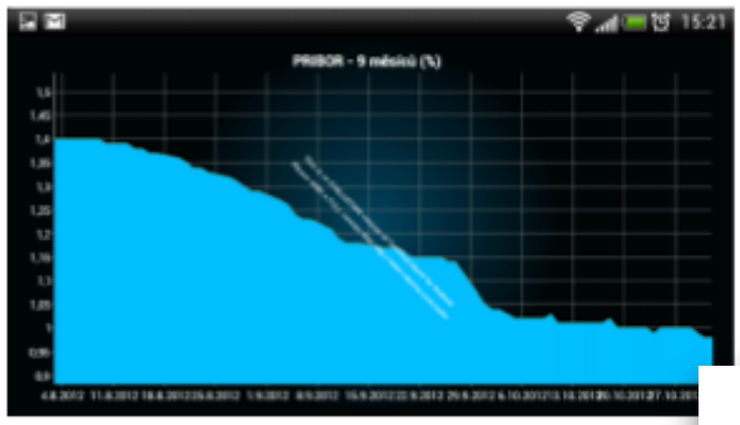
OTHER
MR6500
143,000 JPY

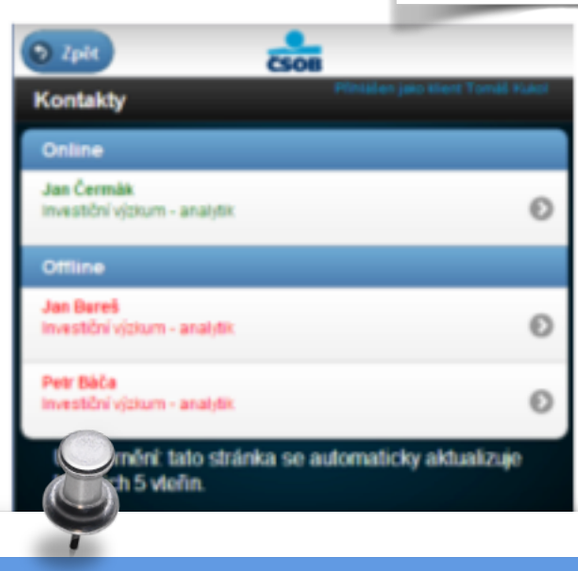
OTHER
GH250-GML
165,000 JPY

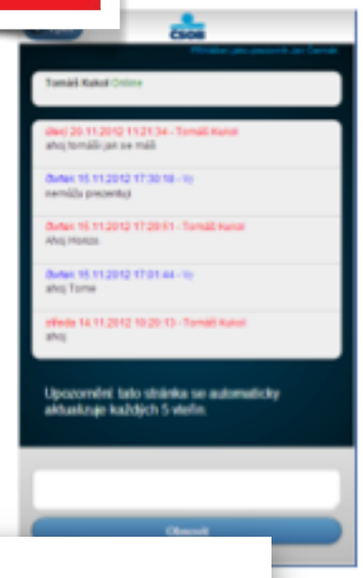
FinWorks in South Africa

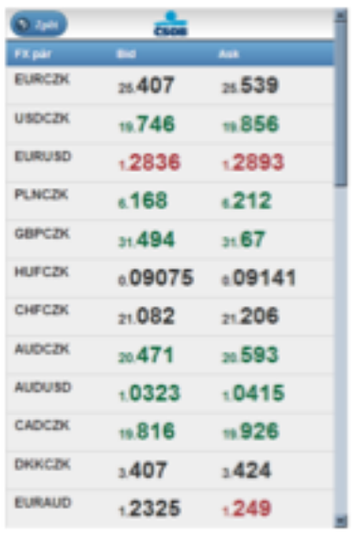
ČSOB

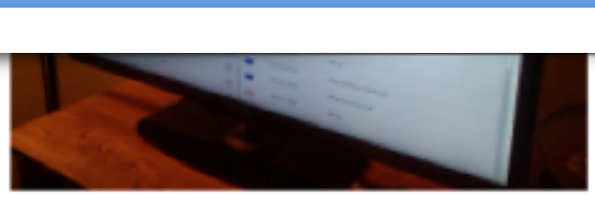


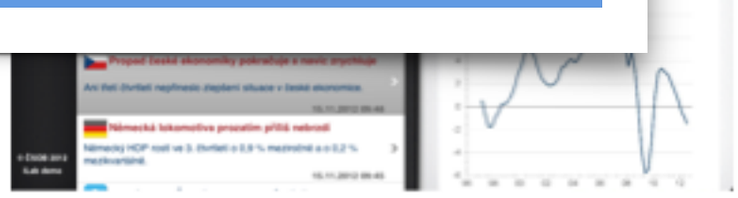








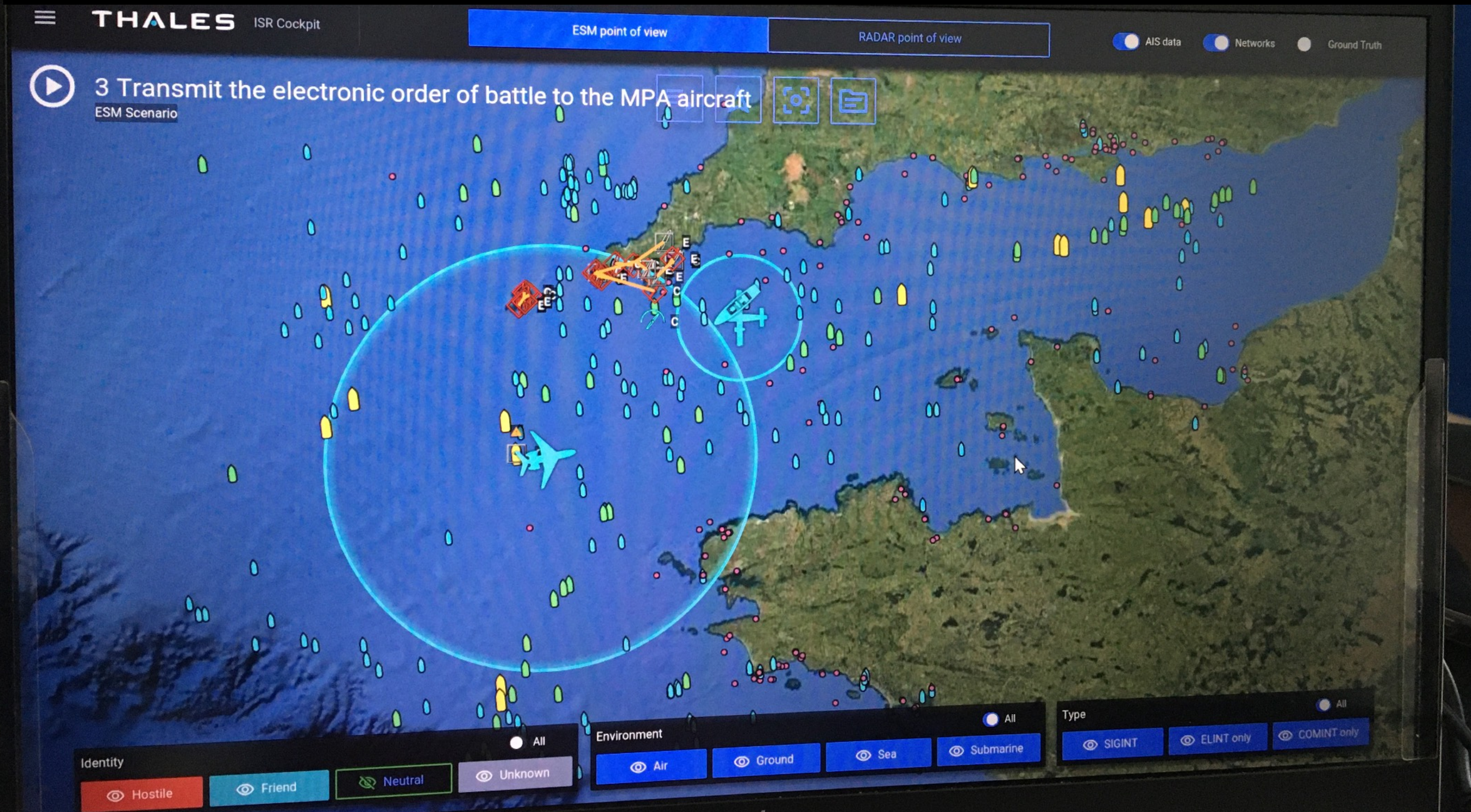




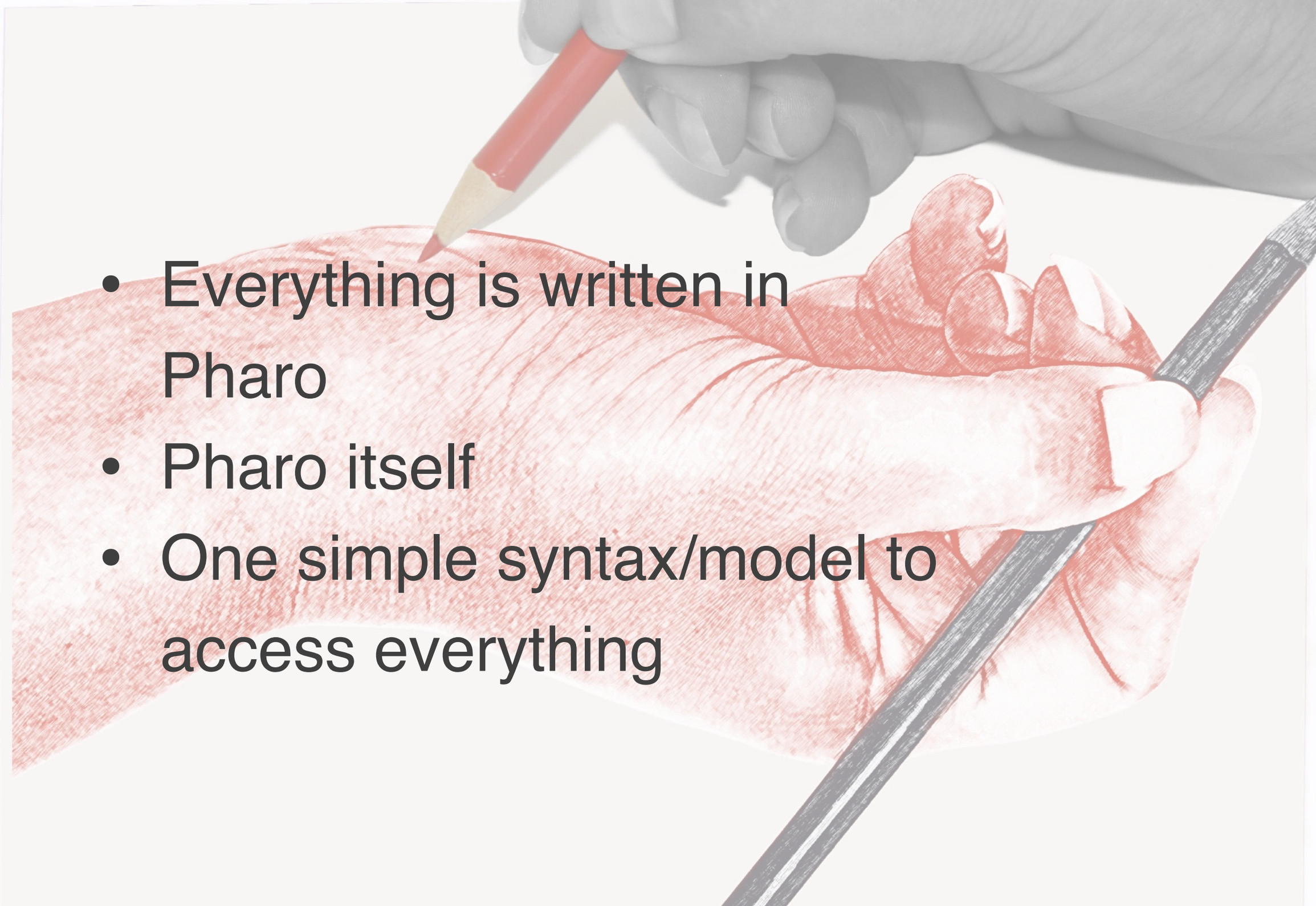
Developed in 3 months one dev
Sold to another bank
Java team estimated 2 year effort

Sales support @ Thales

Geo-Based Maps for Defense



Fully Written in Itself



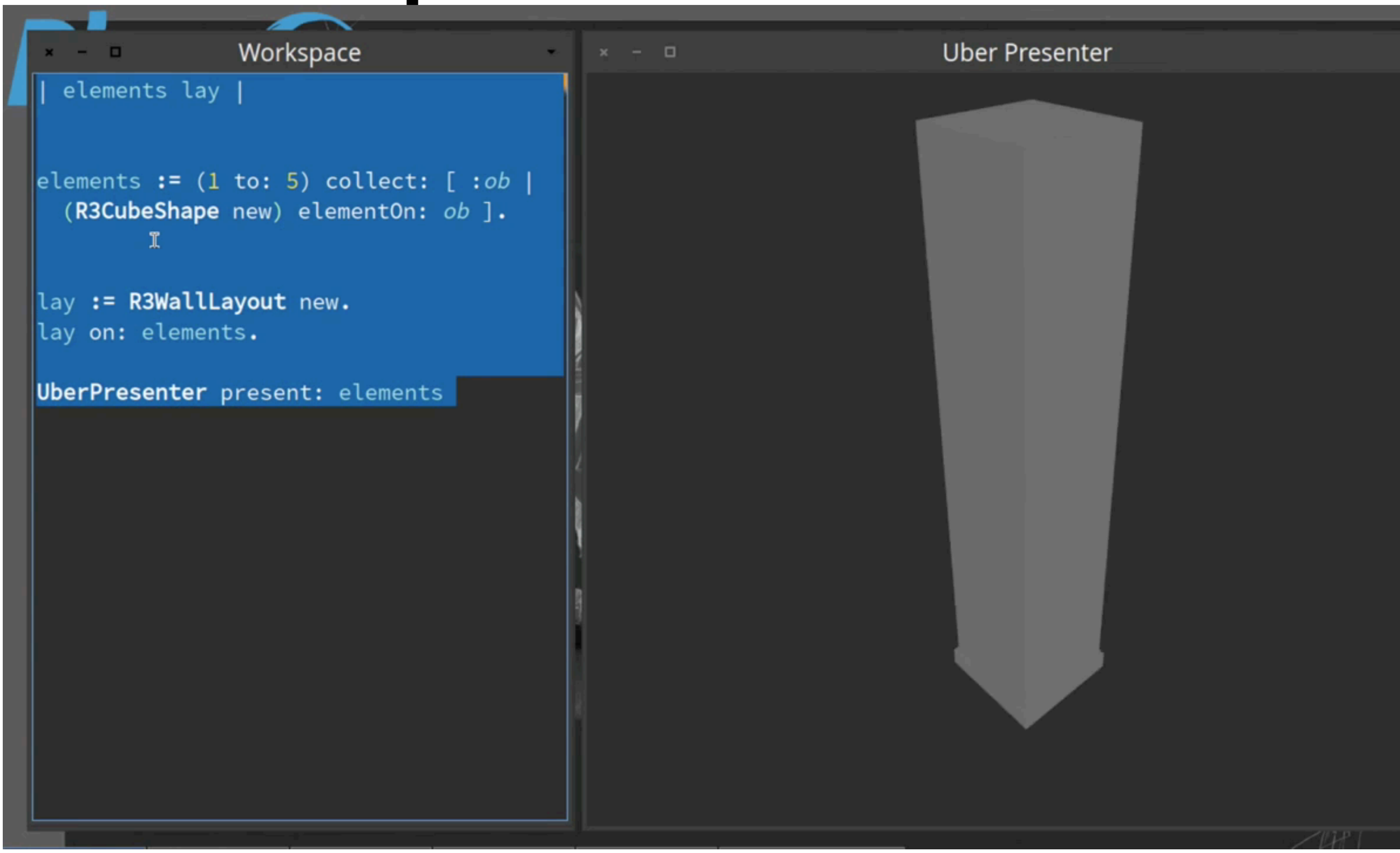
- Everything is written in Pharo
- Pharo itself
- One simple syntax/model to access everything

Immersive

- Pharo is not a blackbox
- Pharo is fully inspectable and reflective
- You can get immersed in objects

Immersed and interacting with live objects

An example of immersion

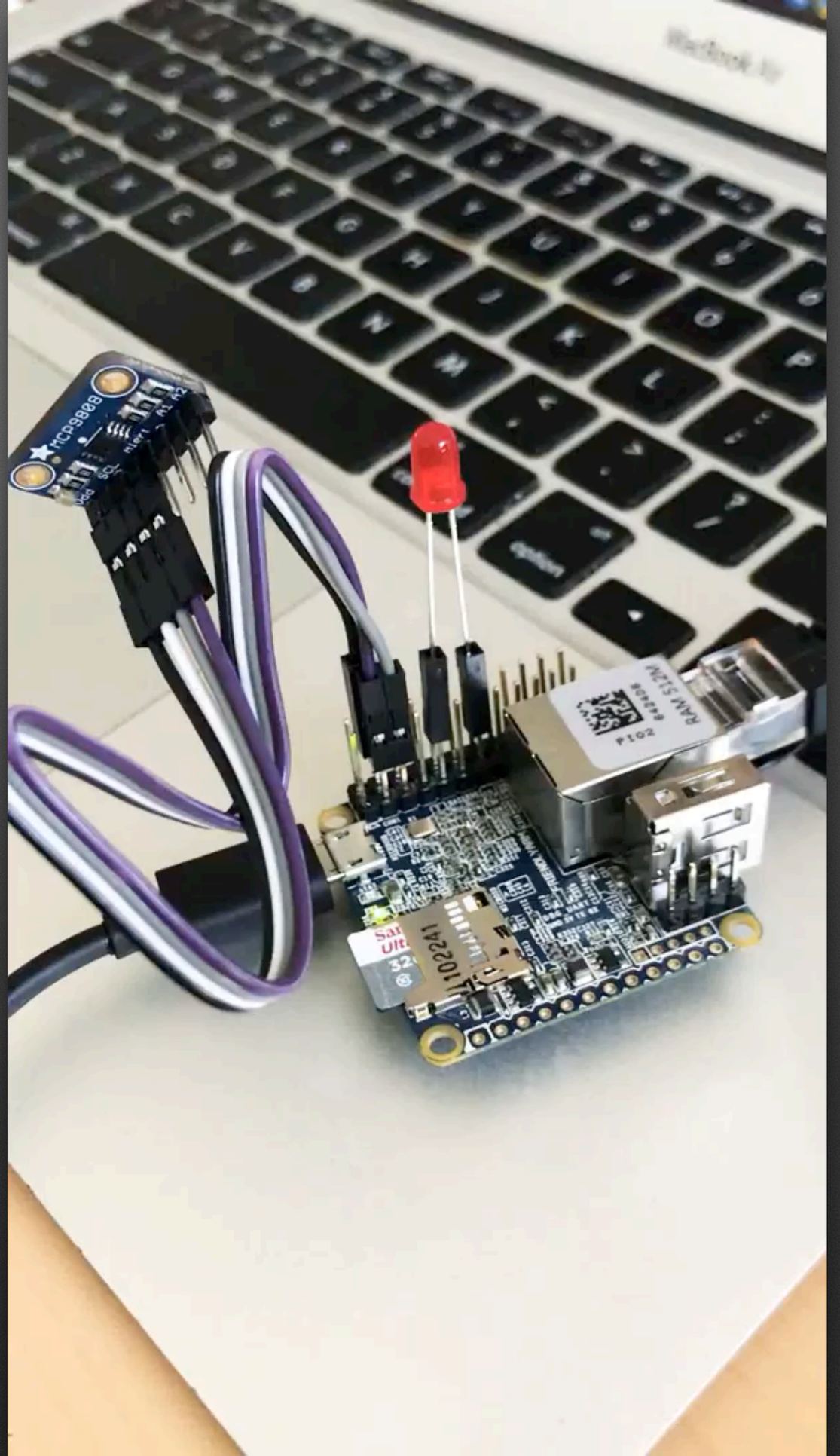


Can do the same with web apps, sockets, networks, sensors, live programming….

Can code in the debugger!

Scripting live the coffee machine

Controlling physical devices…



Training yourself and having fun doing it

Three minutes of theory

Pure OOP

A simple computational model

- Create objects via message sends
- Send messages
- Return, store, and retrieve objects
- Define & execute lambda expression

```
d := Date today
```

```
[ :x | x + 2 ] value: 5

> 7
```

Pharo object model

- Instance variables: protected
- Methods: public and virtual
- Single inheritance

A Simple and Uniform Model

- **Everything** is an object (instance of a class)
- Classes are objects too!
- One single method lookup (no difference for classes or instances)

Syntactic Elements

- comment "a comment"
- character $c $# $@
- string 'lulu' 'l''idiot' symbol (unique string) #mac #+
- literal array #(12 23 36) dynamic array { 1+ 2 . 3 . 4 - 1}
- integer 1, 2r101
- real 1.5 6.03e-34,4, 2.4e7
- boolean true, false
- undefined nil

Constructs

- Temporary variable declaration:
- Variable assignment:
- Separator: message . message
- Return: ^ expression
- Block (lexical closures)

```
| var |

var := 3.14

self foo. self bar

^ Float pi
```

```
[ :x | x + 2 ] value: 5

> 7
```

Three kinds of messages

- Unary message — receiver selector

```
9 squared
```

- Binary message — receiver selector argument

```
1 + 2
```

- Keyword message — receiver key1: arg1 key2: arg2
  - In Java 2.betweenAnd(10,20)

```
2 between: 10 and: 20
```

Composition

(Msg) > Unary > Binary > Keywords

(3 squared + 1) * 2 max: 10

Can we get some help to learn?

Tools are our friends

ProfStef go

one simple
on boarding tutorial

Prof Stef

Next lesson previous lesson go to a lesson

```
1 "Hello! I'm Professor Stef.
2
3 I'm here to  help you learn Pharo.
4 So let's go to the first lesson. Select the text below, right-click
  and choose 'Do it'"
5
6 ProfStef next
```

Line: 6:14 +L

Basic types: Numbers (7/30)

Next lesson | previous lesson | go to a lesson

```
"You now know how to execute Pharo code.

Now let's talk about basic objects.

1, 2, 100, 2/3 ... are Numbers, and respond to many messages executing mathematical expressions.
Execute (Print It) these ones:"

   2.

   20 factorial.

   1000 factorial / 999 factorial.

   (1/3).

   (1/3) + (4/5).

   (1/3) asFloat.

   1 class.

   1 class maxVal class.

   (1 class maxVal + 1) class.
```

Line: 6:1

Message syntax: Binary messages (14/30)

Next lesson | previous lesson | go to a lesson

```
"Binary messages have the following form:

    anObject aBinarySelector anotherObject

where a binary selector is from a list of non alphabetical characters e.g. ~ = , < > + - / *
& @ %. A binary selector can be for example == or <="

3 * 2.

Date today + 3 weeks.

false | false.

true & true.

true & false.

10 @ 100.

10 <= 12.

'ab' , 'cd'.

Date today < Date yesterday.
```

Line: 6:1

Interactively go step by step over the key points

Get ready fast!

Students got it in couple of hours

Shift + Enter = Spotter

Poin

Environment

Fetch all 01/25

Point	class
PointArray	class
PointArrayTest	class
PointTest	class
PointerLayout	class
PointerUtils	class
BaselineOfDebugPoints	class
BreakDebugPoint	class
BreakpointIconStyler	class
CDPointClassParserTest	class
ClyActiveBreakpointsQuery	class
ClyActiveBreakpointsQueryTest	class
ClyAddAndConfigureBreakPointCommand	class
ClyAddBreakPointCommand	class
ClyAddWatchDebugPointCommand	class
ClyBreakpointMethodGroupProvider	class
ClyBrowseBreakpointsCommand	class
ClyClassWithBreakpointsMock	class
ClyClassWithoutBreakpoints	class
ClyDebuggingPointsCommand	class
ClyDebuggingPointsMenuGroup	class

```
1 Object << #Point
2    slots: { #x . #y };
3    tag: 'BasicObjects';
4    package: 'Kernel'
```

?

How to find information?

Libraries are large

You know what you want

You do not know how to express it

With Finder

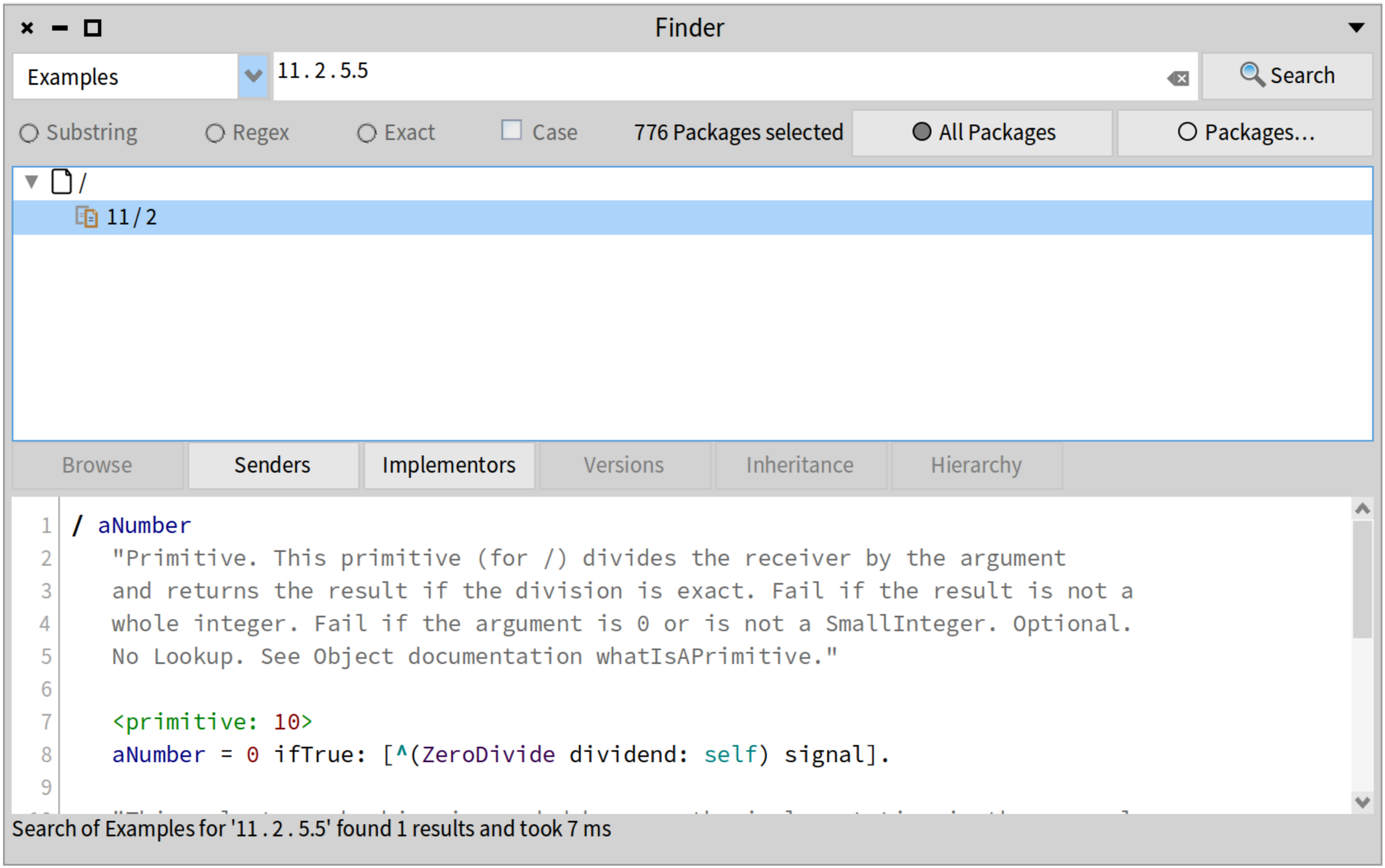
You give inputs outputs

You get methods

Example

What is the message that sent to 11 with 2 returns 5.5?

The message that sent to 11 with 2 returns 5.5



The message that sent to 11 with 2 returns 5

Finder

Examples | 11.2.5 | Search

Substring | Regex | Exact | Case | 776 Packages selected | All Packages | Packages...

- //
  - 11 // 2
- quo:

Browse | Senders | Implementors | Versions | Inheritance | Hierarchy

```
1  // aNumber
2     "Primitive. Divide the receiver by the argument and answer with the
3     result. Round the result down towards negative infinity to make it a
4     whole integer. Fail if the argument is 0 or is not a SmallInteger.
5     Essential. No Lookup. See Object documentation whatIsAPrimitive. "
6
7     <primitive: 12>
8     ^ super // aNumber    "Do with quo: if primitive fails"
```

Search of Examples for '11 . 2 . 5' found 2 results and took 7 ms

How to concatenate strings?

Finder

Examples | 'ab' . 'cd' . 'abcd' | Search

Substring | Regex | Exact | Case | 776 Packages selected | All Packages | Packages...

- ,
  - 'ab' , 'cd'
- copyWithAll:

Browse | Senders | Implementors | Versions | Inheritance | Hierarchy

```
1  , otherCollection
2     "Concatenate two Strings or Collections."
3     "#(2 4 6 8) , #(who do we appreciate) >>> #(2 4 6 8 who do we appreciate)"
4     "((2989 storeStringBase: 16) copyFrom: 4 to: 6) , ' boy!' >>> 'BAD boy!'"
5
6     ^ self copyReplaceFrom: self size + 1
7           to: self size
8           with: otherCollection
```

Search of Examples for "'ab' . 'cd' . 'abcd'" found 2 results and took 14 ms

All the messages that returns true

Finder

Examples

$0 . true

Search

Substring Regex Exact Case 776 Packages selected All Packages Packages…

- isCompletionCharacter
- isLiteral
- tokenish
- shouldBePrintedAsLiteral
- isOctetCharacter
- isDecimalDigit
- isCharacter
  - $0 isCharacter

Browse Senders Implementors Versions Inheritance Hierarchy

```
1 isCharacter
2
3     ^ true
```

Search of Examples for '$0 . true' found 11 results and took 4 ms

Xtreme Test Driven Design

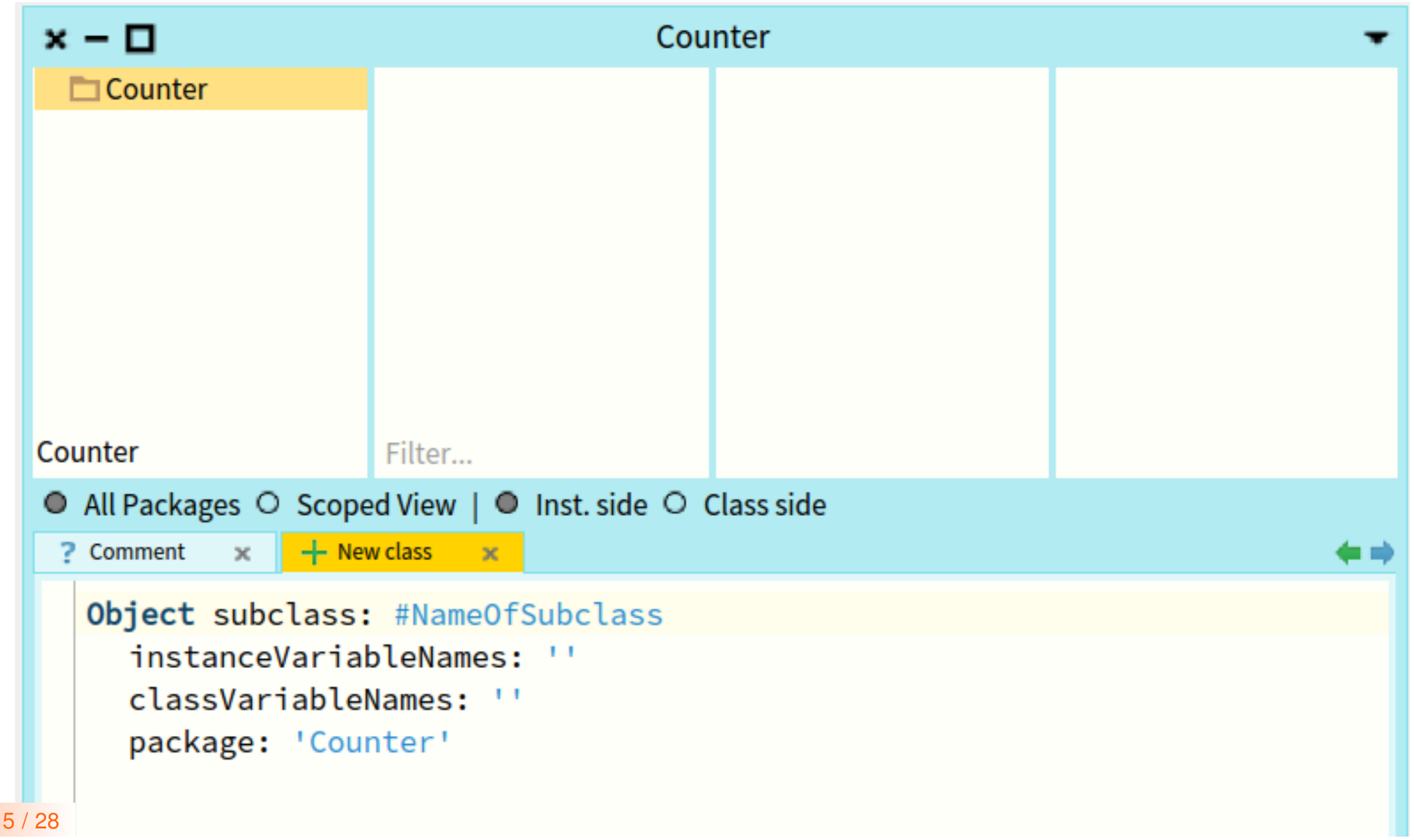
So productive - Feel of power

Code in the debugger while TDD

XTDD: A powerful flow

- Write a failing test
- Execute that FAILING test
- Code in the debugger with hot update and live objects at your fingers
- Resume and continue until the test gets green

An empty package



An empty test case class

CounterTest

Counter | CounterTest | instance side

Counter

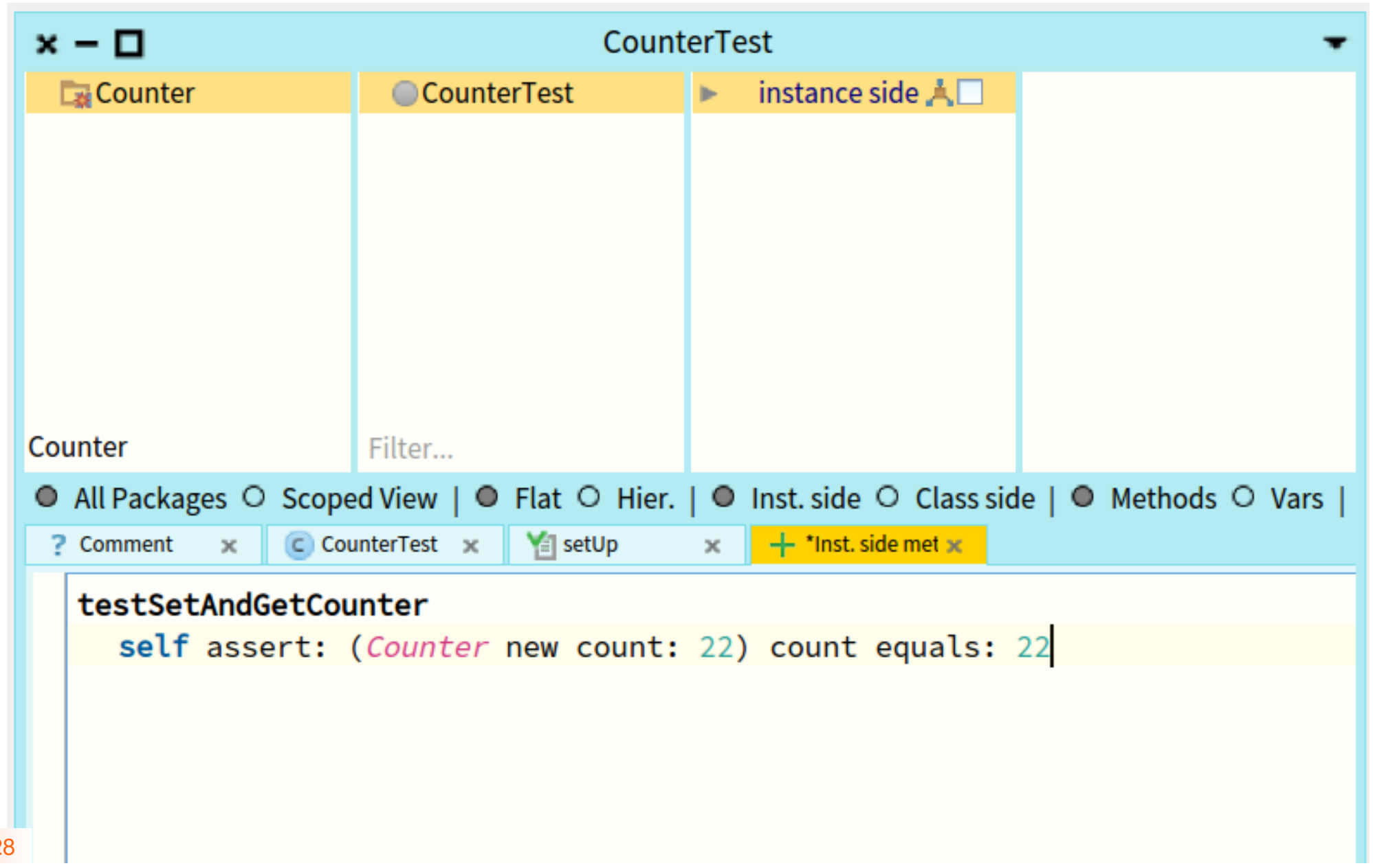
Filter...

All Packages | Scoped View | Flat | Hier. | Inst. side | Class side | Methods | Vars

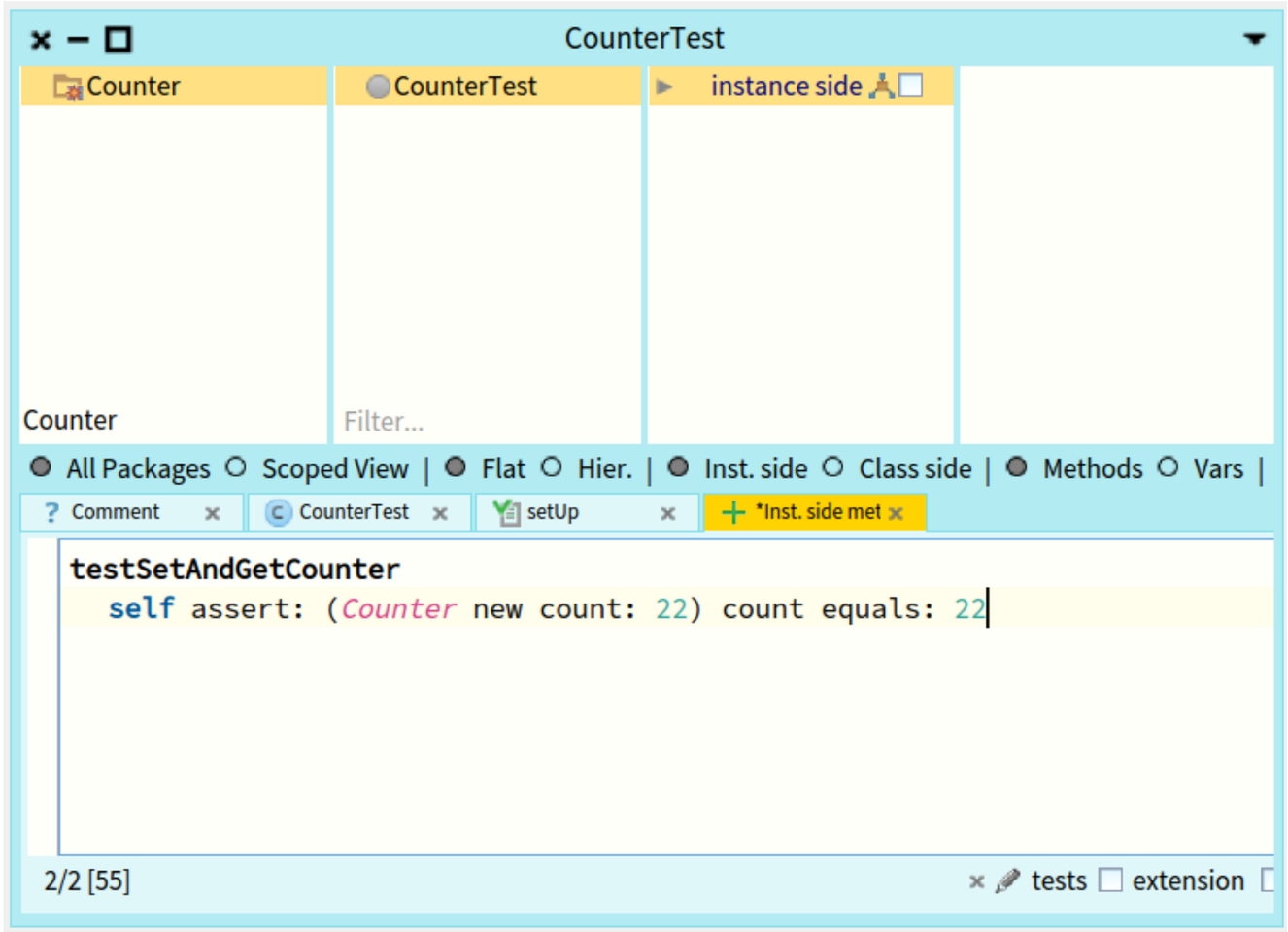
New class | Comment | CounterTest | setUp | Inst. side meth

```
TestCase subclass: #CounterTest
   instanceVariableNames: ''
   classVariableNames: ''
   package: 'Counter'
```

A first test


CounterTest
Counter
CounterTest
instance side
Counter
Filter...
All Packages
Scoped View
Flat
Hier.
Inst. side
Class side
Methods
Vars
Comment
CounterTest
setUp
*Inst. side met
testSetAndGetCounter
self assert: (Counter new count: 22) count equals: 22

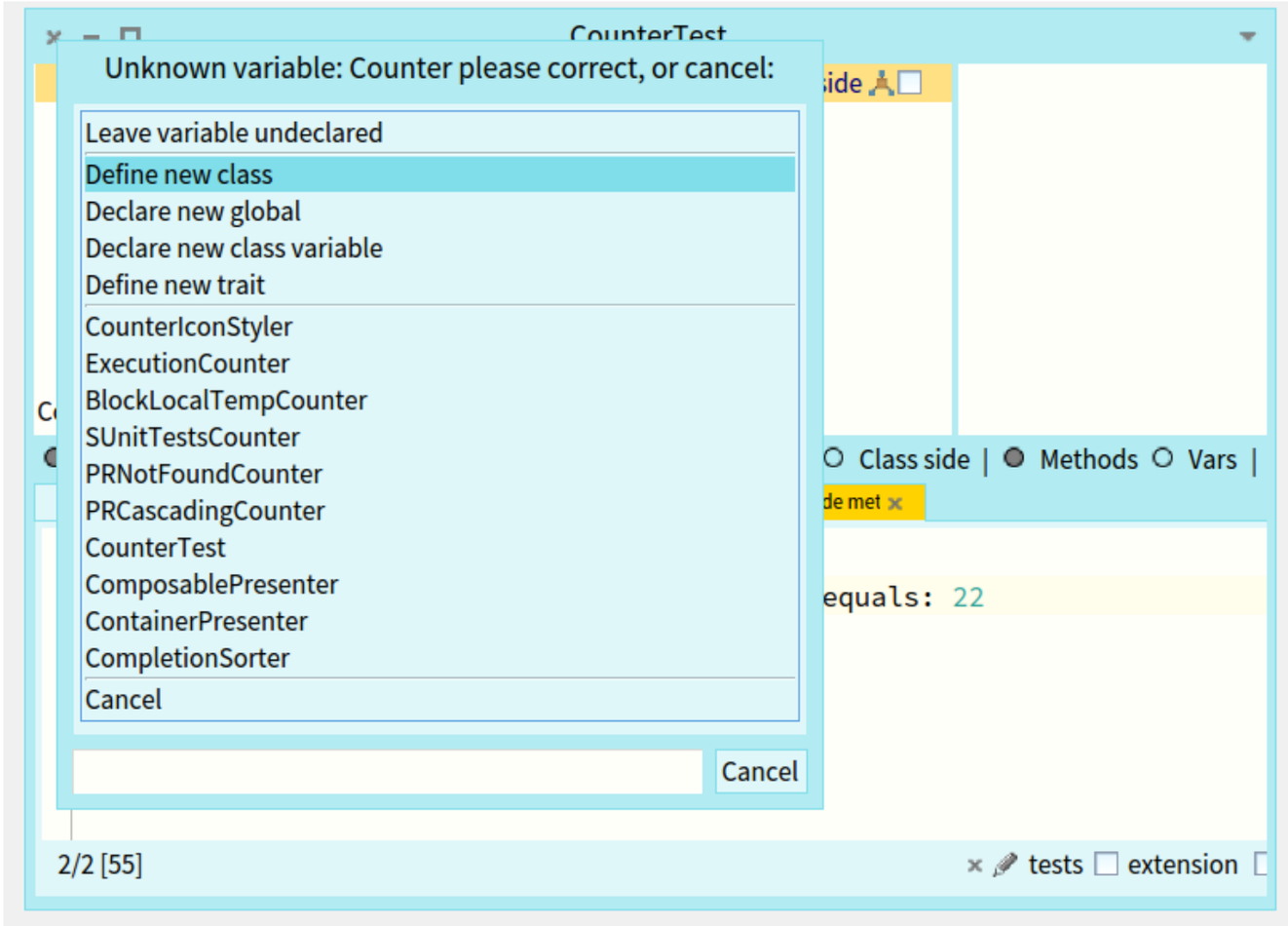
A first test (II)



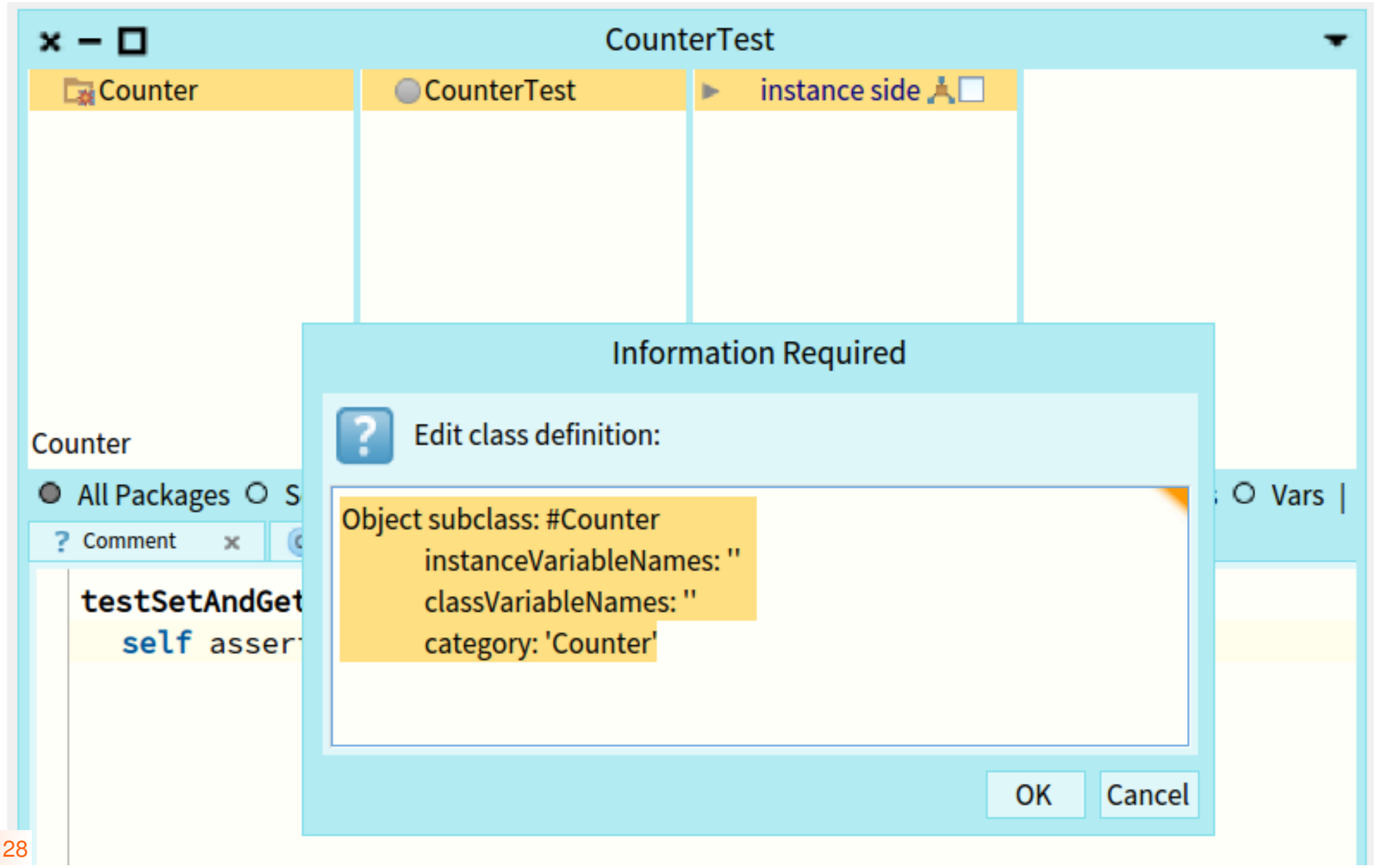
- Method is about to be compiled
- The system knows the class does not exist!

Define a class

- At compile time...



Define a class (II)



Test defined but not executed

CounterTest>>testSetAndGetCounter

Counter | Counter ! | instance side | testSetAndGetCounter

CounterTest | tests

Counter

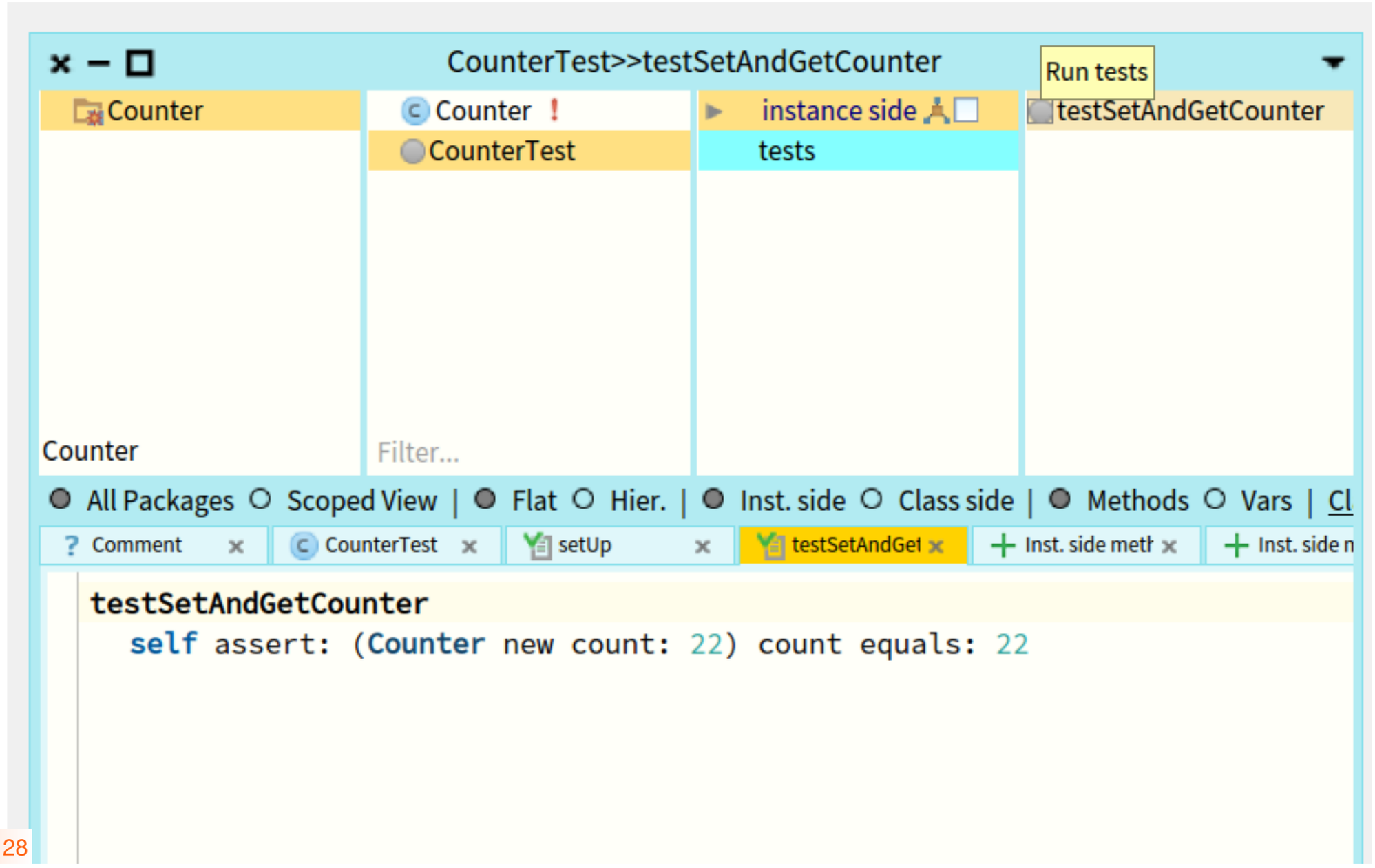
Filter...

All Packages | Scoped View | Flat | Hier. | Inst. side | Class side | Methods | Vars | Cl

Comment | CounterTest | setUp | testSetAndGet | Inst. side meth | Inst. side n

```
testSetAndGetCounter
    self assert: (Counter new count: 22) count equals: 22
```

Running the test



First Error

Instance of Counter did not understand #count:

Bytecode GT

Stack

+ Create | Proceed | Restart | Step into | Step over | Step through

Class	Method	Other	Package
CounterTest	testSetAndGetCounter		Counter
CounterTest(TestCase)	performTest		SUnit-Core
CounterTest(TestCase)	runCase	[self setUp. self performTest	SUnit-Core
FullBlockClosure(BlockClosure)	ensure:		Kernel

Source

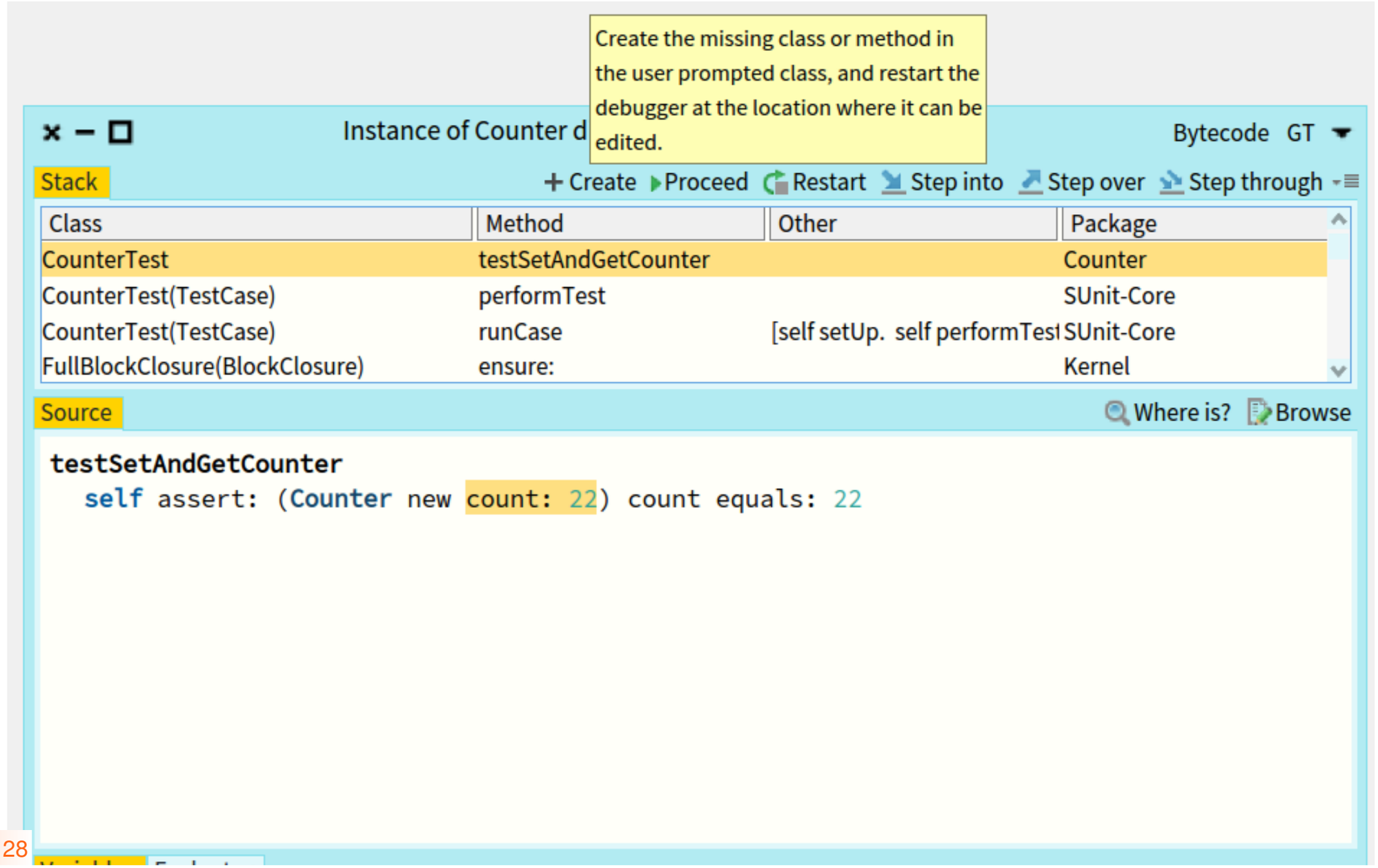
Where is? | Browse

```
testSetAndGetCounter
  self assert: (Counter new count: 22) count equals: 22
```

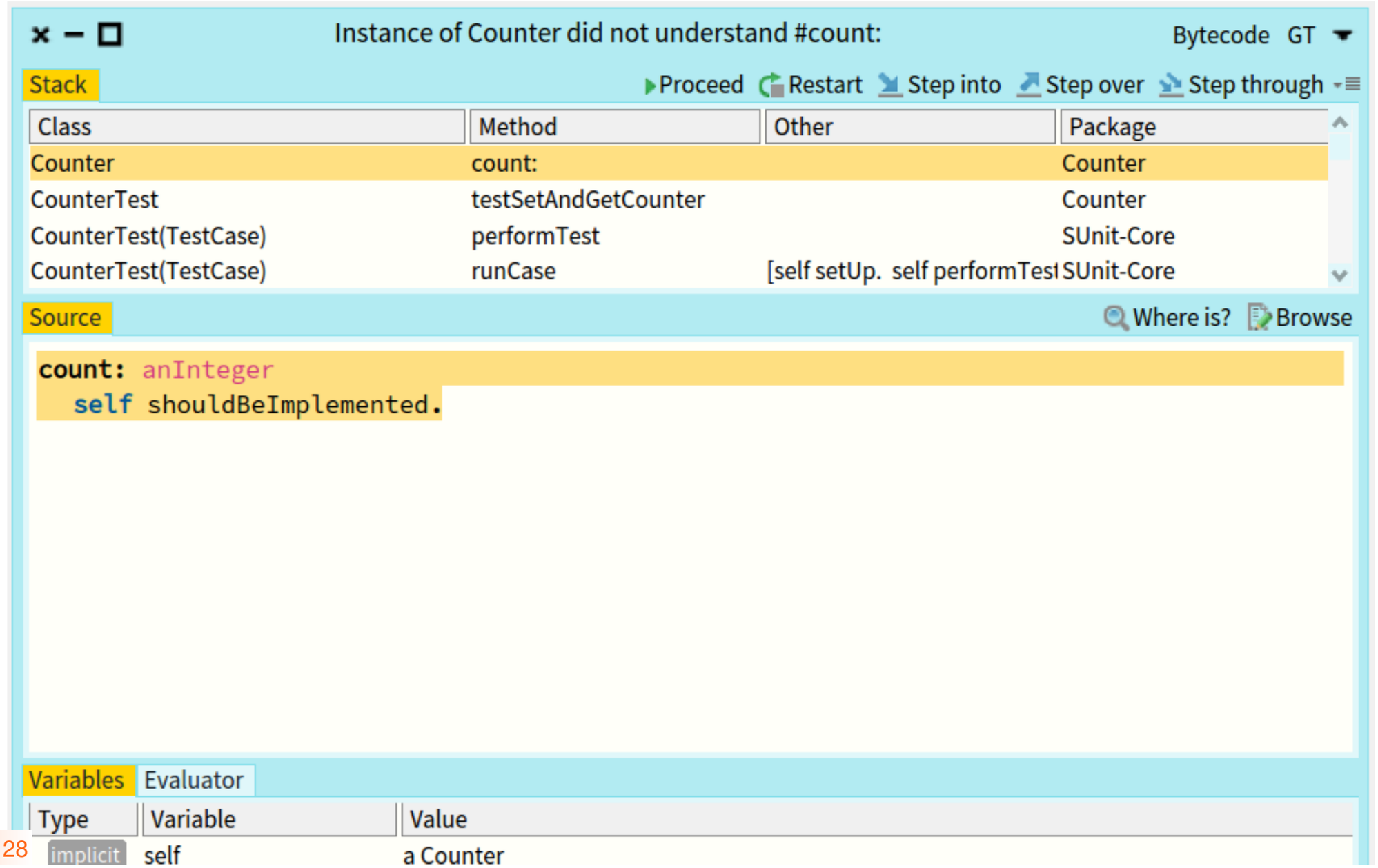
Variables | Evaluator

Type	Variable	Value
implicit	self	CounterTest>>#testSetAndGetCounter
attribute	expectedFails	an Array [0 items] ()
attribute	testSelector	#testSetAndGetCounter
implicit	thisContext	CounterTest>>testSetAndGetCounter

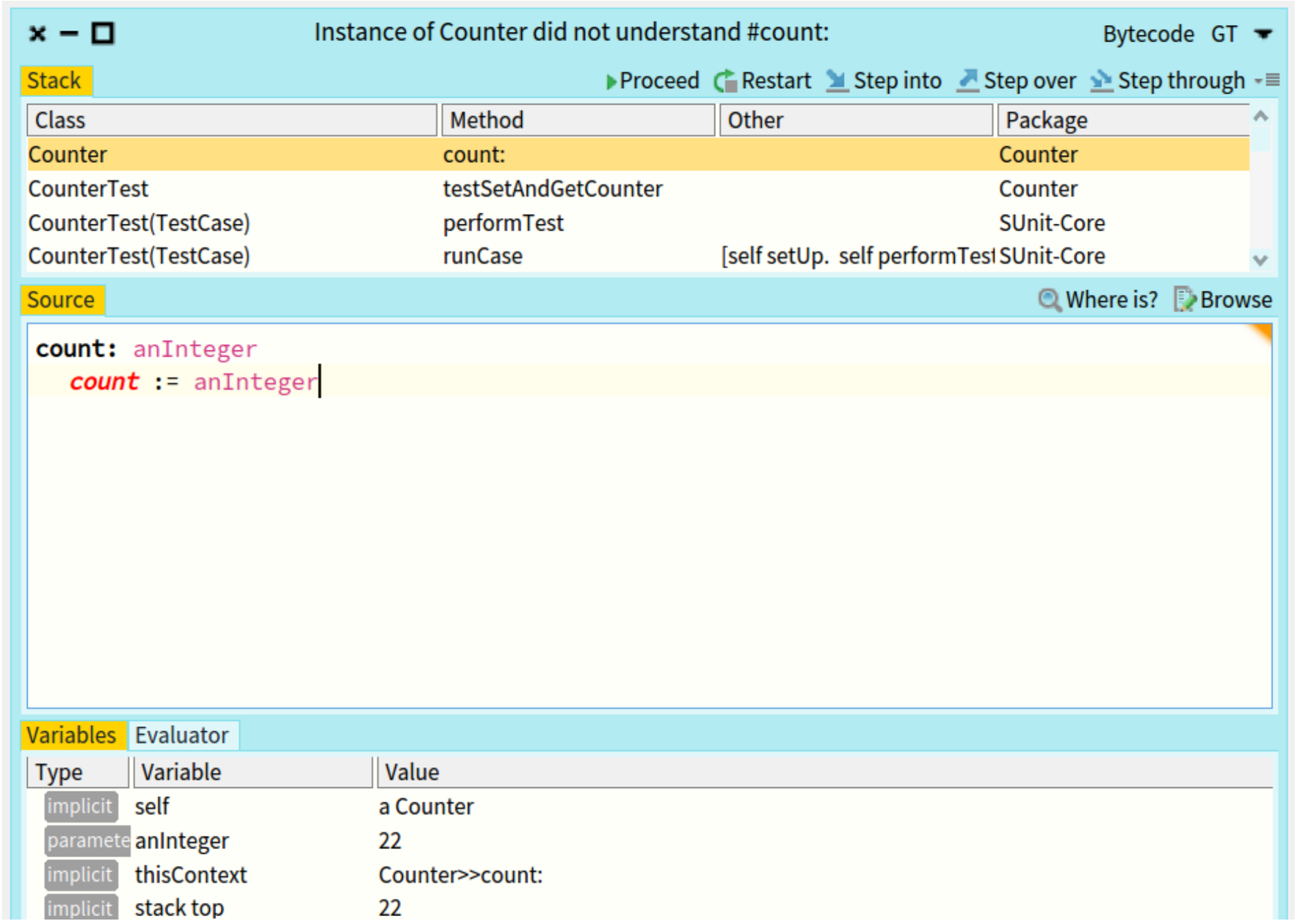
Create a method on the fly



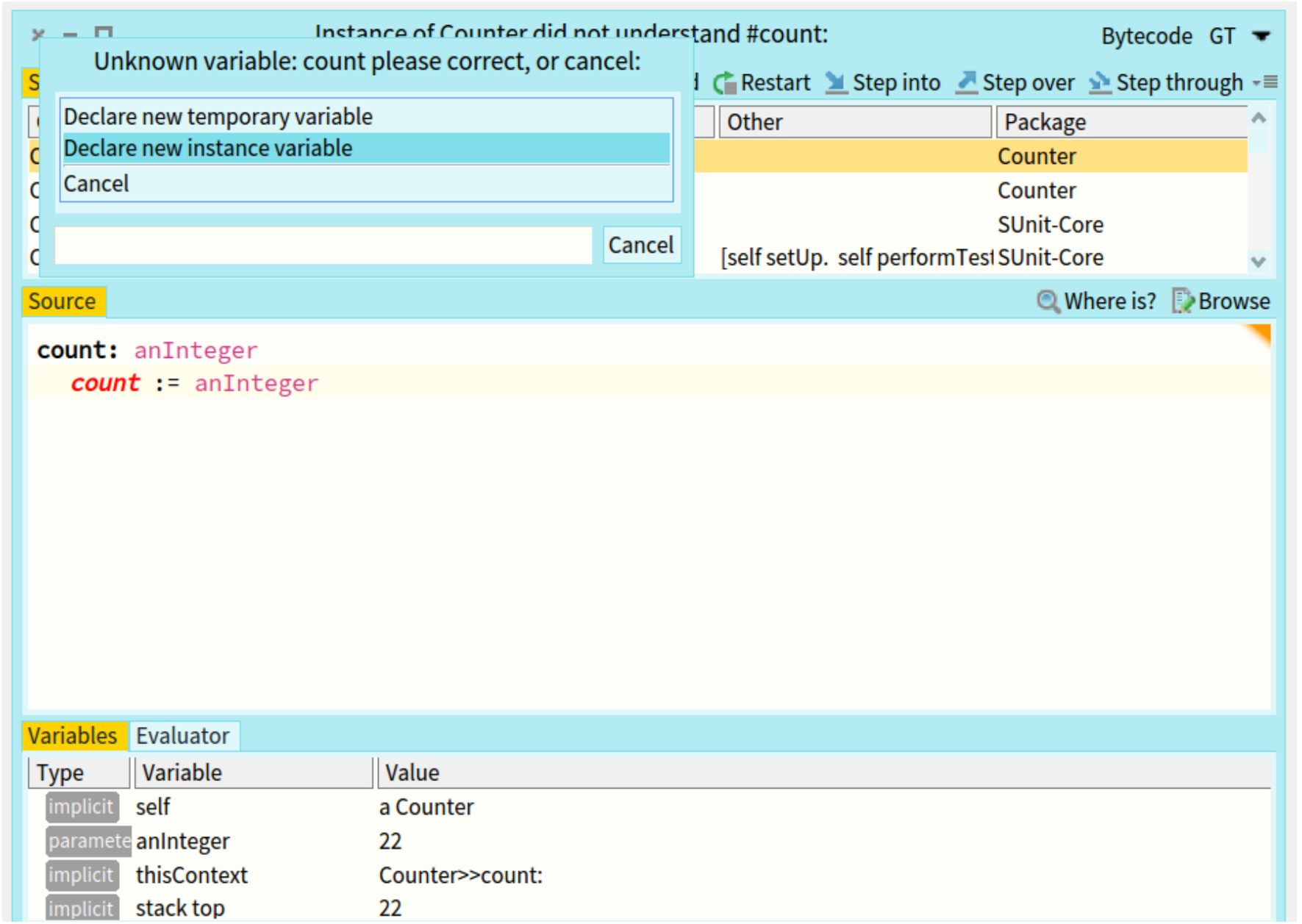
Create a method on the fly (II)



Edit the method in the debugger



Add an instance variable on the fly



Compile....

Instance of Counter did not understand #count:

Bytecode GT

Stack

Proceed Restart Step into Step over Step through

Class	Method	Other	Package
Counter	count:		Counter
CounterTest	testSetAndGetCounter		Counter
CounterTest(TestCase)	performTest		SUnit-Core
CounterTest(TestCase)	runCase	[self setUp. self performTest	SUnit-Core

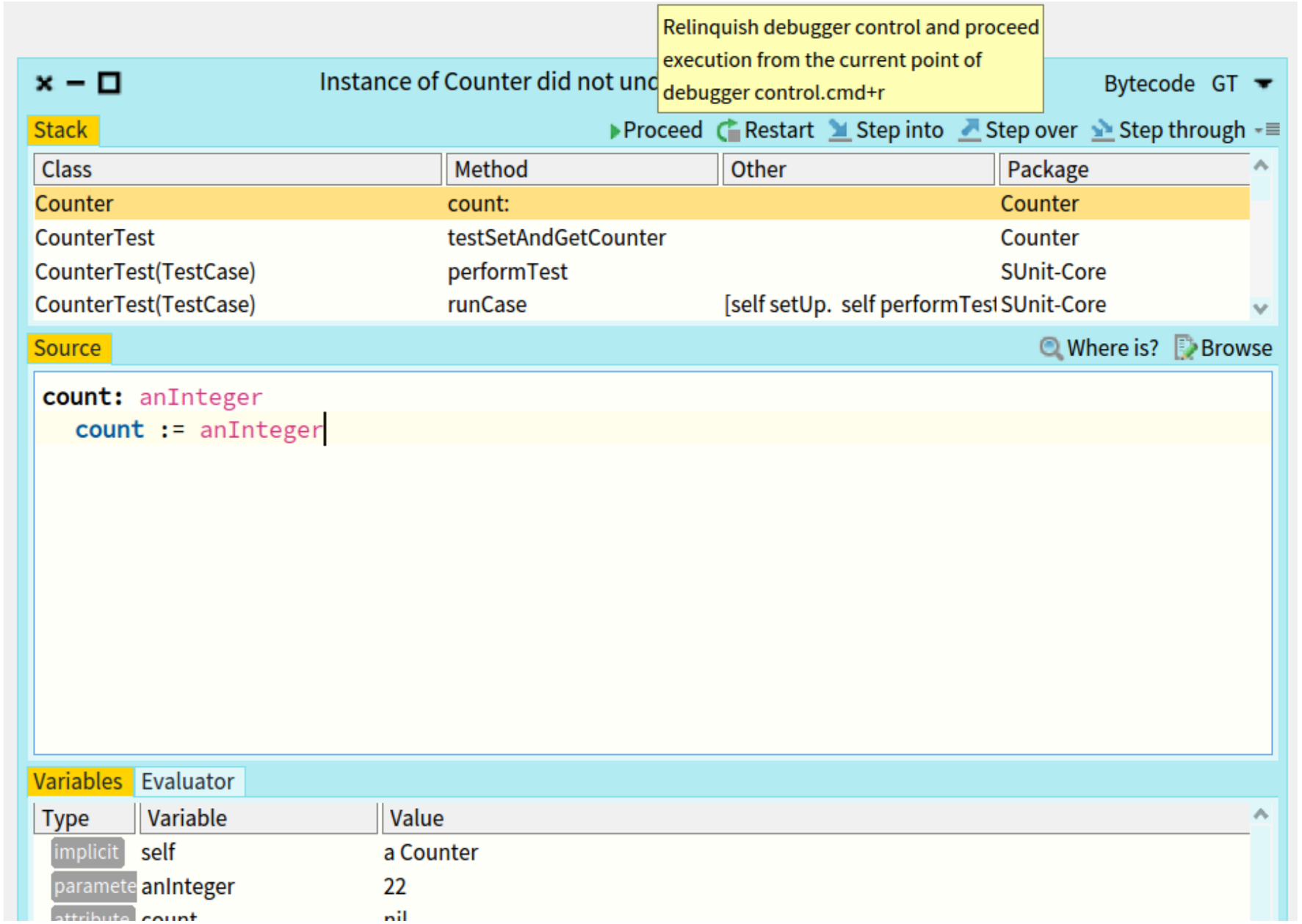
Source

Where is? Browse

```
count: anInteger
   count := anInteger
```

Evaluator

Continue the execution...



Same story....

Instance of Counter did not understand #count — Bytecode GT

Stack — + Create | Proceed | Restart | Step into | Step over | Step through

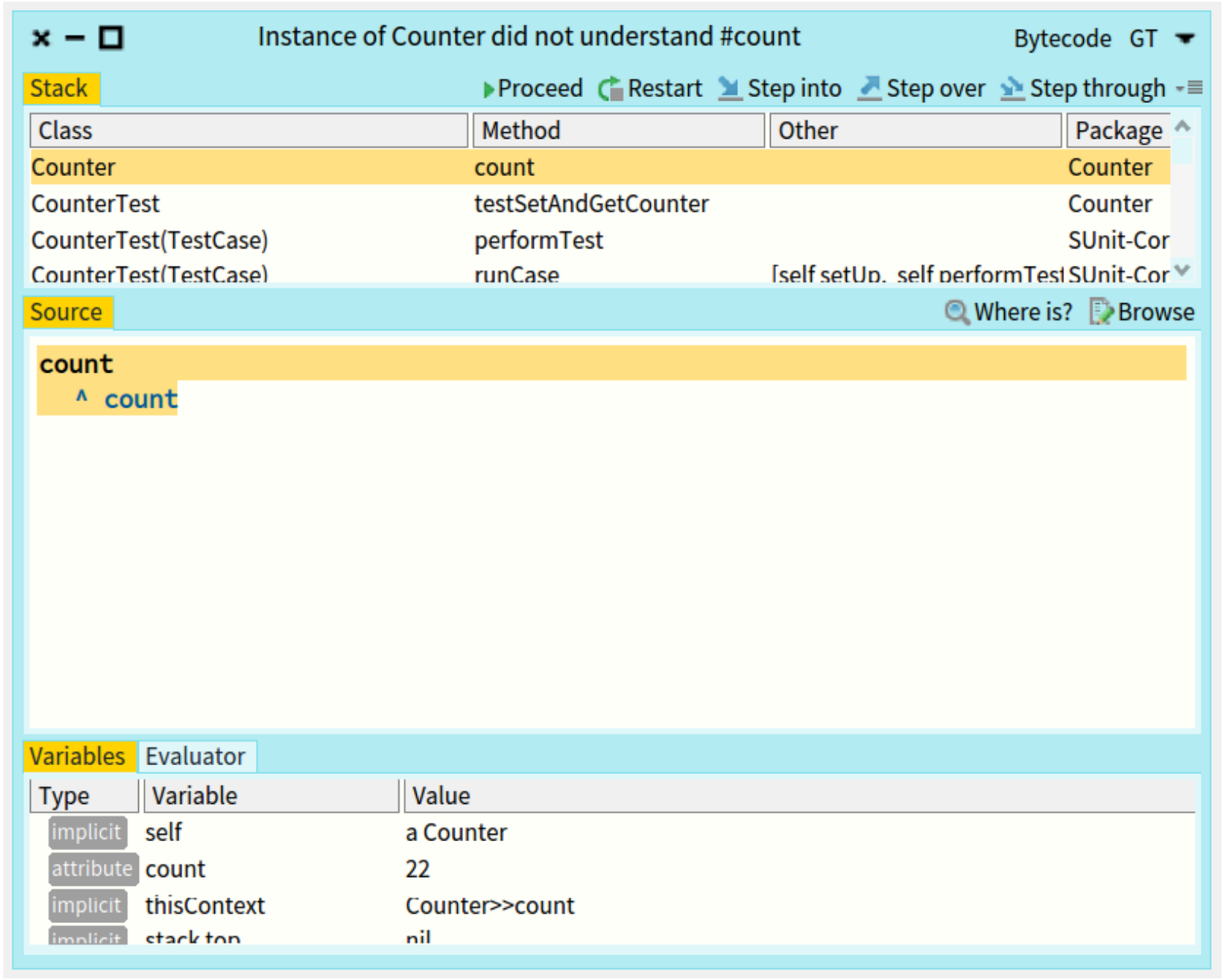
Class	Method	Other	Package
CounterTest	testSetAndGetCounter		Counter
CounterTest(TestCase)	performTest		SUnit-Cor
CounterTest(TestCase)	runCase	[self setUp. self performTest	SUnit-Cor
FullBlockClosure(BlockClosure)	ensure:		Kernel

Source — Where is? | Browse

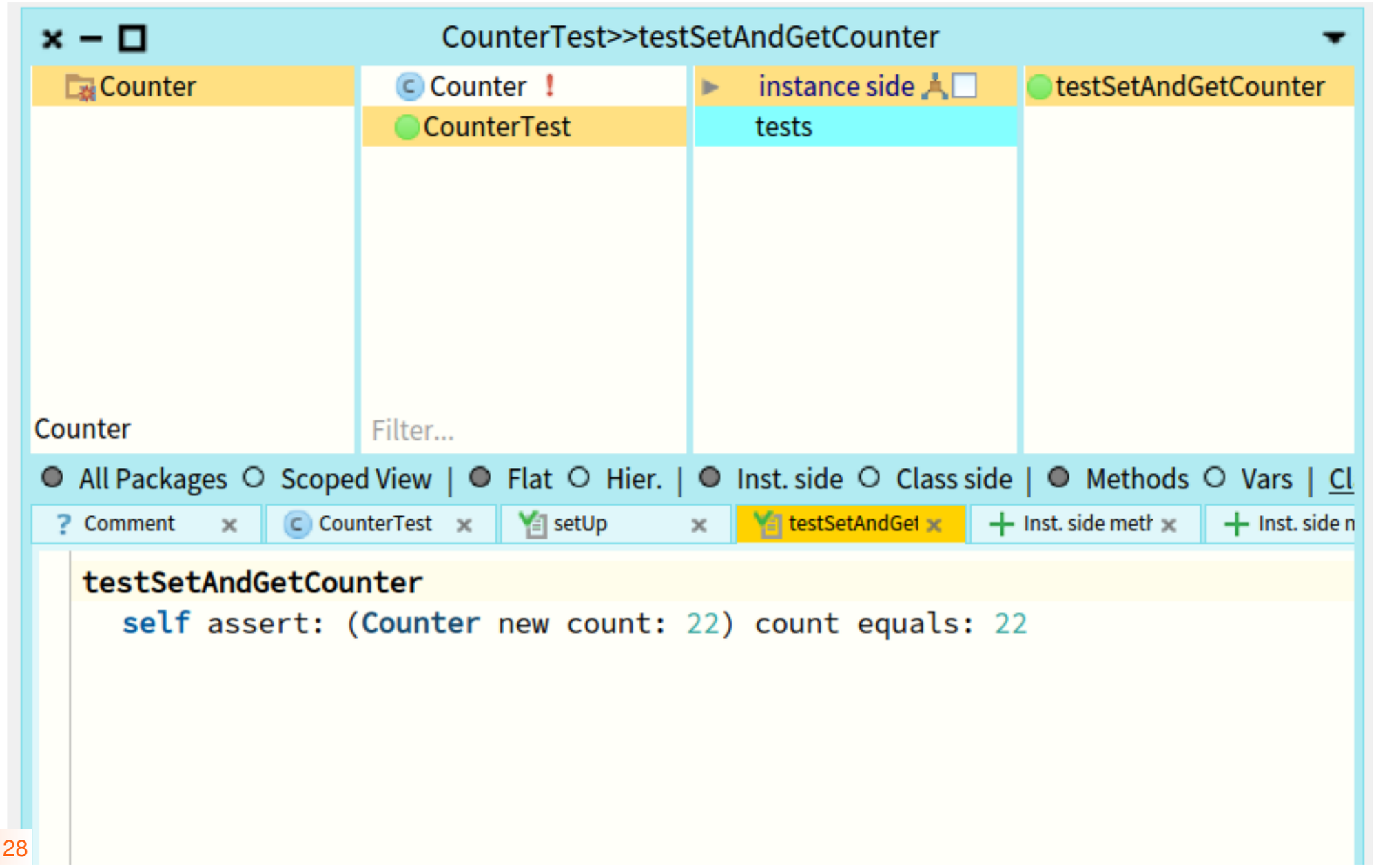
```
testSetAndGetCounter
   self assert: (Counter new count: 22) count equals: 22
```

 Variables | Evaluator

Debugger also precompiles methods



Test is green



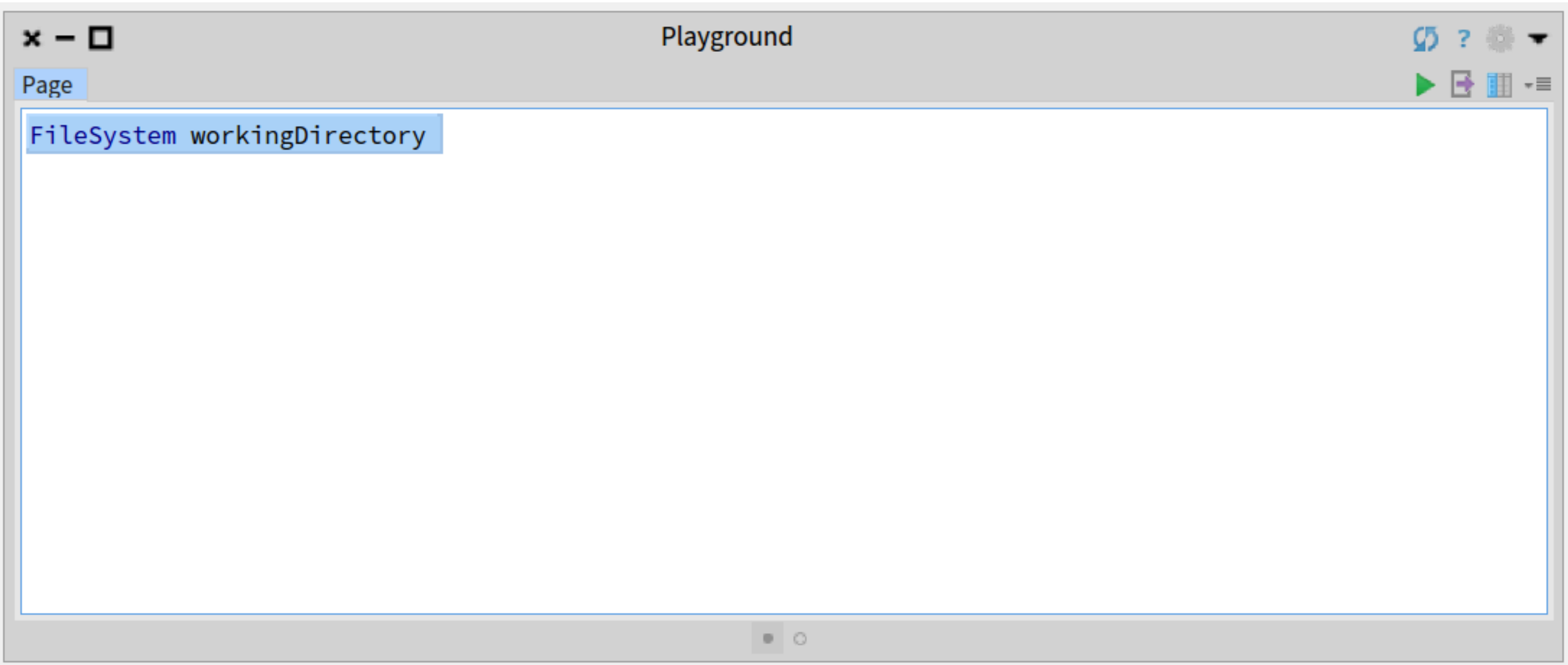
Step back

- The test and its context drove us to code
- Interact with objects: **Access** the **exact** information during the execution **flow**
- Access invariant/relation violations on the **spot**
- Compile on the fly

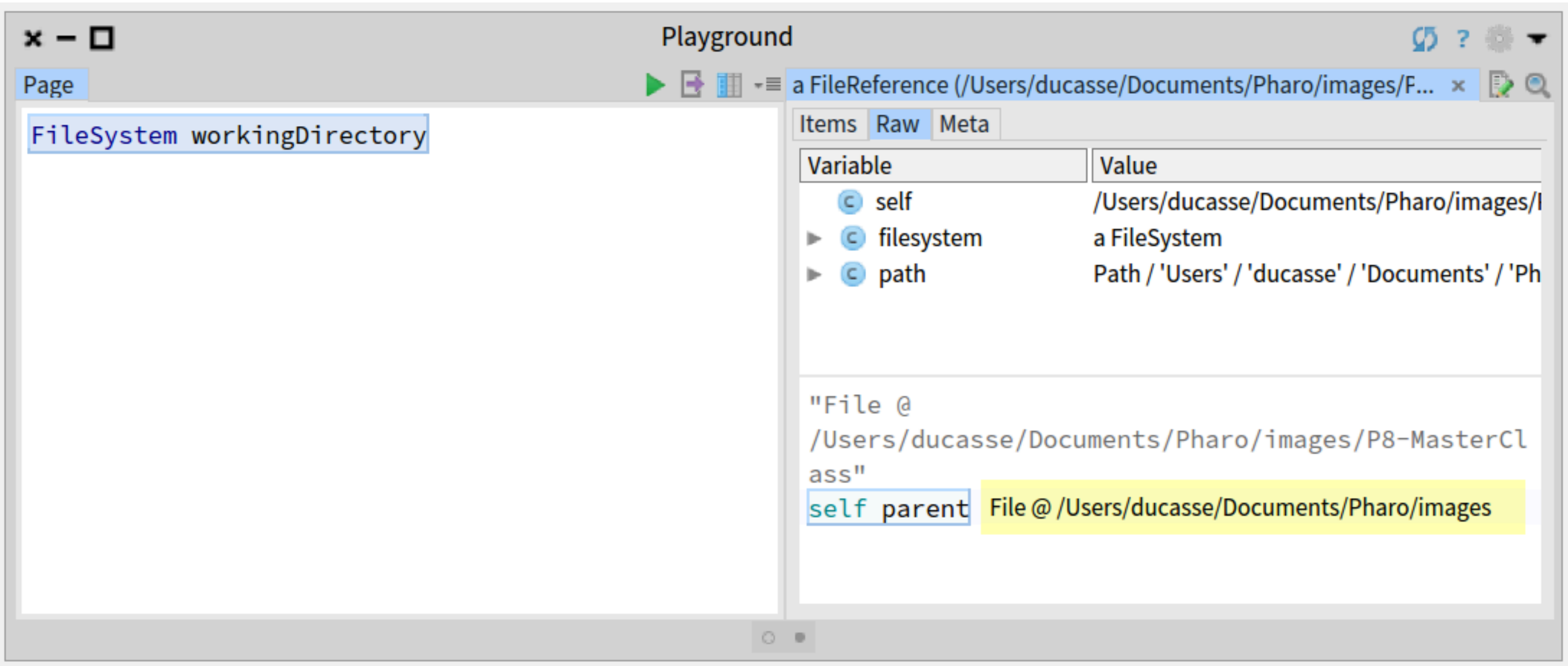
Tools are our friends

But they should adapt to our process

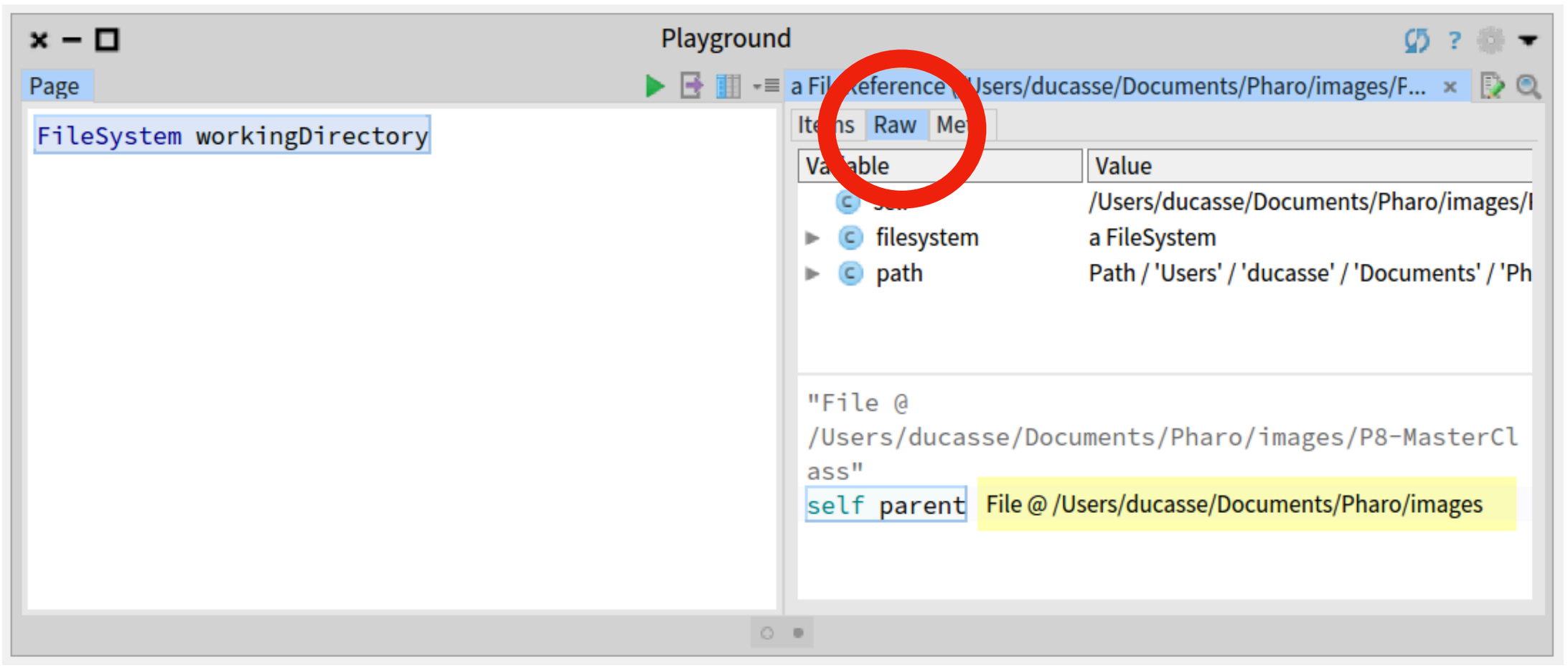
Looking at a file reference



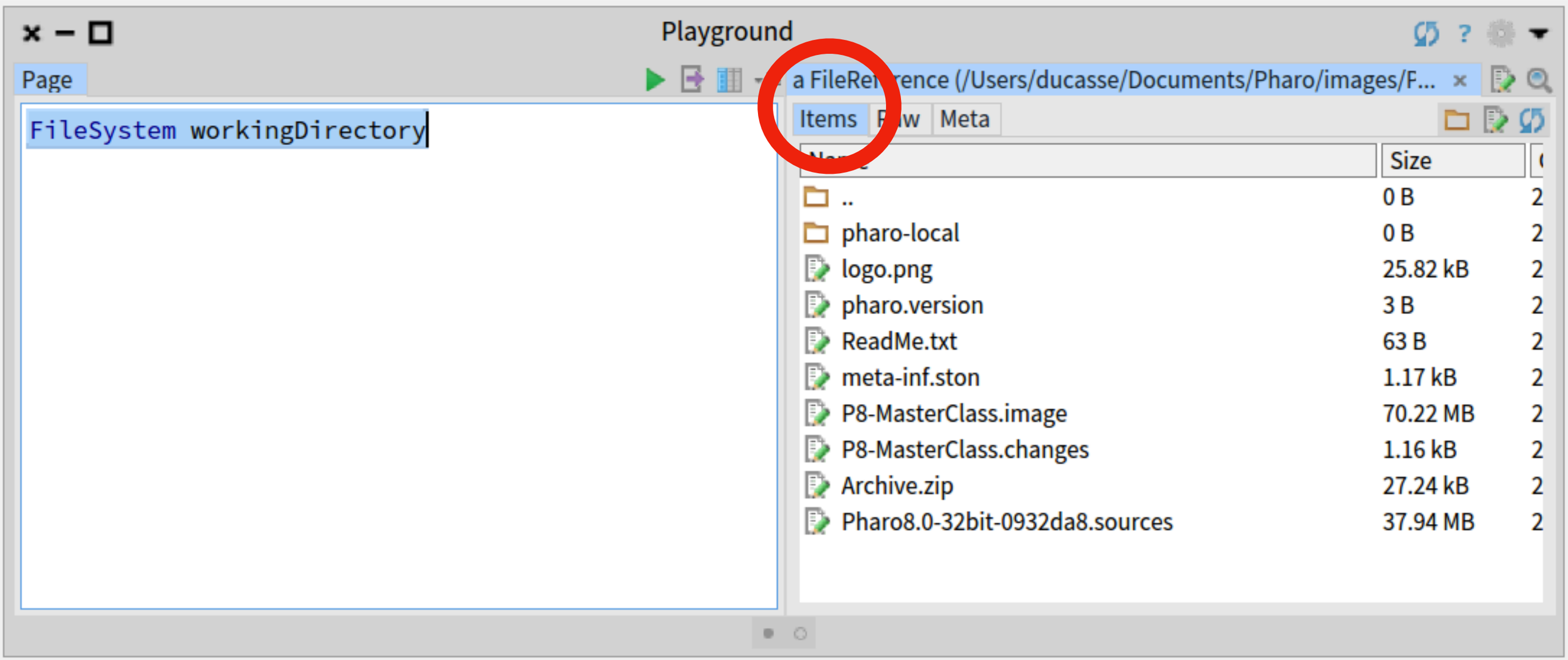
OK I have a file reference: a dull object



The default behavior inspector just shows the implementation

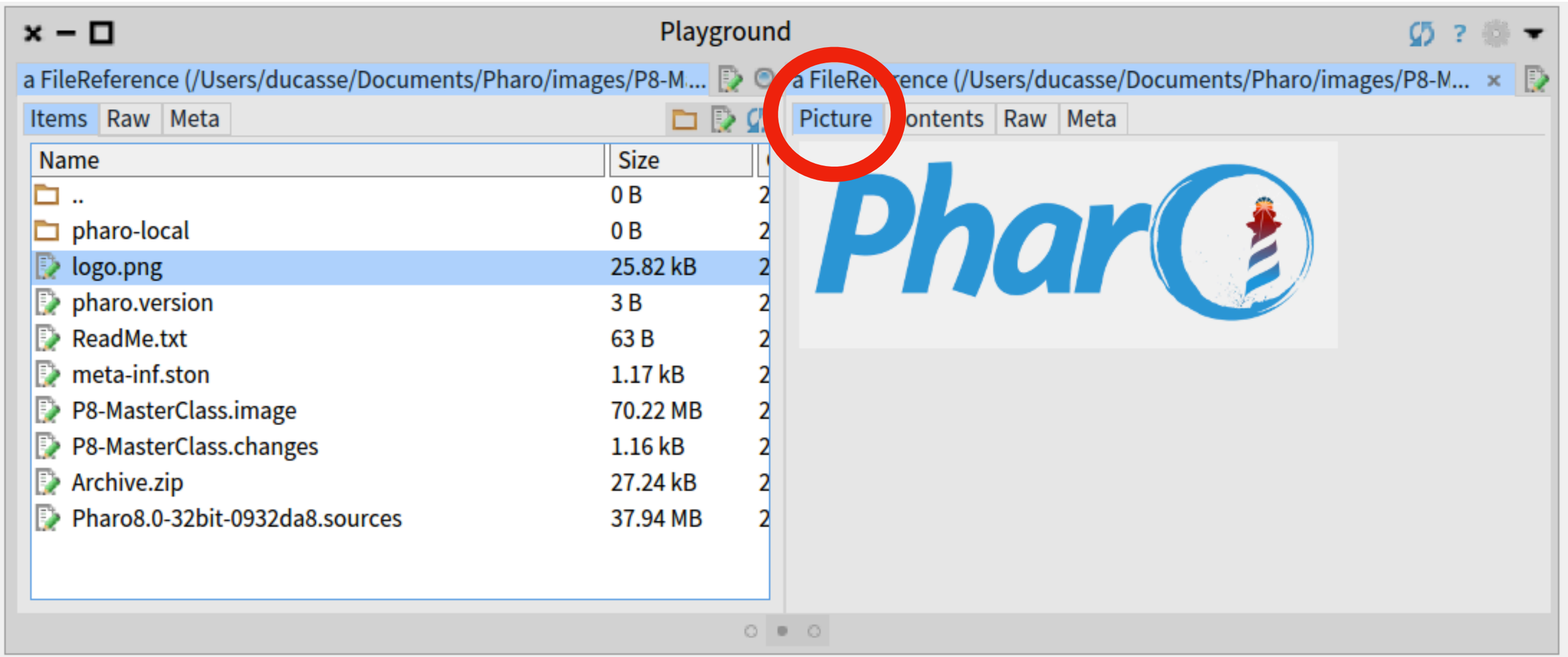


Oh! a file browser in my inspector!



You can create new behavior in the inspector for your objects in your domain!

Ah this png ;)


Playground
a FileReference (/Users/ducasse/Documents/Pharo/images/P8-M...
Items
Raw
Meta
Name
Size
..
0 B
pharo-local
0 B
logo.png
25.82 kB
pharo.version
3 B
ReadMe.txt
63 B
meta-inf.ston
1.17 kB
P8-MasterClass.image
70.22 MB
P8-MasterClass.changes
1.16 kB
Archive.zip
27.24 kB
Pharo8.0-32bit-0932da8.sources
37.94 MB
Picture
Raw
Meta

Looking inside that PNG file

Playground

a FileReference (/Users/ducasse/Documents/Pharo/images/P8-M...

Items | Raw | Meta

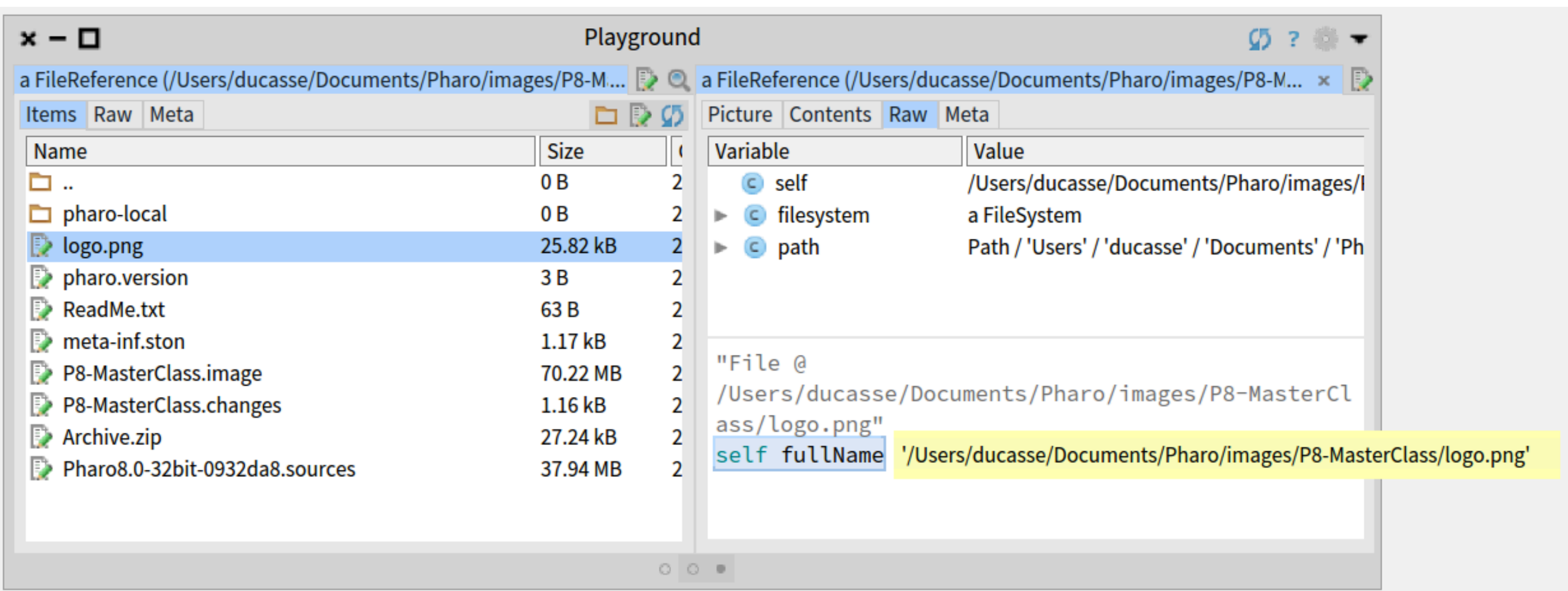
Name	Size	
..	0 B	2
pharo-local	0 B	2
logo.png	25.82 kB	2
pharo.version	3 B	2
ReadMe.txt	63 B	2
meta-inf.ston	1.17 kB	2
P8-MasterClass.image	70.22 MB	2
P8-MasterClass.changes	1.16 kB	2
Archive.zip	27.24 kB	2
Pharo8.0-32bit-0932da8.sources	37.94 MB	2

a FileRef ence (/U ers/ducasse/Documents/Pharo/images/P8-M...

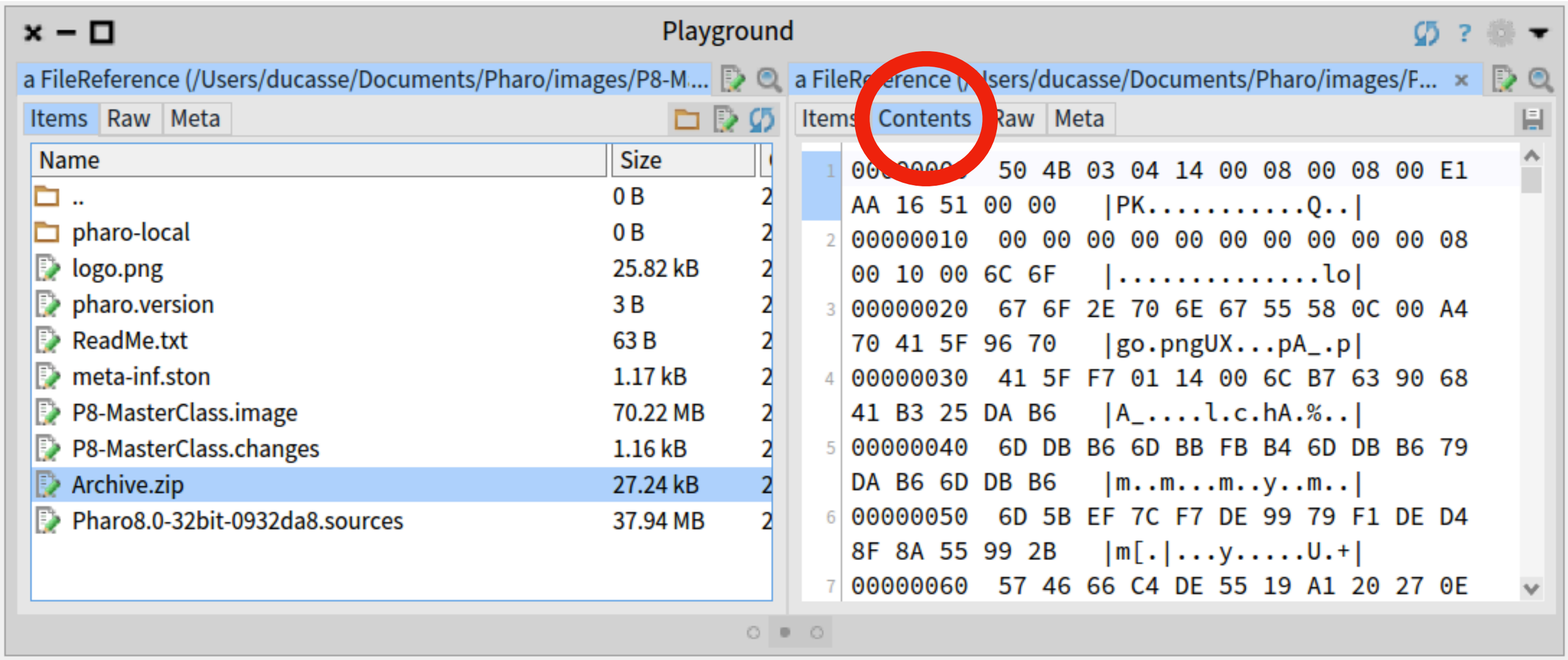
Picture | Contents | Raw | Meta

```
1 0000[illegible]  89 50 4E 47 0D 0A 1A 0A 00 00 00
  0D 49 48 44 52   |.PNG........IHDR|
2 00000010  00 00 01 77 00 00 00 90 08 06 00
  00 00 F3 F6 2B   |...w...........+|
3 00000020  70 00 00 0A D1 69 43 43 50 49 43
  43 20 50 72 6F   |p....iCCPICC Pro|
4 00000030  66 69 6C 65 00 00 48 89 95 97 07
  54 53 69 16 C7   |file..H....TSi..|
5 00000040  BF F7 D2 43 42 4B 08 45 4A E8 4D
  90 5E A5 84 1E   |...CBK.EJ.M.^...|
6 00000050  40 41 3A D8 08 49 48 42 09 21 05
  15 3B 32 38 02   |@A:..IHB.!..;28.|
7 00000060  23 8A 8A 08 96 01 1D 8A 28 38 16
```

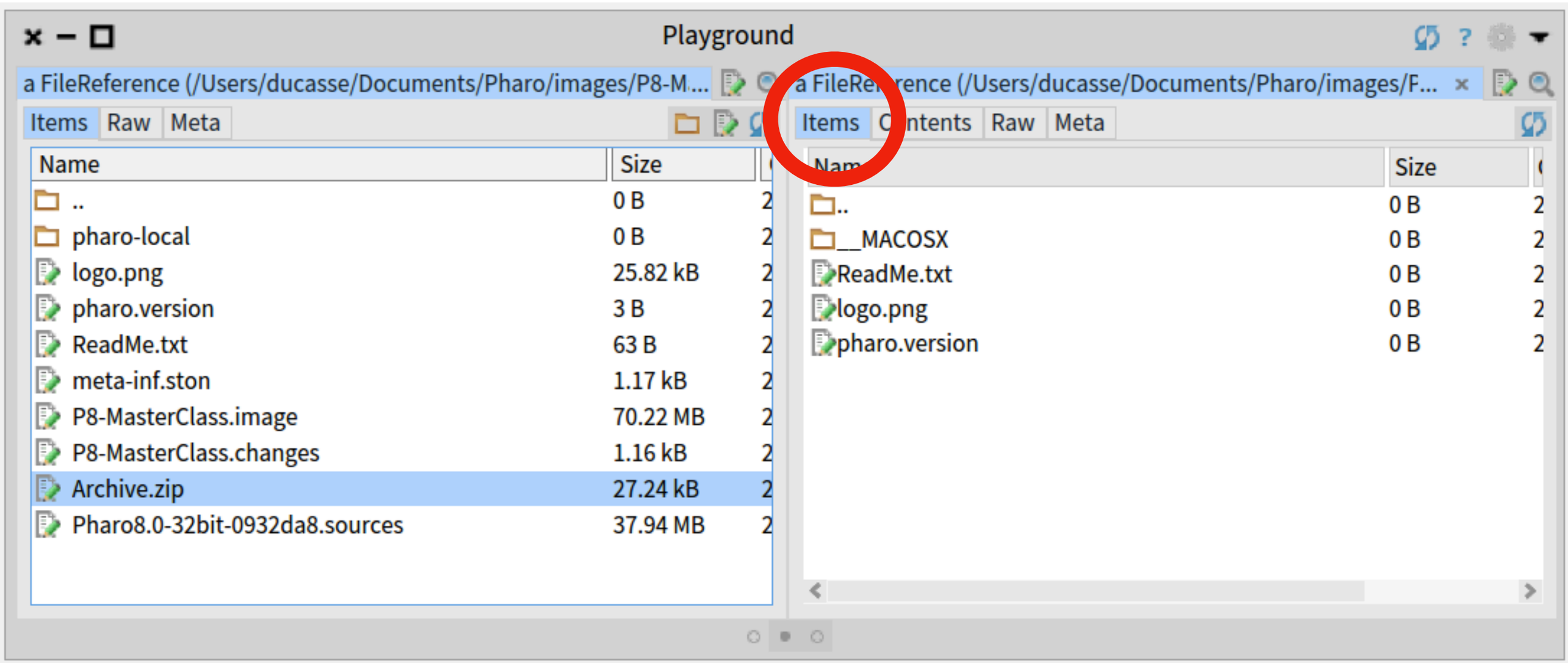
But still a file reference!



See! an archive ‘.zip’

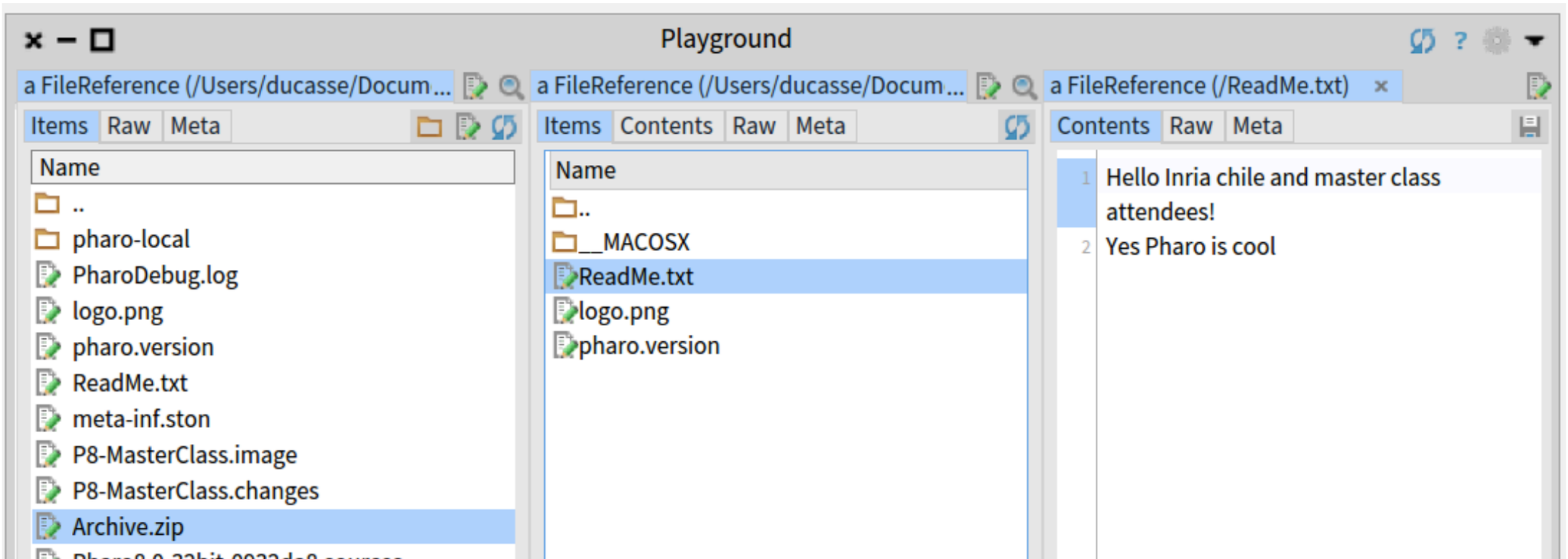


Kind of clear…



An object can expose multiple interactive views!

- **Users** get the best view for **their** tasks!
- **Users** can add **views** to **their** domain objects



It is cool but it is not magic
You can define your own
Yes you can define your own!

Implementing a pane!

Method: AbstractFileReference>>inspectionPng:

Public
Extensions
FileSystem-Core-Tests
FileSystem-Disk
FileSystem-Disk-Tests
FileSystem-Memory
FileSystem-Memory-Tests
FileSystem-Path
FileSystem-PlatformSpecific-T
FileSystem-Tests-Attributes
FileSystem-Zip
Filter...

AbstractFileReference
FileReference
FileSystem
FileSystemDirectoryEntry
DiskDirectoryEntry
DiskSymlinkDirectoryEntry
MemoryDirectoryEntry
FileSystemPermission
FileVersionner
NewFileVersionner
Filter...

instance side
extensions
Deprecated14
Native-Browser
NewTools-Inspector-Extensic
NewTools-SettingsBrowser
NewTools-Spotter-Extension
System-Settings-Core
Zinc-Resource-Meta-Core
ToDeprecate
accessing
comparing

asZnUrl
inspectionCompressedItemsConte
inspectionContents:
inspectionContentsContext:
inspectionFuelContext:
inspectionGif:
inspectionGifContext:
inspectionItems:
inspectionItemsContext:
inspectionJpeg:
inspectionJpegContext:
inspectionPng:

All Packages | Scoped View | Projects | Flat | Hier. | Inst. side | Class side | Methods | Vars | Class refs. | Implementors | Ser

AbstractFileRef | Comment | inspectionPng: | Inst. side methc

```
inspectionPng: aBuilder
    <inspectorPresentationOrder: 0 title: 'Picture'>

    ^ aBuilder newImage
        autoScale: true;
        image: (self binaryReadStreamDo: [ :stream | PNGReadWriter formFromStream: stream ]);
        yourself
```

1/7 [1] -- 7/7 [11] Edit package: NewTools-Inspector-Extensions | Extension | Format | Line numbers | Word Wrap

Implementing a pane!

```
inspectionPng: aBuilder
    <inspectorPresentationOrder: 0 title: 'Picture'>
    ^ aBuilder newImage
            autoScale: true;
            image: (self binaryReadStreamDo:
[ :stream | PNGReadWriter formFromStream:
stream ]);
            yourself
```

Tweaking and manipulating data

https://github.com/SquareBracketAssociates/booklet-CountryTutorial

The world map challenge…

svg = xml

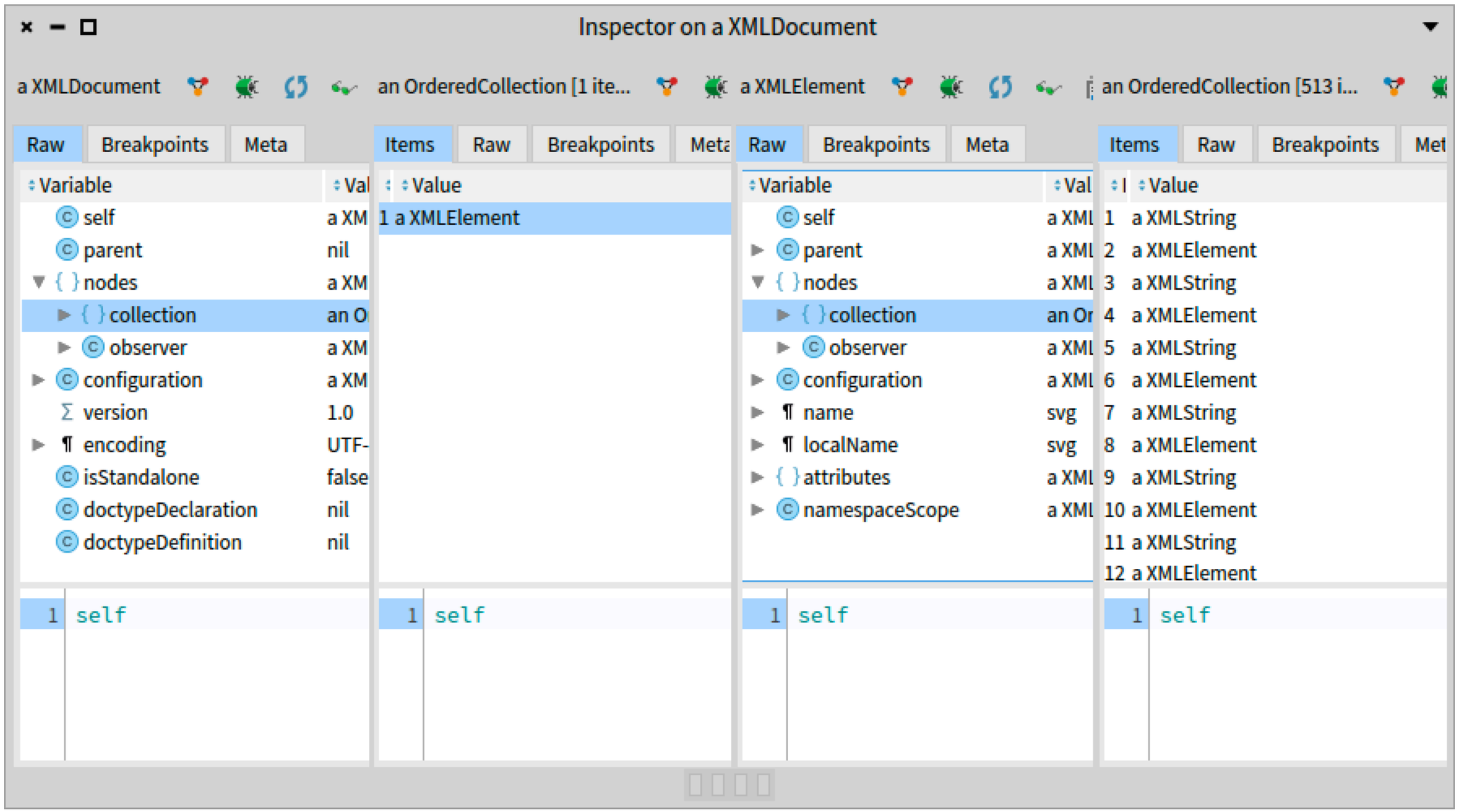
An exercise of one hour…

- DOM navigation
- Turn data into **objects**
- Enhance object **printing**
- Script **visualization** to get **more** information
- **Extend** inspector with **new** behavior
- Support tool definition

How to get an inspector on the DOM?

```
(XMLDOMParser
    parse: 'world.svg' asFileReference readStream contents)

    document nodes inspect
```

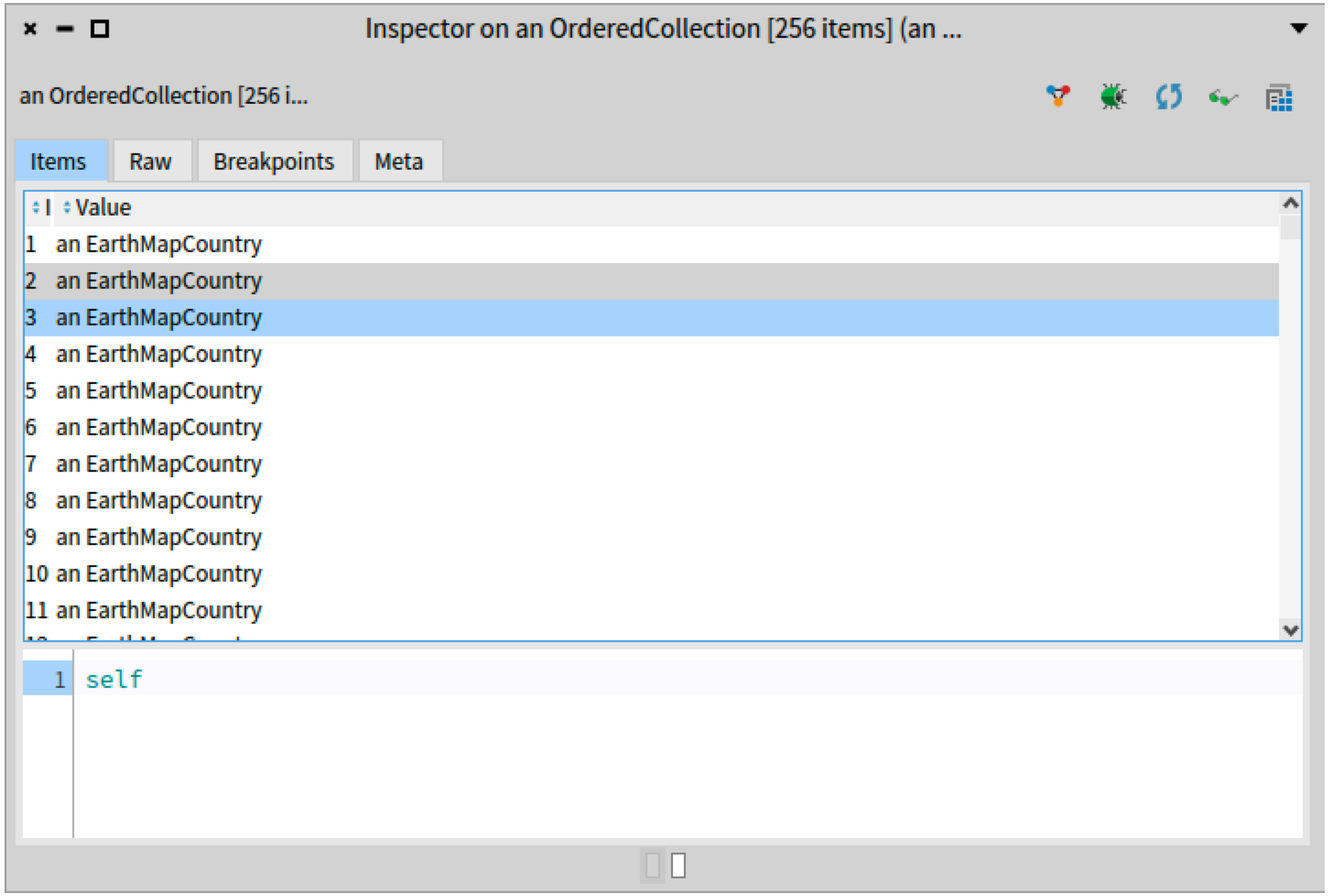
Navigating SVG File



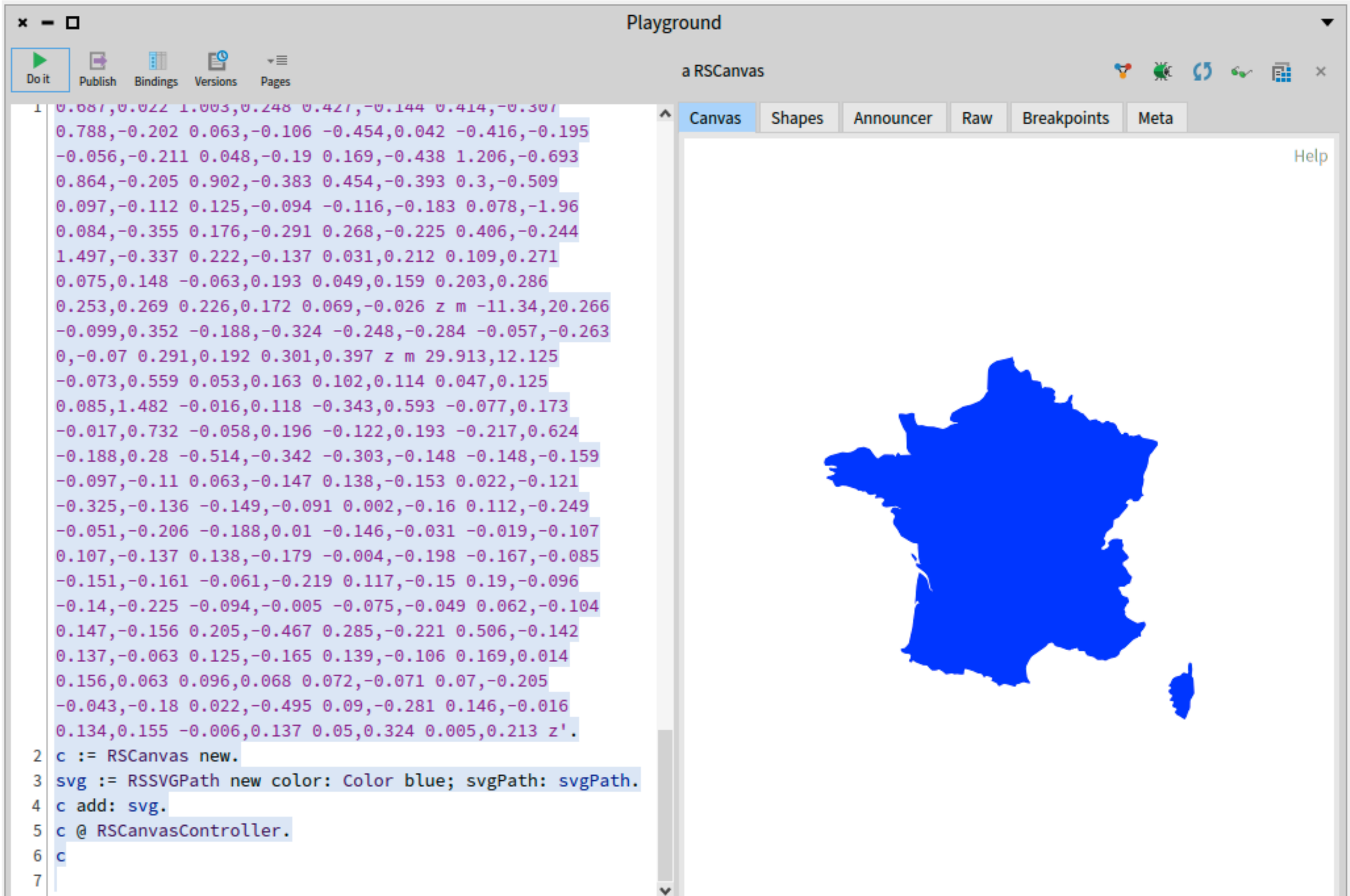
Turning data into little objects

```
Object << #EarthMapCountry
        slots: { #svgPath . #name . #code };
        package: ‘SummerSchool'

EarthMapCountry >> fromXML: aXMLElement
        svgPath := aXMLElement attributeAt: 'd'.
        name := aXMLElement attributeAt: 'title'.
        code := aXMLElement attributeAt: 'id'.
```

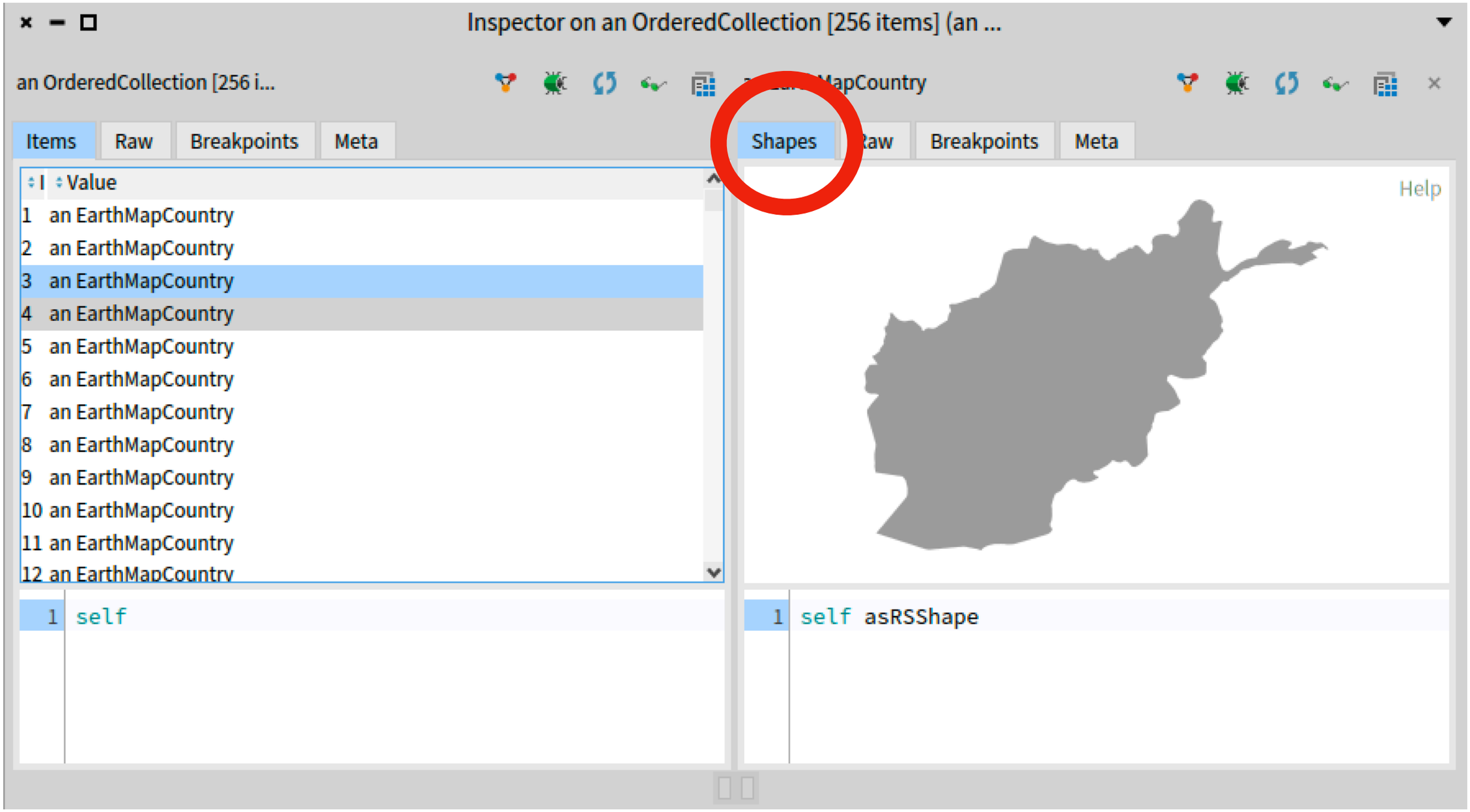
Once we turn XML into objects



Scripting visualization with Roassal



Now a list on steriods!

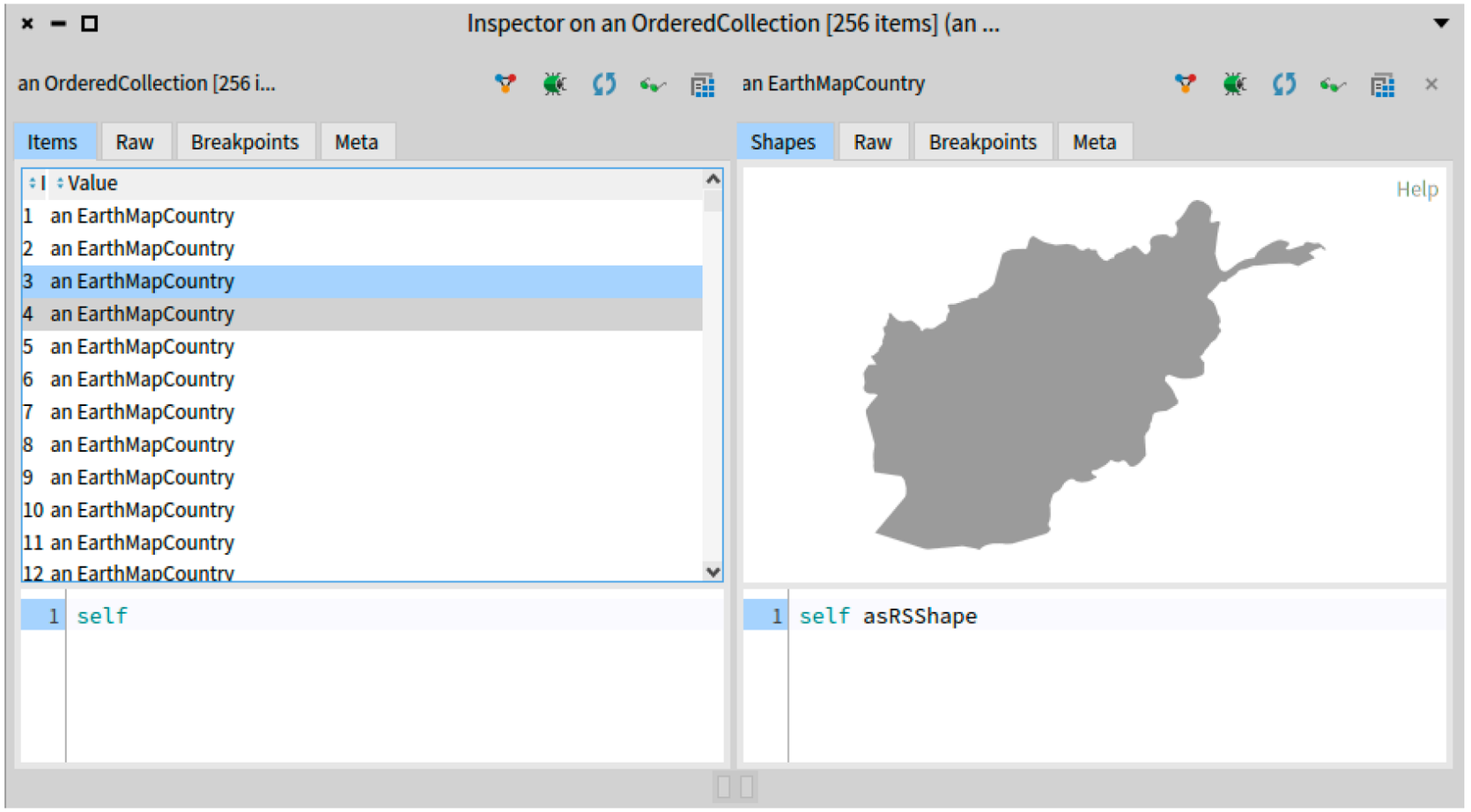


Extending the inspector

```
EarthMapCountry >> asRSShape
        ^ RSSVGPath new svgPath: svgPath
```

```
EarthMapCountry >> inspectorShape: aBuilder
    <inspectorPresentationOrder: 0 title: 'Shape'>
    | canvas |
    canvas := RSCanvas new.
    canvas add: self asRSShape.
    ^ SpRoassalInspectorPresenter new canvas: canvas
```

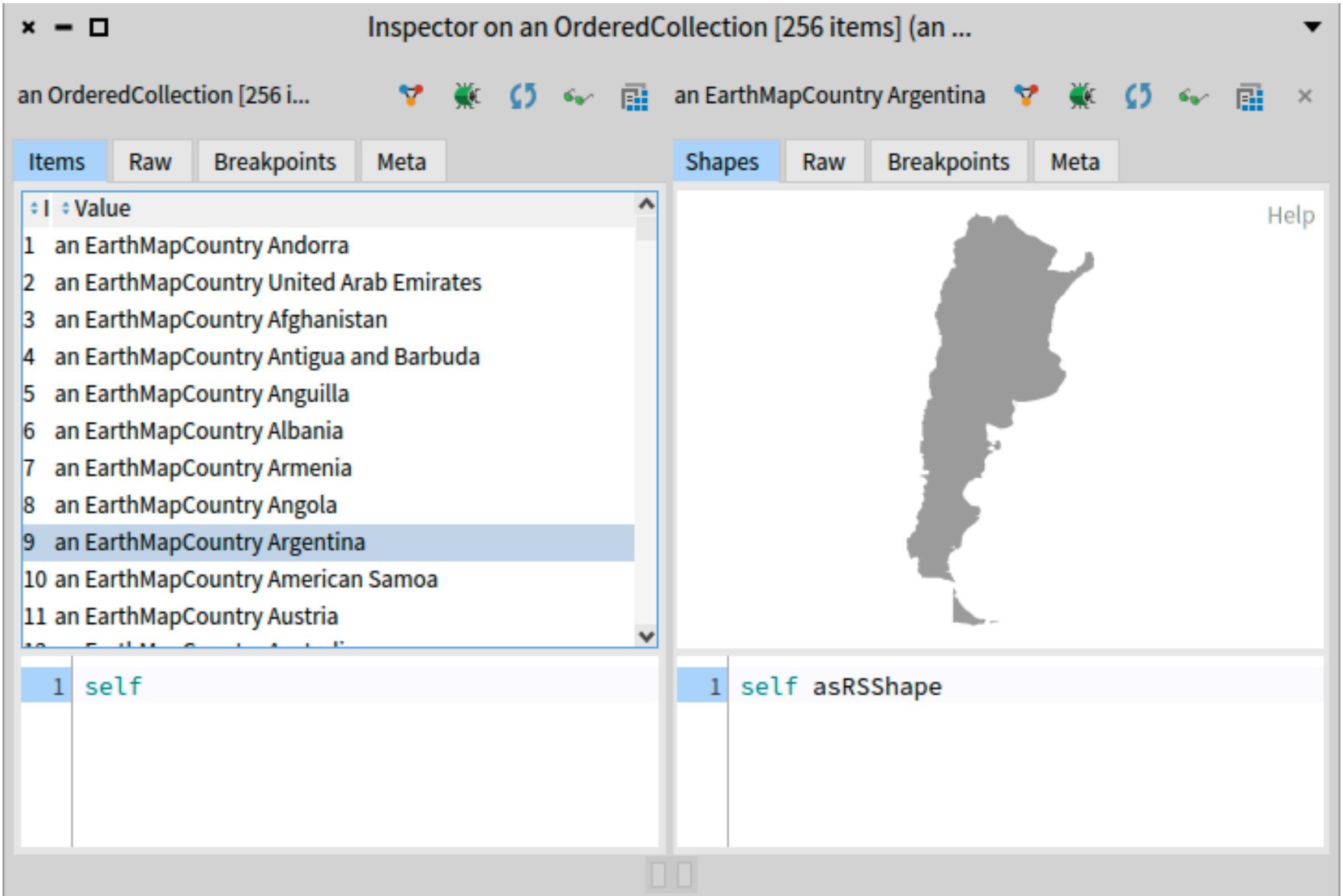
I would like to see the name of the countries



printOn: as easy as

```
EarthMapCountry >> printOn: aStream
    super printOn: aStream.
    aStream nextPutAll: ' ', name
```

With printOn: defined



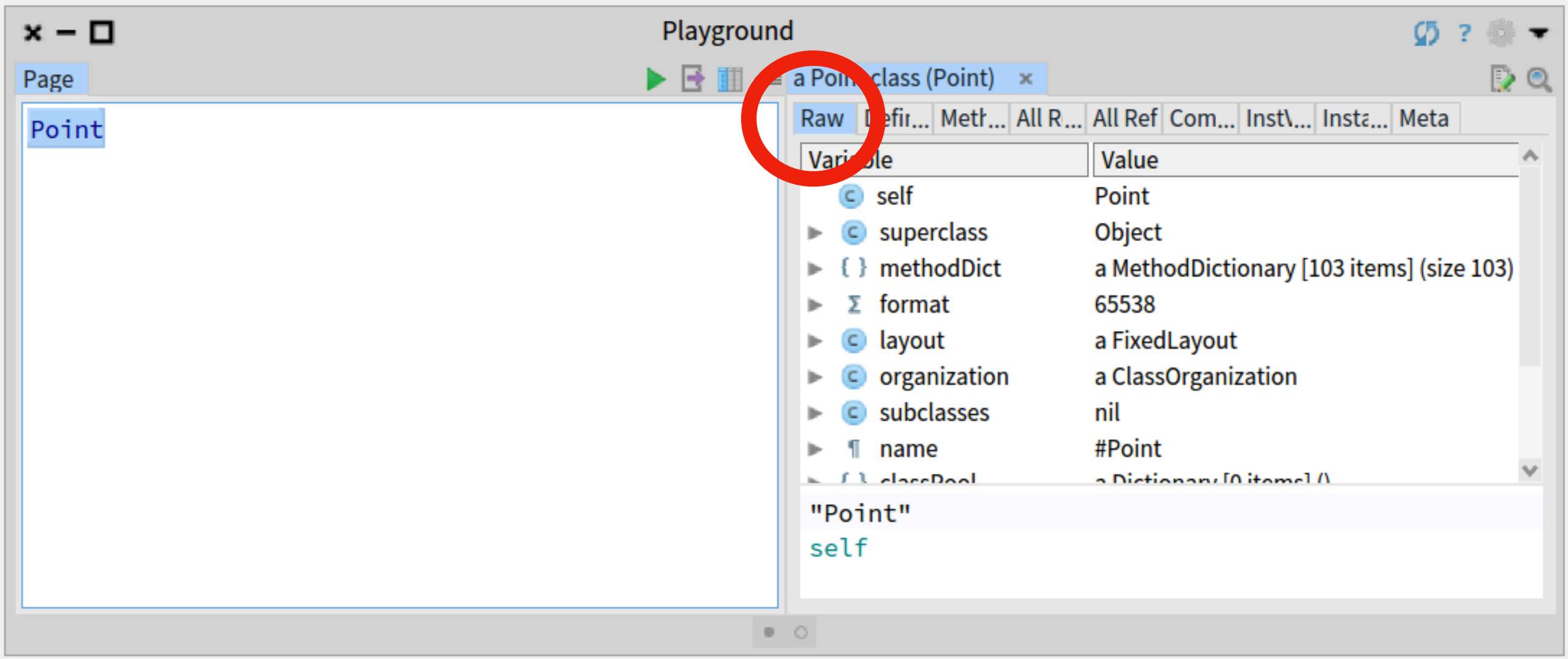
Step back

in one hour
get from files to objects
with powerful views
and a mini browser of
countries

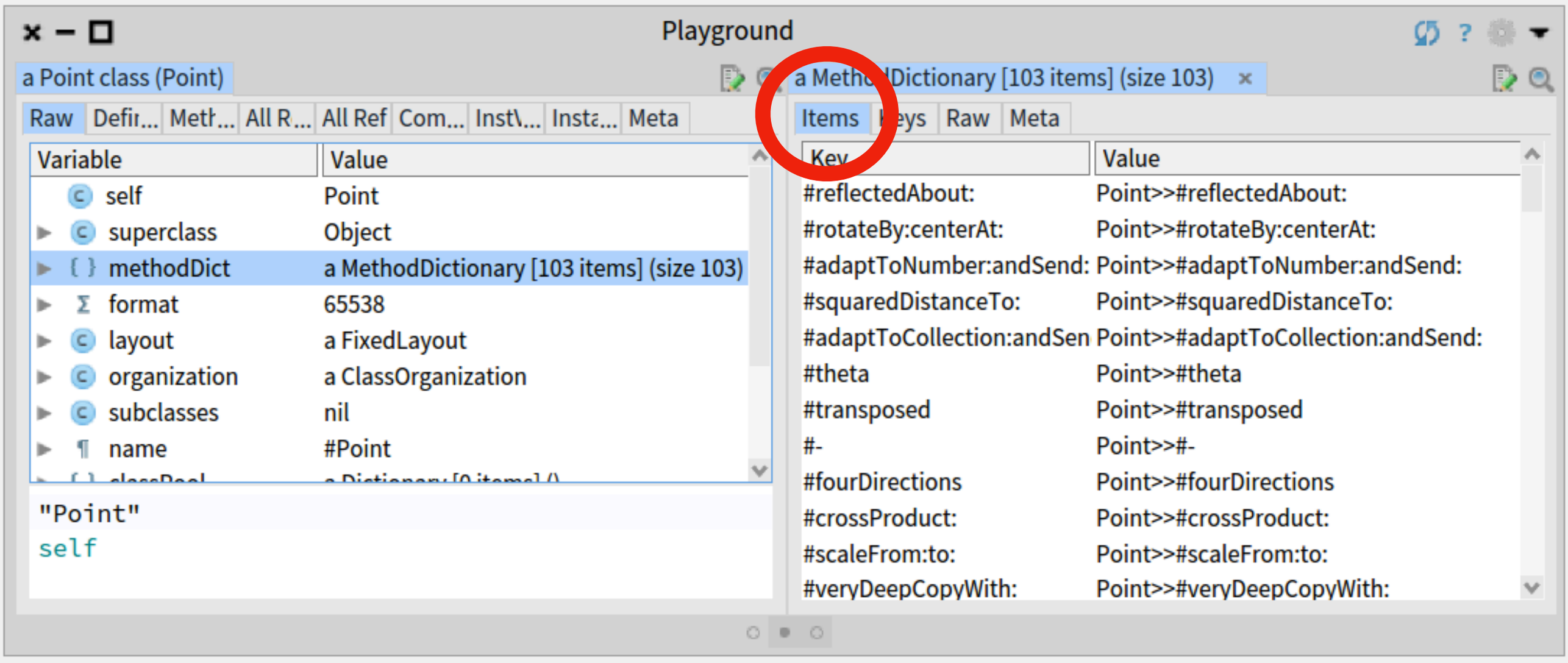
Deeper…

JULES VERNE
VOYAGE AU CENTRE DE LA TERRE
PAT BOONE · JAMES MASON · ARLENE DAHL · DIANE BAKER
CINEMASCOPE
CHARLES BRACKETT · HENRY LEVIN · WALTER REISCH et CHARLES BRACKETT

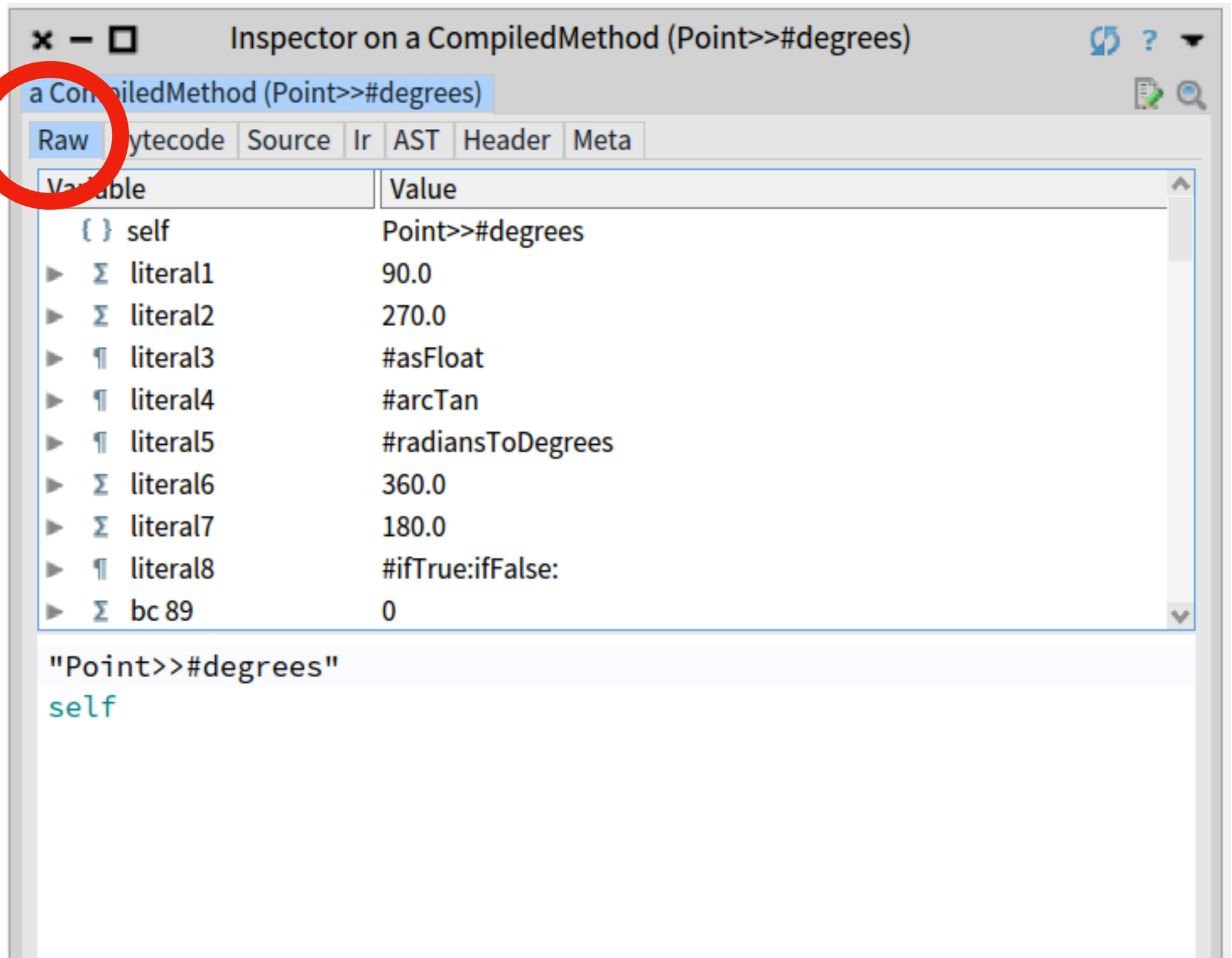
A class is an object we can inspect!



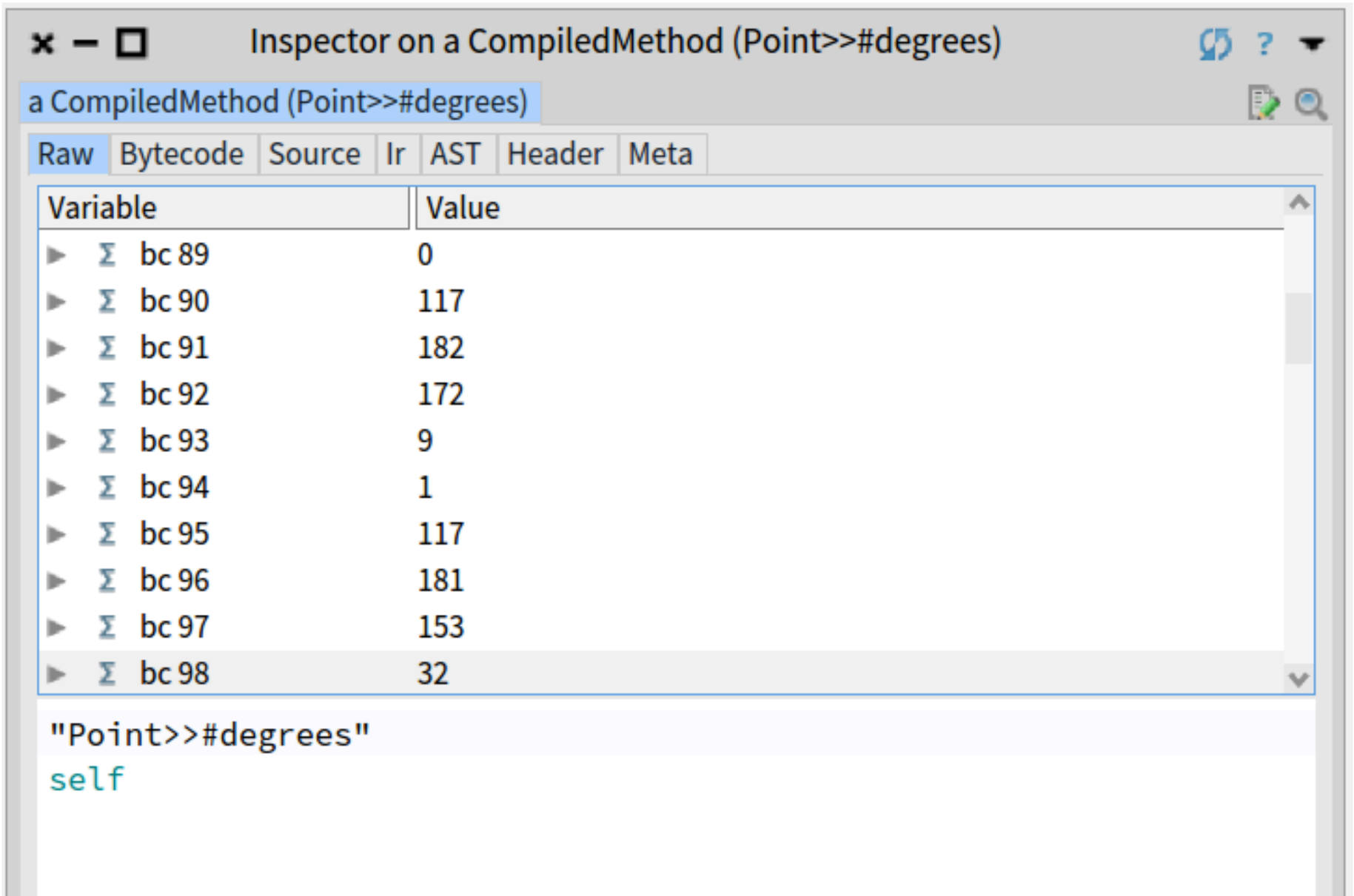
“A class has a method dictionary” they said



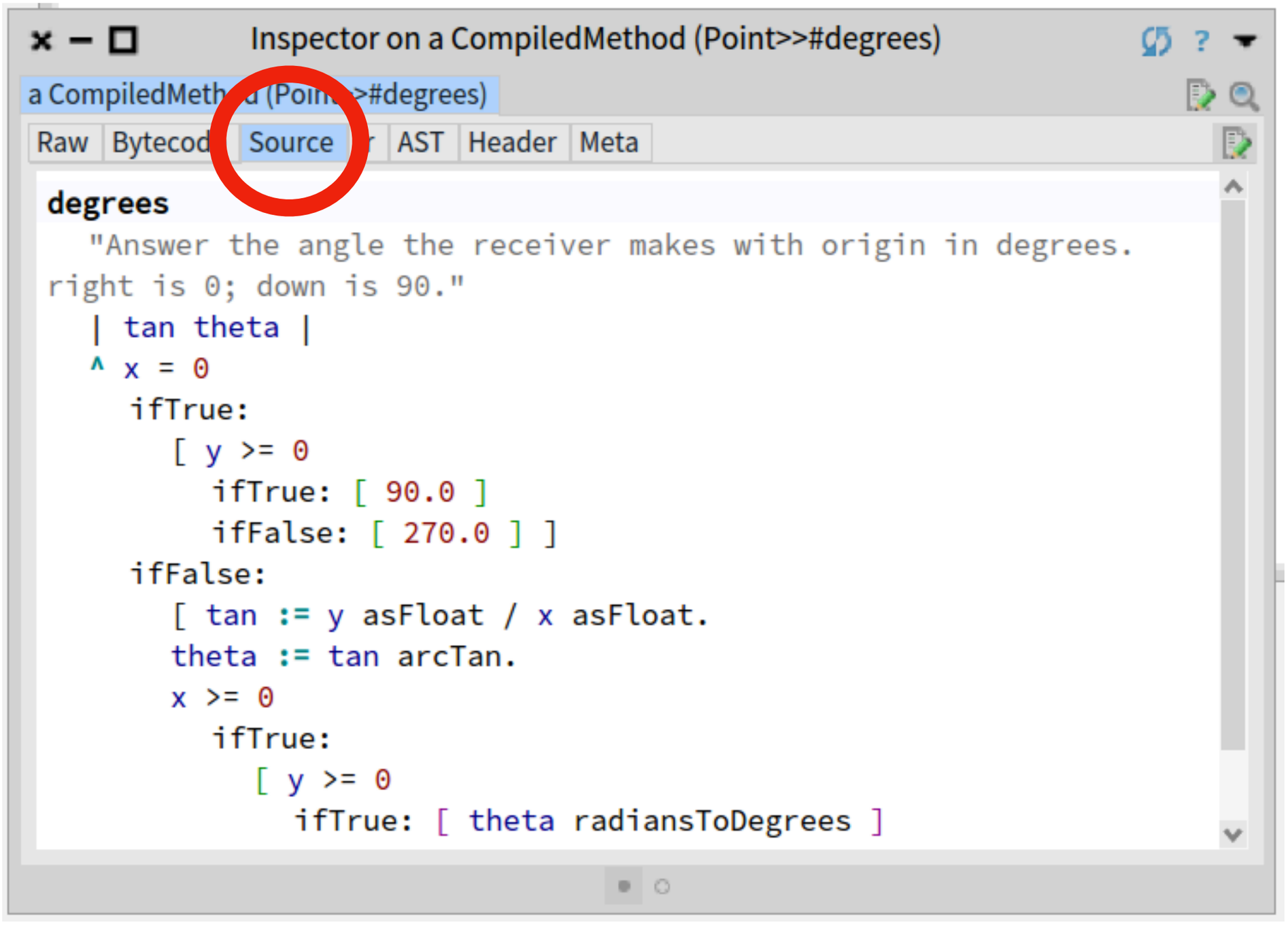
Dissecting one method object



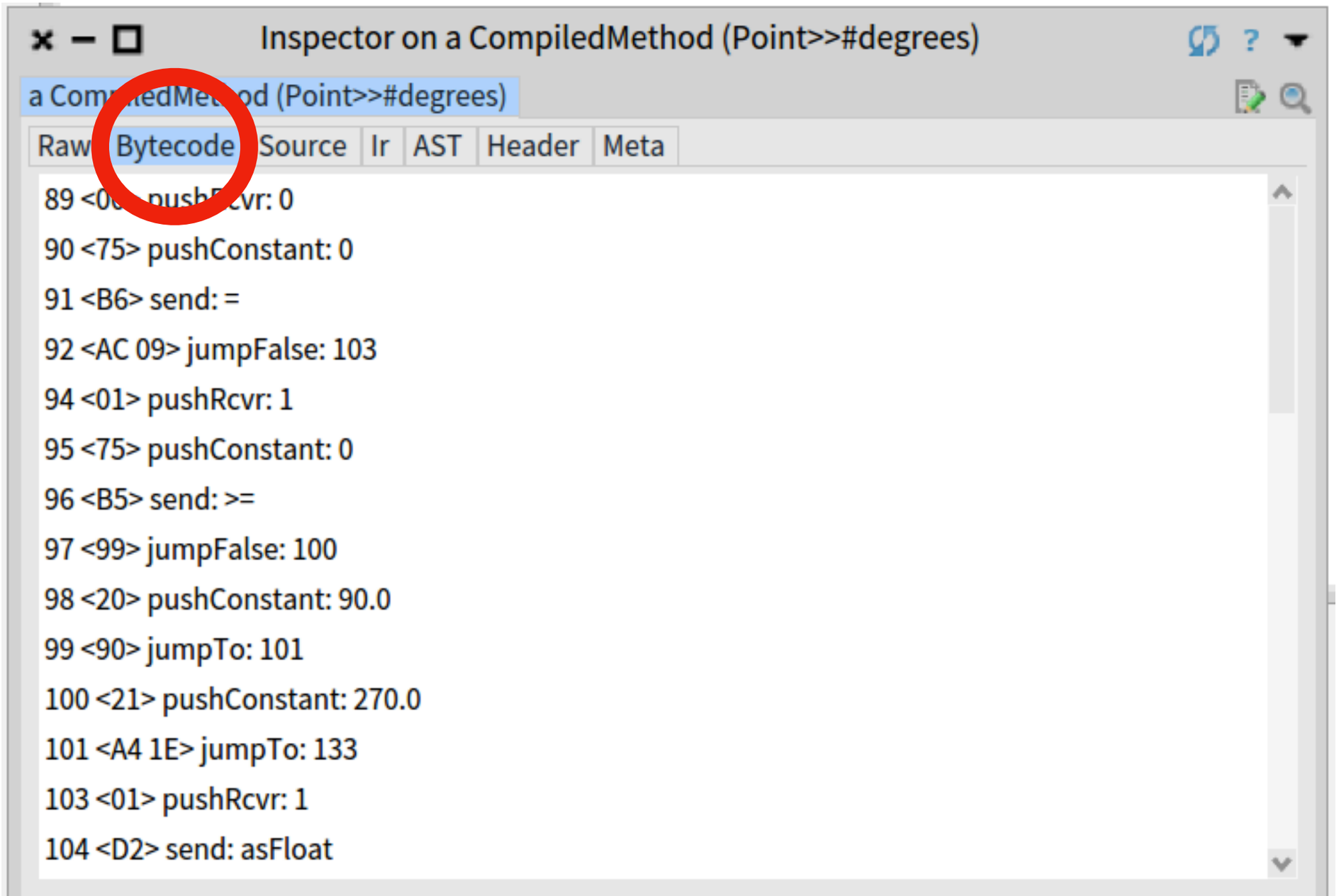
I do not want to be a decompiler!



It looks like a method



Numbers are not that obscure finally



Mapping to the right abstraction helps

from AST to Source

Inspector on a CompiledMethod (Point>>#degrees)

a CompiledMethod (Point>>#degrees) | a RBMessageNode (RBMessageNode(y >= 0))

Raw | Bytecode | Source | Ir | AST | Header | Meta

Raw | Tree | Scopes | Source cc... | AST Dump | Meta

- RBMethodNode(degrees "Answer the angle the receiver makes with)
  - RBSequenceNode(| tan theta | ^ x = 0 ifTrue: [y >= 0 ifTrue:)
    - RBTemporaryNode(tan)
    - RBTemporaryNode(theta)
    - RBReturnNode(^ x = 0 ifTrue: [y >= 0 ifTrue: [90.0]
      - RBMessageNode(x = 0 ifTrue: [y >= 0 ifTrue: [90.0]
        - RBMessageNode(x = 0)
          - RBInstanceVariableNode(x)
          - RBLiteralValueNode(0)
        - RBBlockNode([y >= 0 ifTrue: [90.0] ifFalse: [270.0]])
          - RBSequenceNode(y >= 0 ifTrue: [90.0] ifFalse: [270
            - RBMessageNode(y >= 0 ifTrue: [90.0] ifFalse: [270.
              - RBMessageNode(y >= 0)
              - RBBlockNode([90.0])
              - RBBlockNode([270.0])

```
degrees
  "Answer the angle the receiver makes with
origin in degrees. right is 0; down is 90."
  | tan theta |
  ^ x = 0
    ifTrue:
      [ y >= 0
        ifTrue: [ 90.0 ]
        ifFalse: [ 270.0 ] ]
    ifFalse:
      [ tan := y asFloat / x asFloat.
      theta := tan arcTan.
      x >= 0
        ifTrue:
          [ y >= 0
            ifTrue: [ theta radiansToDegrees ]
            ifFalse: [ 360.0 + theta
radiansToDegrees ] ]
```

Yes pushRcvr: 1 means the second field!

from bytecode to source

Inspector on a CompiledMethod (Point>>#degrees)

a CompiledMethod (Point>>#degrees)

Raw | Bytecode | Source | Ir | AST | Header | Meta

```
89 <00> pushRcvr: 0
90 <75> pushConstant: 0
91 <B6> send: =
92 <AC 09> jumpFalse: 103
94 <01> pushRcvr: 1
95 <75> pushConstant: 0
96 <B5> send: >=
97 <99> jumpFalse: 100
98 <20> pushConstant: 90.0
99 <90> jumpTo: 101
100 <21> pushConstant: 270.0
101 <A4 1E> jumpTo: 133
103 <01> pushRcvr: 1
104 <D2> send: asFloat
105 <00> pushRcvr: 0
106 <D2> send: asFloat
```

a SymbolicBytecode (94 <01> pushRcvr: 1)

Raw | Source | SourceNode | Meta

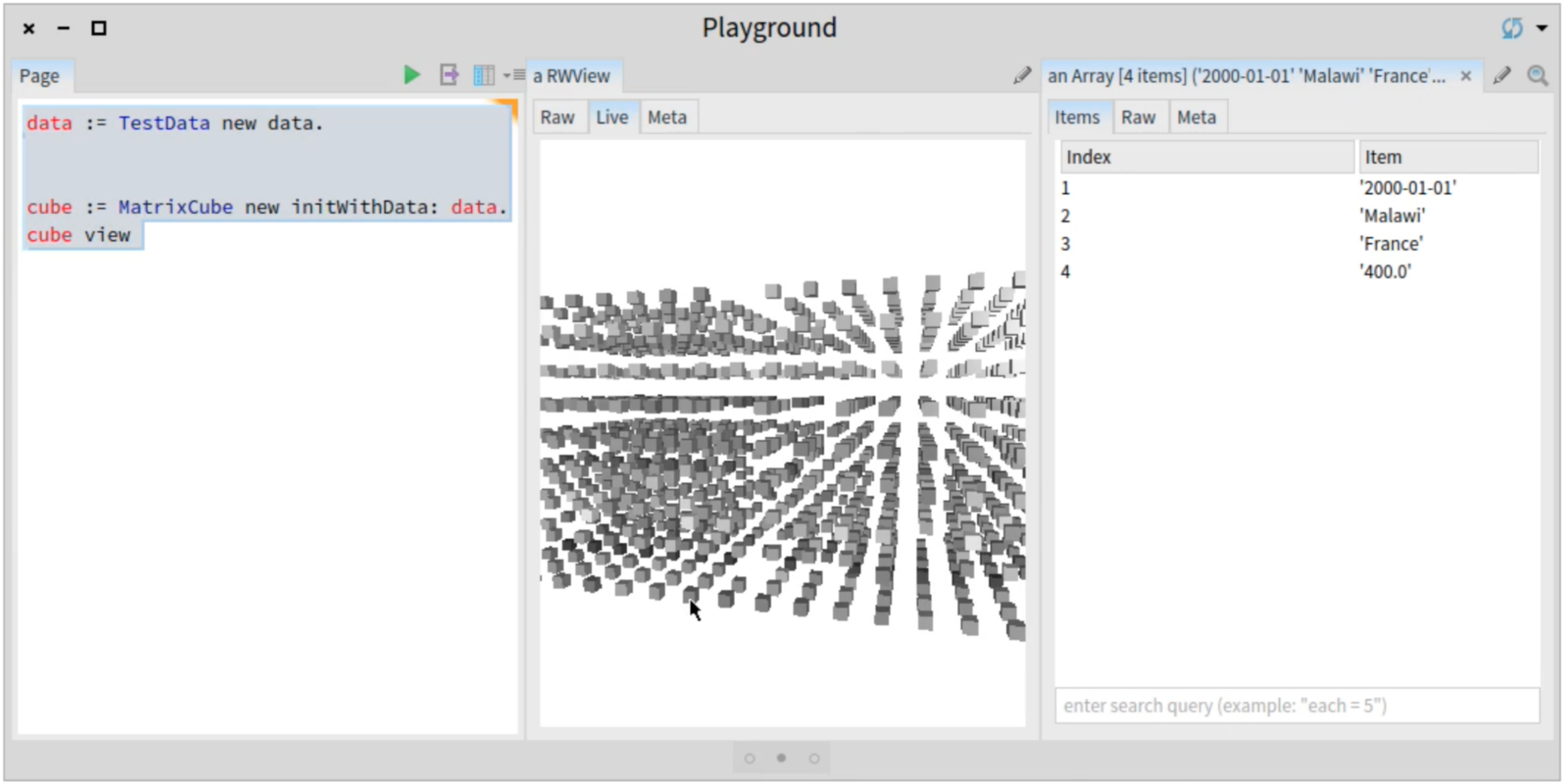
```
origin in degrees. right is 0; down is 90."
  | tan theta |
  ^ x = 0
    ifTrue:
      [ y >= 0
        ifTrue: [ 90.0 ]
        ifFalse: [ 270.0 ] ]
    ifFalse:
      [ tan := y asFloat / x asFloat.
      theta := tan arcTan.
      x >= 0
        ifTrue:
          [ y >= 0
            ifTrue: [ theta radiansToDegrees ]
            ifFalse: [ 360.0 + theta
radiansToDegrees ] ]
        ifFalse: [ 180.0 + theta
radiansToDegrees ] ]
```

Step back

see complex and obscure objects
at different levels of abstractions
with a bridge between them

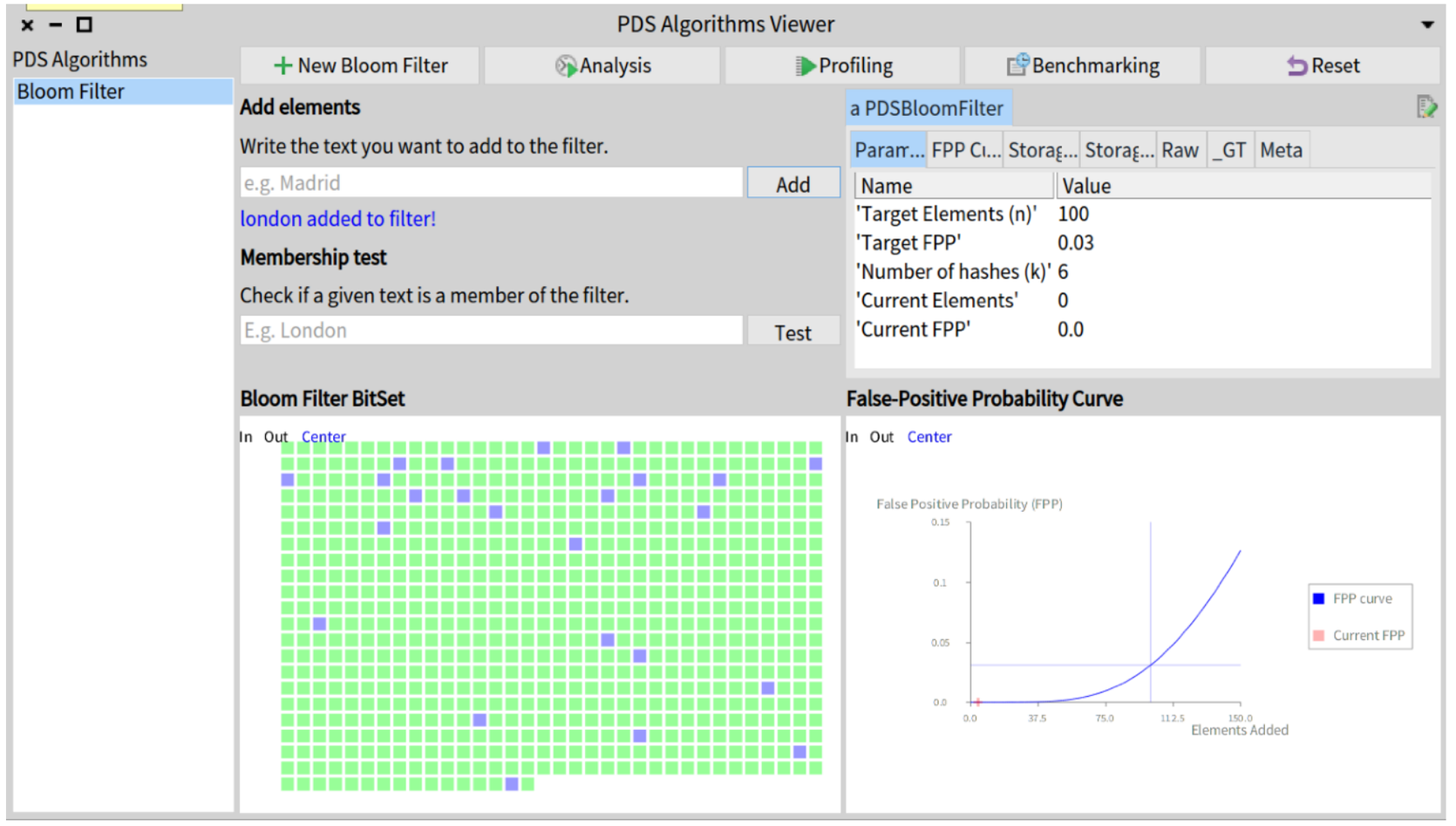
Examples of developer’ extensions

E.g. Inspecting live a temporal database

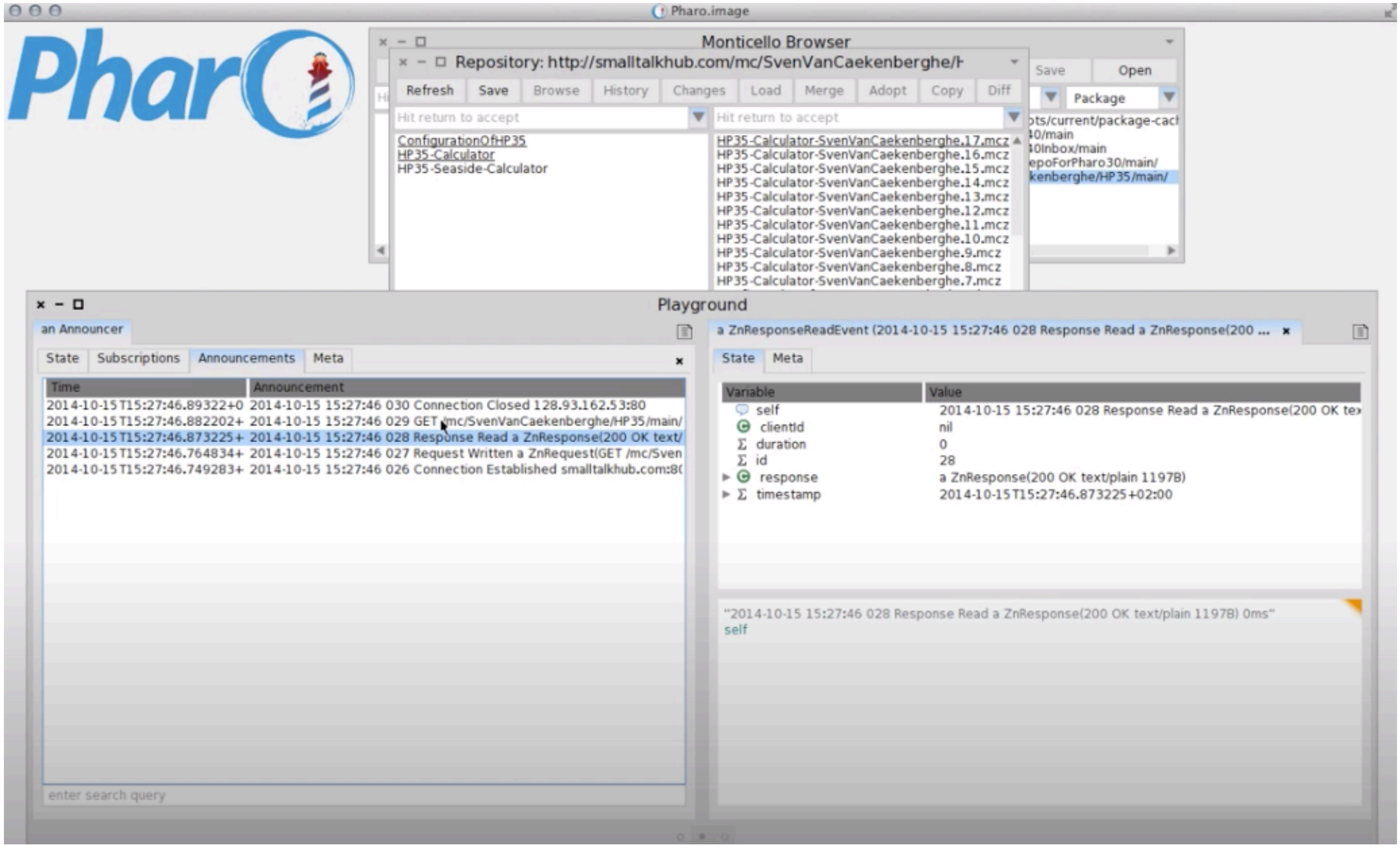


Probabilistic Data Structure

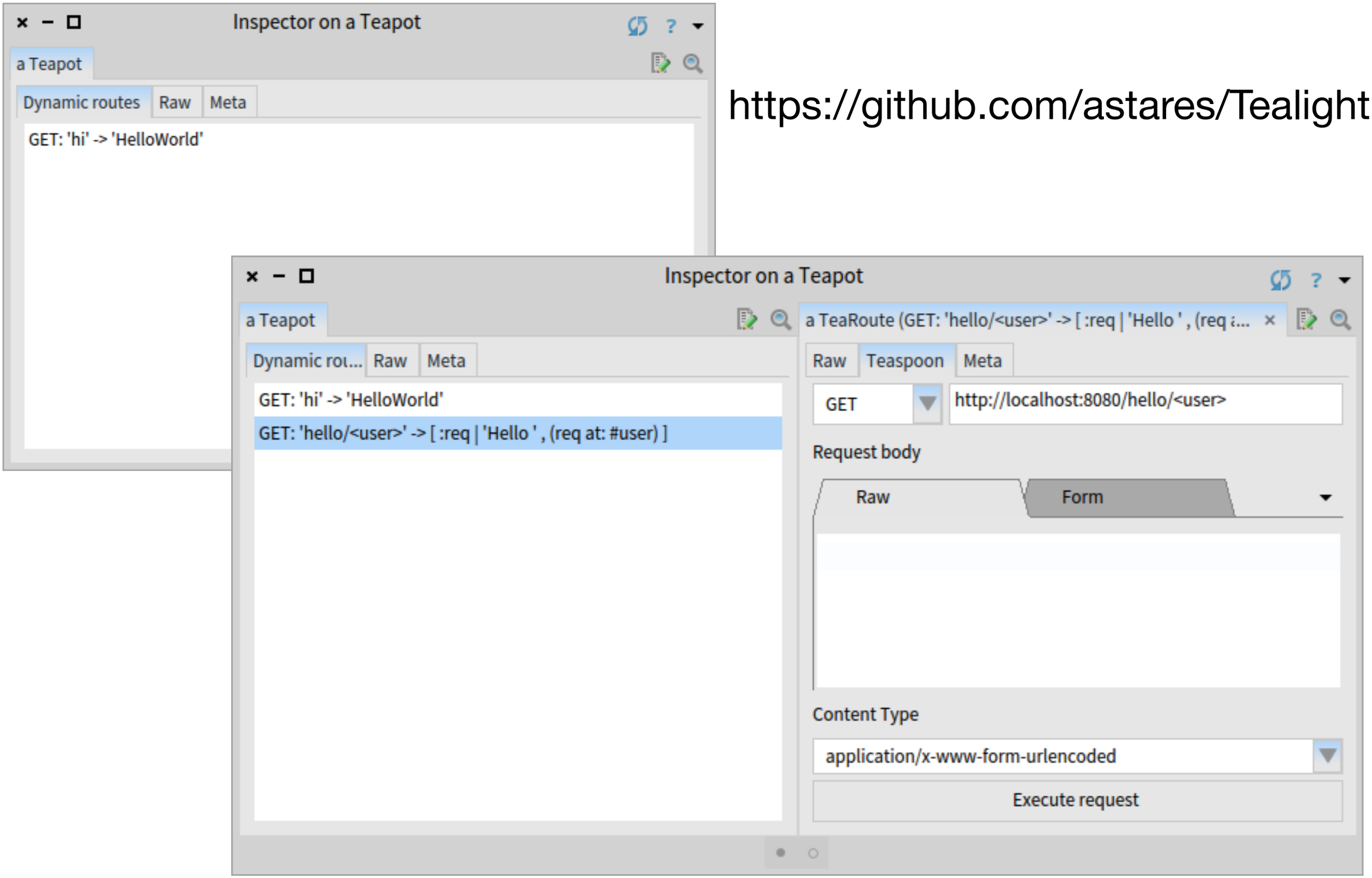
- https://github.com/osoco/PharoPDS
- Defined new data structure
- **And tools to understand it**



HTTP traffic analysis



HTTPS routing



https://github.com/astares/Tealight

Handling data

Pharo Browse Debug Sources System Library Windows Help

Inspector on a DataFrame [150 items] (a DataSeries('sepal length (cm)'->5.1 'sepal width (cm)'->3.5 'petal length (cm)'->1.4 'petal width (cm)'->0.2 'species'->'setosa') a DataSeries('sepal length (cm)'->4.9 'sepal

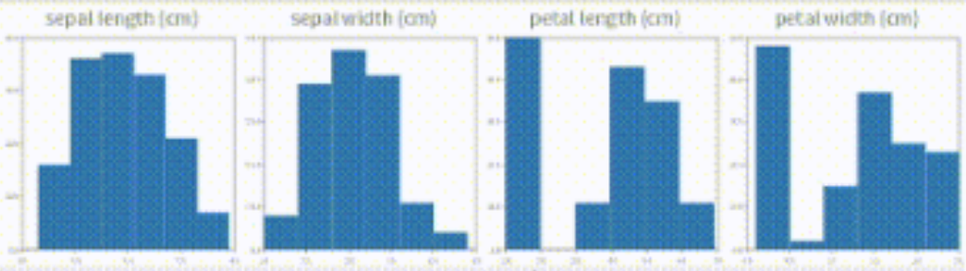
a DataFrame [150 items] (a ...

Inspector | Items | Raw | Breakpoints | Meta

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7000000000000002	0.4	setosa
4.6	3.4	1.4	0.30000000000000004	setosa
5.0	3.4	1.5	0.2	setosa
4.4	2.9	1.4	0.2	setosa

Do it | Publish | Bindings

Statistic	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
1st Quartile	5.1	2.8	1.6	0.30000000000000004	0
3rd Quartile	6.4	3.3	5.1	1.8	0
Median	5.8	3.0	4.35	1.3	0
Minimum	4.3	2.0	1.0	0.1	0
Maximum	7.9	4.4	6.9	2.5	0
Variance	0.6856935123042504	0.18997941834451895	3.1162778523489942	0.5810062639821028	0
Standard deviation	0.8280661279778628	0.43586628493669816	1.765298233259467	0.7622376689603465	0

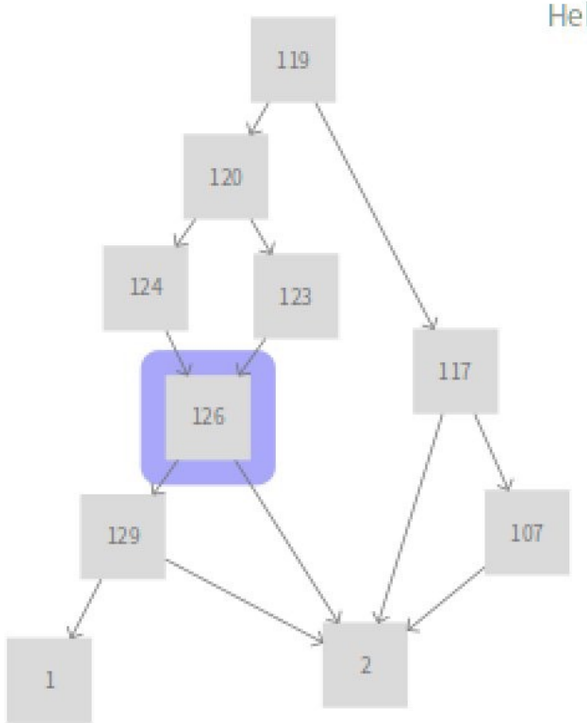


Help

Type	Value
Dimensions	(150@5)
Has categorical	true
Has nil	false

Welcome | Playground | Inspector on a DataFrame [1... | Inspector on a DataFrame [1...

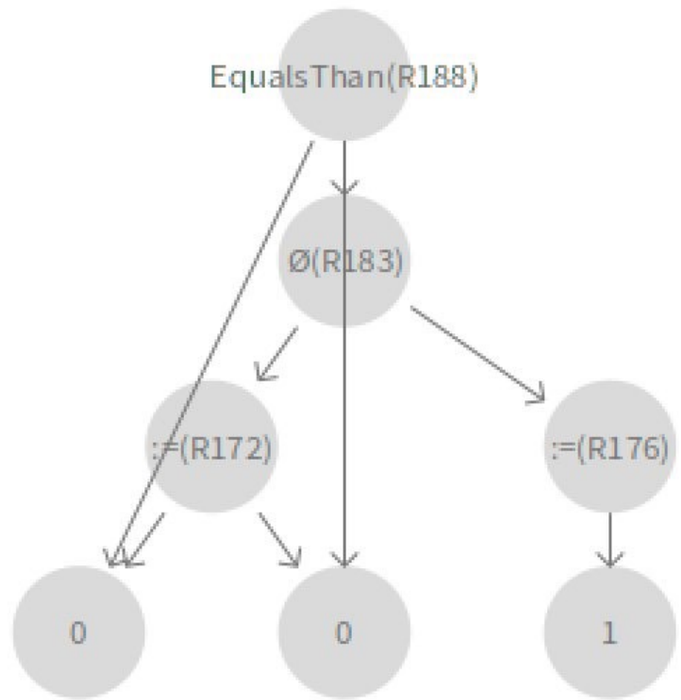
Better than reading basic blocks in assembly



Help

R183 := Ø R172 R176
R188 := EqualsThan R183 0
Jump if R188 true -> 2 if false -> 129

R183 := Ø R172 R176
R188 := EqualsThan R183 0
Jump if R188 true -> 2 if false -> 129



Pharo is empowering you!

One example: DJ Lucretio

- Music producer
- C++ programmer then a Pharoer
- Live concerts with Pharo
- Trainings
- Building music tools and libraries

Concerts and trainings

Four years on the banks

ON STAGE

Ohm & Tresor, Berlin 2023

System Shanghai -ICLC 2024 & 2025

LAUT Barcelona, ICLC 2025

ESUG 2023 – 2025

Robot Festival, Bologna 2025

Audio Developer Conference, Bristol 2025

ince 2025 every performance is all Pharo.
No external audio engine.

IN THE CLASSROOM — PISA

40 *participants, ages 11–60 with no coding experience.*

Four environments, comparable sessions:
Coypu, Strudel, Mercury, Sonic Pi.

Coypu came first *for ease of use, intuitiveness and enjoyment.*

Informal, single event , no questionnaire , biased
/ a structured study has to be planned planned with the University of Padova.

Inri

Specific tools and libraries

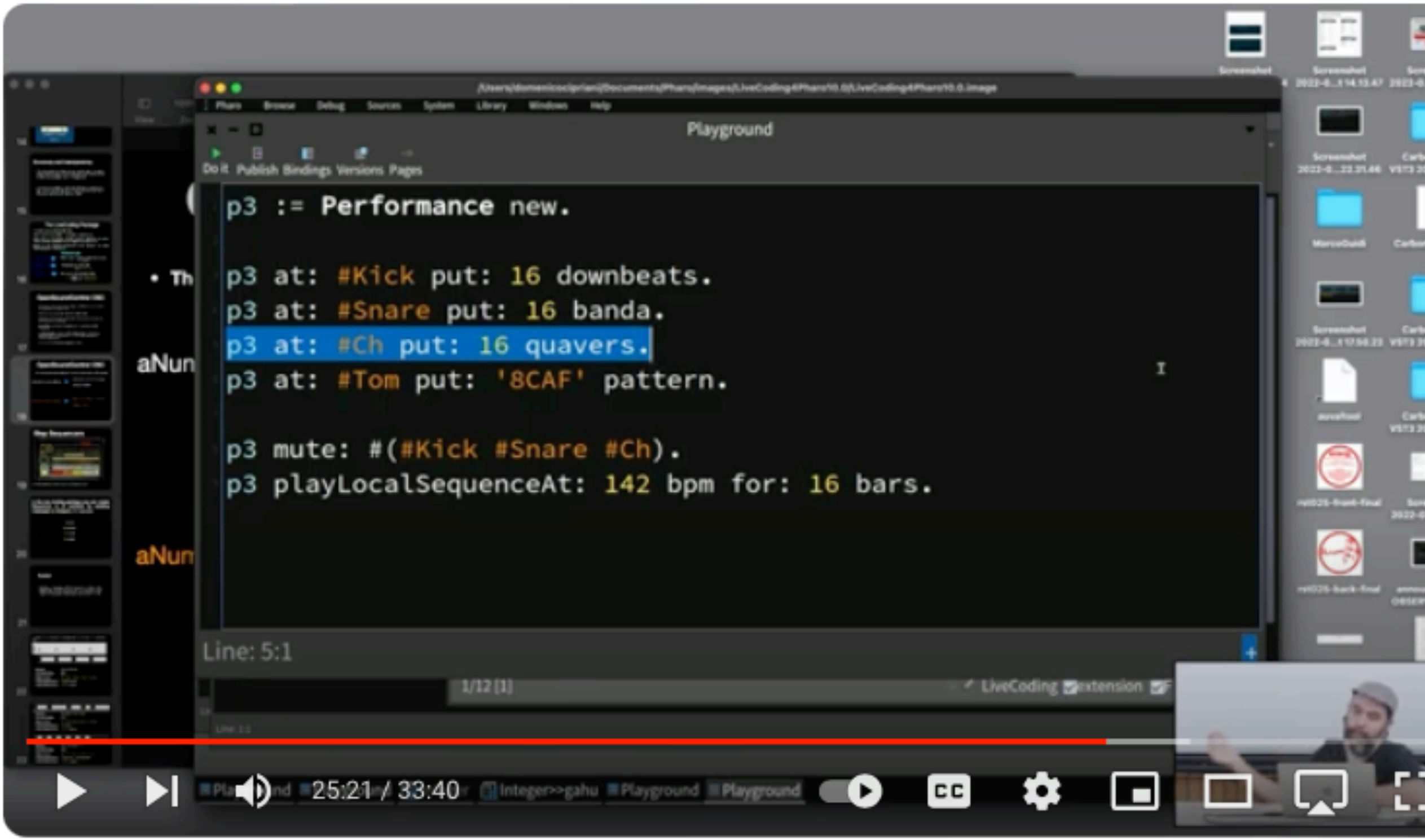
COYPU (2022) :

Programming music on-the-fly with Pharo.

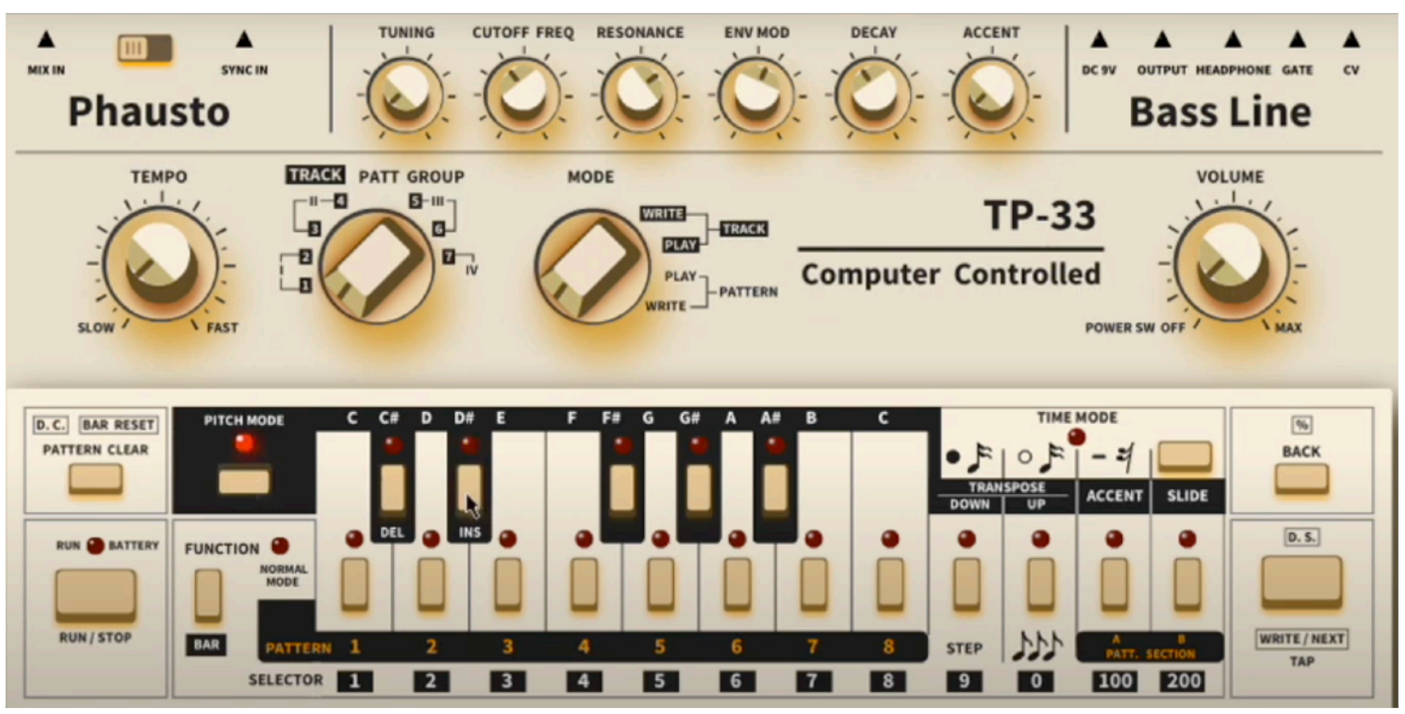
PHAUSTO (2024) :

sound synthesis (embedding Faust) within Pharo.

Special workspace for Live Music



New UI Elements



Thanks to Jayesh from India

Pharo 11.0 - 64bit (work in progress).image

Pharo Browse Debug Sources System Library Windows Help

Playground

Do it Publish Bindings Versions Pages

```
p := Performance uniqueInstance .
p performer: PerformerKyma new.

p freq: 138 bpm.

'9090' hexBeat to:  #kick.
'0202' hexBeat to: #Snare.
16 upbeats to: #ch.
'0002' hexBeat to: #oh.

16 rests, 8 banda to: #timbale.


4 breves  notes: #(36 40 ); to: #Wobble.
'62/16 , 64/16 , 65/16, 67/16 ' asDirtNotes to: #Pad.

p solo: #Pad.


p playFor: 256 bars.

f := VerticalFaderForKyma newWithAddress: 'WobbleLFOFreq' range: #(0.3 10).
f openInWindow

p stop.
```

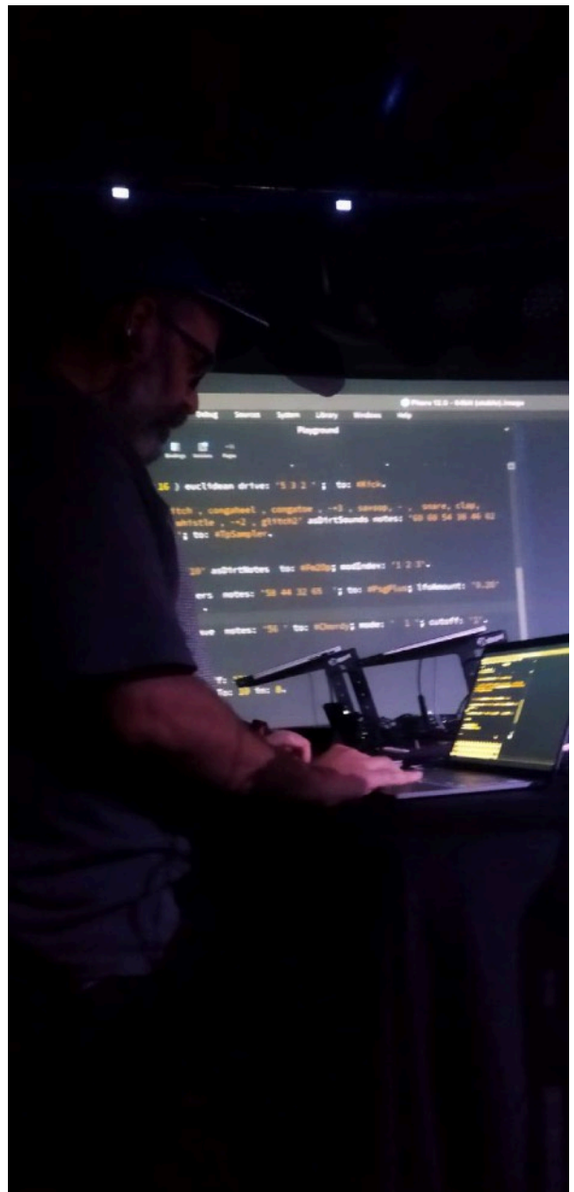
ne: 2:1 +L

Scripting visual language

Some Links

- https://www.youtube.com/watch?v=4lhoYmI_ivs
- https://www.youtube.com/watch?v=S2Dff90aYDI
- https://github.com/lucretiomsp

ESUG 2024 https://www.youtube.com/watch?v=zP1GVVrydOs



More powerful tools

Micro commits
Control flow break points
Automatic deprecation
Hot update on the fly
Customizable debuggers

Micro commits

Control flow break points

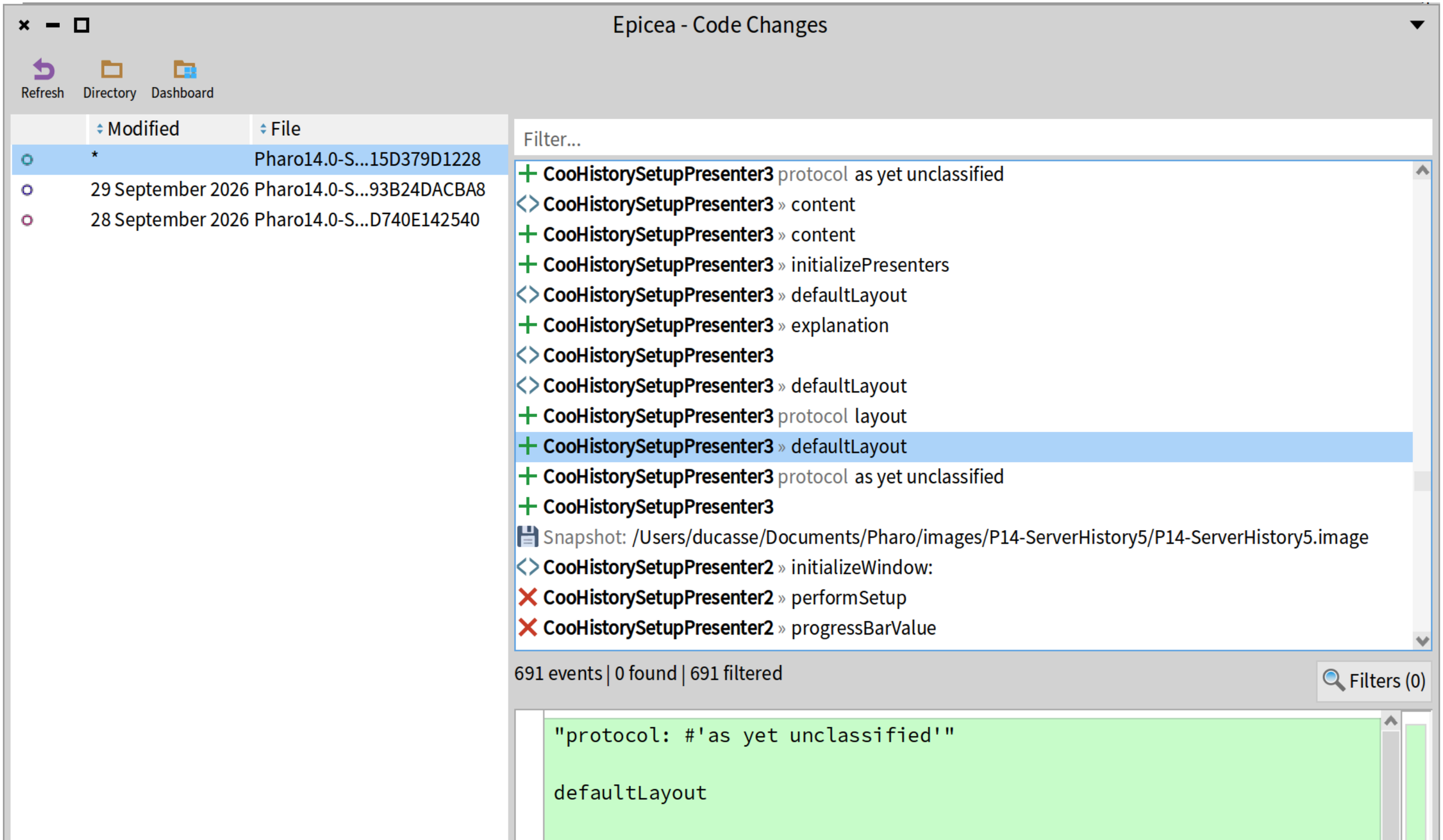
Automatic deprecation

Hot update on the fly

Customizable debuggers

Micro commits

All your compilation is logged



All your method versions at hand

Recent versions of CooHistoryEventRecorder2 class>>#completionSettingsOn:

Browse | References | Senders | Implementors | Revert | Commit history | Copy to clipboard | Run Tests | Side By Side

Location	Selector	Package	Timestamp
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:42:22.731844+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:38:08.759844+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:33:26.970894+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:33:03.655833+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:30:50.535174+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:30:33.167292+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:29:32.136464+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:25:42.256364+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:24:02.364971+02:00
CooHistoryEventRecorder2 class (settings)	completionSettingsOn:	[]	2026-10-07T18:09:47.287038+02:00

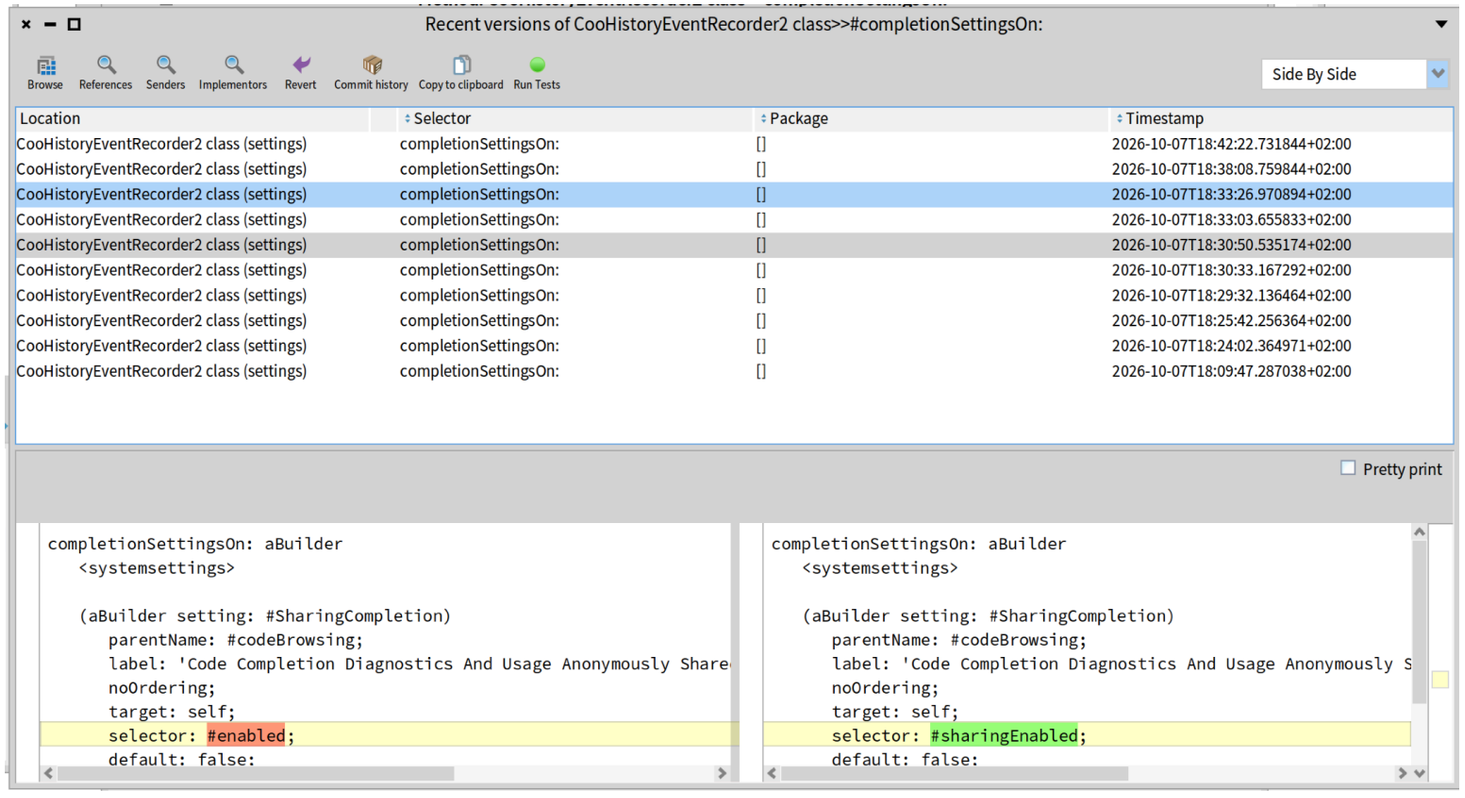
Pretty print

```
completionSettingsOn: aBuilder
   <systemsettings>

   (aBuilder setting: #SharingCompletion)
      parentName: #codeBrowsing;
      label: 'Code Completion Diagnostics And Usage Anonymously Share
      noOrdering;
      target: self;
      selector: #enabled;
      default: false:
```

```
completionSettingsOn: aBuilder
   <systemsettings>

   (aBuilder setting: #SharingCompletion)
      parentName: #codeBrowsing;
      label: 'Code Completion Diagnostics And Usage Anonymously S
      noOrdering;
      target: self;
      selector: #sharingEnabled;
      default: false:
```

Micro commits are cool

You can revert, replay, navigate coding history at method level



Micro commits

Control flow break points

Automatic deprecation

Hot update on the fly

Customizable debuggers

Control flow wishes

I want my breakpoint to only stop when called from THAT test called testMe!

Basically a control flow breakpoint

E.g. Only stop if called from testMe

```
mycode

    self haltIf: #testMe

    ...
```

Implementation

- Use stack reification
- Walk it
- Halt if needed
- (in 5 lines)

```
haltIf: aSelector
    | cntxt |
    cntxt := thisContext.
    [ cntxt isNil ] whileFalse: [
        cntxt selector = aSelector
            ifTrue: [ self halt ].
        cntxt := cntxt sender ]
```

Micro commits
Control flow break points
Automatic deprecation
Hot update on the fly
Customizable debuggers

Imagine

```
User >> main
    m := Math new.
    logger := Logger new.
    logger log: (m log: 8)
```

And you deprecate log:

```
User >> main
    m := Math new.
    logger := Logger new.
    logger log: (m log: 8)
```

which one?

Manually checking calls to migrate

```
Math >> log: aNumber
    " computing log "

Logger >> log: aMessage
    " Deprecated in favor of log:level: "

User >> main
    m := Math new.
    logger := Logger new.
    logger log: (m log: 8)
```

You have to manually check all the log: senders and migrate the good ones

Automatic deprecation rewriting

```
Logger >> log: aMessage
    self
        deprecated: 'Use log:level: instead of log:'
        transformWith:
                '`@rec log: `@arg
                    -> '`@rec log: `@arg level: #info' .
    self log: aMessage level: #info
```

Only correct calls are translated during execution

```
User >> main
    m := Math new.
    logger := Logger new.
    logger log: (m log: 8) level: #info
```

Micro commits
Control flow break points
Automatic deprecation
Hot update on the fly
Customizable debuggers

Default debugger

Can be morphed into a domain specific debugger

Focus where the gold is!

QuittanceGenerator class>>#exampleJustMonthAndRenter

Halt

Bytecode

Stack

Proceed Restart Into Over Through

PDFCellElement	getSubElementsWith:styleSheet:
PDFCellElement(PDFComposite)	generateCodeSegmentsCollectionWi
PDFCellElement(PDFComposite)	generateCodeSegmentWith:styleShe
PDFDataTableElement(PDFComposite)	generateCodeSegmentsCollectionWi [:aSubElement | aSubElement generateCodeSe
Array(SequenceableCollection)	collect:

Source

Where is? Browse

```
generateCodeSegmentsCollectionWith: aPDFGenerator styleSheet: compositeStyleSheet format: aFormat
    ^ (self getSubElementsWith: aPDFGenerator styleSheet: compositeStyleSheet)
        collect: [ :aSubElement |
            aSubElement
                generateCodeSegmentWith: aPDFGenerator
                styleSheet: (aSubElement buildCompositeStyleSheetFrom: compositeStyleSheet)
                format: aFormat ]
```

Variables

Type	Variable	Value
implicit	self	a PDFCellElement
parameter	aFormat	a PDFA4Format
parameter	aPDFGenerator	a PDFGenerator
parameter	compositeStyleSheet	a StyleSheet

```
        dimension: 80 mm @ 20 mm;
```

debug it

Bytecode GT

Parser Stack | Through Action | Into Reduce | Next Token

Symbol	Value
Expression	a JSUnaryExpressionNode
")"	{)(238498,238498,#(23))}
"{"	{{(238499,238499,#(25))}
StatementList	an OrderedCollection(a JSExpressionStatementNode a JSI
"("	{((238678,238678,#(28))}
"function"	{function(238679,238686,#(53 82))}
Identifieropt	nil
"("	{((238687,238687,#(28))}
FormalParameterListopt	an Array(an OrderedCollection() an OrderedCollection())
")"	{)(238688,238688,#(23))}
"{"	{{(238689,238689,#(25))}
SourceElements	an OrderedCollection(a JSVariableStatementNode a JSVar
MemberExpression	a JSNameExpressionNode
"."	{.(274447,274447,#(38))}

Possible Actions

Symbol	Action
<identifier>	Shi
IdentifierName	Shi

GLR

1

A specific parser compiler debugger

Input | Step to Cur | Next Character

```
            return t._stash(this); // dojo.NodeList
        }

    });

    nl.events = [
        // summary:
        //          list of all DOM events used in NodeList
        "blur", "focus", "change", "click", "error", "keydown", "keypress",
        "keyup", "load", "mousedown", "mouseenter", "mouseleave", "mousemove",
        "mouseout", "mouseover", "mouseup", "submit"
    ];

    // FIXME: pseudo-doc the above automatically generated on-event functions

    // syntactic sugar for DOM events
    d.forEach(nl.events, function(evt){
```

Nam	Value
State	default

Stack | Proceed | Restart | Into | Over | Through

JSParser(SmaCCGLRParser)	performParsingStep	
JSParser(SmaCCGLRParser)	performParsingLoop	
JSParser(SmaCCParser)	parse	
JSParser class(SmaCCParser class)	parseStream:startingAt:	
JSParser class(SmaCCParser class)	parseFile:	[self parseStrean
BlockClosure	ensure:	
JSParser class(SmaCCParser class)	parseFile:	
UndefinedObject	DoIt	
CompiledMethod	valueWithReceiver:arguments:	
RubSmalltalkEditor	debug:receiver:in:	[aCompiledMetho
BlockClosure	newProcess	[self value.Proces

Source | Where is? | Browse

```
(tryAllStates and: [ self isEOFToken ])
    ifTrue: [ ^ self processEOFToken ].
currentToken isPattern
    ifTrue: [ ^ self splitForPatternToken ].
[ actions := self actionsForCurrentToken.
actions isEmpty
    ifTrue: [ self killState.
        currentToken := nil.
        ^ self ].
2 to: actions size do: [ :i | self splitWithAction: (actions at: i) ].
action := actions first.
action = self acceptAction
    or: [ self performAction: action.
        currentToken isNil ] ] whileFalse.
action = self acceptAction
    ifTrue: [ currentState markAccepted ]
```

Halt in OCDBox>>name:

ack

Class	Method	Context
DBox	name:	UndefinedObject>>DoIt
defined Object	DoIt	CompiledMethod>>valueWithRec
mpiledMethod	valueWithReceiver:arguments:	[aCompiledMethod
bSmalltalkEditor	debug:receiver:in:	[self value.

ceed | Into | Over | Through | Run to | Restart | Return | Where is? | Create

lt

```
1 name: anObject
2     self halt.
3     name := anObject
```

Bytecode | Breakpoints | Sindari

Type	Target	Method
Breakpoint	self	Reflecti
Breakpoint	self	OCDBox>>initialize
Breakpoint	self	OCDBox>>name:
Breakpoint	self	OCDBox>>name:
Halt	OCDBox	OCDBox>>name:
Breakpoint	OCDBox	OCDBox>>name:
Breakpoint	self	OCDBox>>removeElement:
Breakpoint	self	OrderedCollection>>removeI
Halt	StHaltCacheTest	StHaltCacheTest>>testInitial
Halt	StHaltCacheTest	StHaltCacheTest>>testInitial

```
1 name: anObject
2     self halt.
3     name := anObject
```

Specific view

Bytecode | Breakpoints | Sindarin

```
33 <4C> self
34 <80> send: halt
35 <D8> pop
36 <40> pushTemp: 0
37 <C9> popIntoRcvr: 1
38 <58> returnSelf
```

ceiver in: a StDebuggerContext (OCDBox>>name:)

an OCDBox

Variable	Value
[arg] anObject	'i'
self	an OCDBox
{ } elements	an OrderedCollection [0 items] ()
name	''
{ } Temps	a Dictionary [1 item] (#anObject->'i')
anObject	'i'
stackTop	an OCDBox

Raw | Breakpoints | Meta

Type	Target	Method
Halt	OCDBox	OCDBox>>name:

```
1
```

```
1 "an OCDBox"
2 self
```

Variable	Value
stackTop	an OCDBox
0 [anObject]	'i'
{ } rcvr: 0 [elements]	an OrderedCollection [0 iten
rcvr: 1 [name]	''

object centric + adv scripting (AST + call stack)

Rece n: a Debugg Context (OCDBox>>name:)

an OCDBox

Variable	Value
[arg] anObject	'i'

Raw | Breakpoints | Meta

Variable	Value

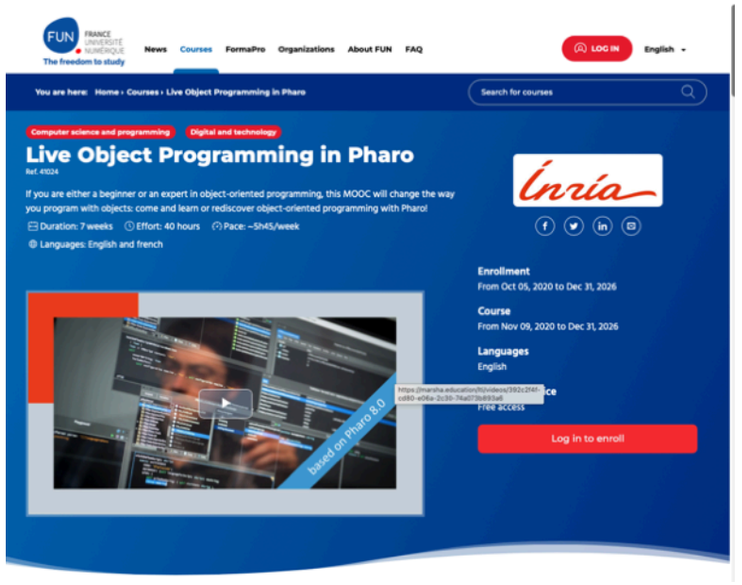
What about teaching?

One of the best vehicule for teaching

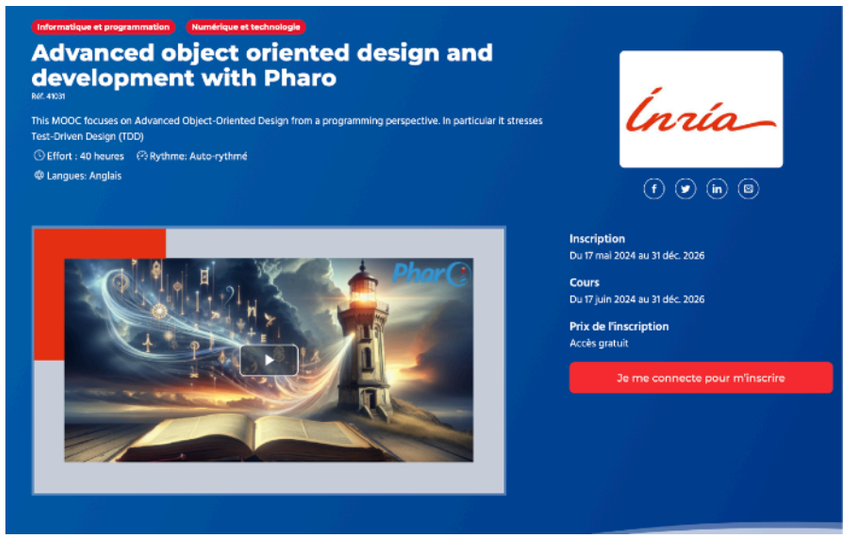
- Immersive environment to fully understand object-oriented programming
- Two free ready to use excellent MOOCs
- Used in several universities worldwide

Free excellent moocs

https://mooc.pharo.org



https://advanced-design-mooc.pharo.org/

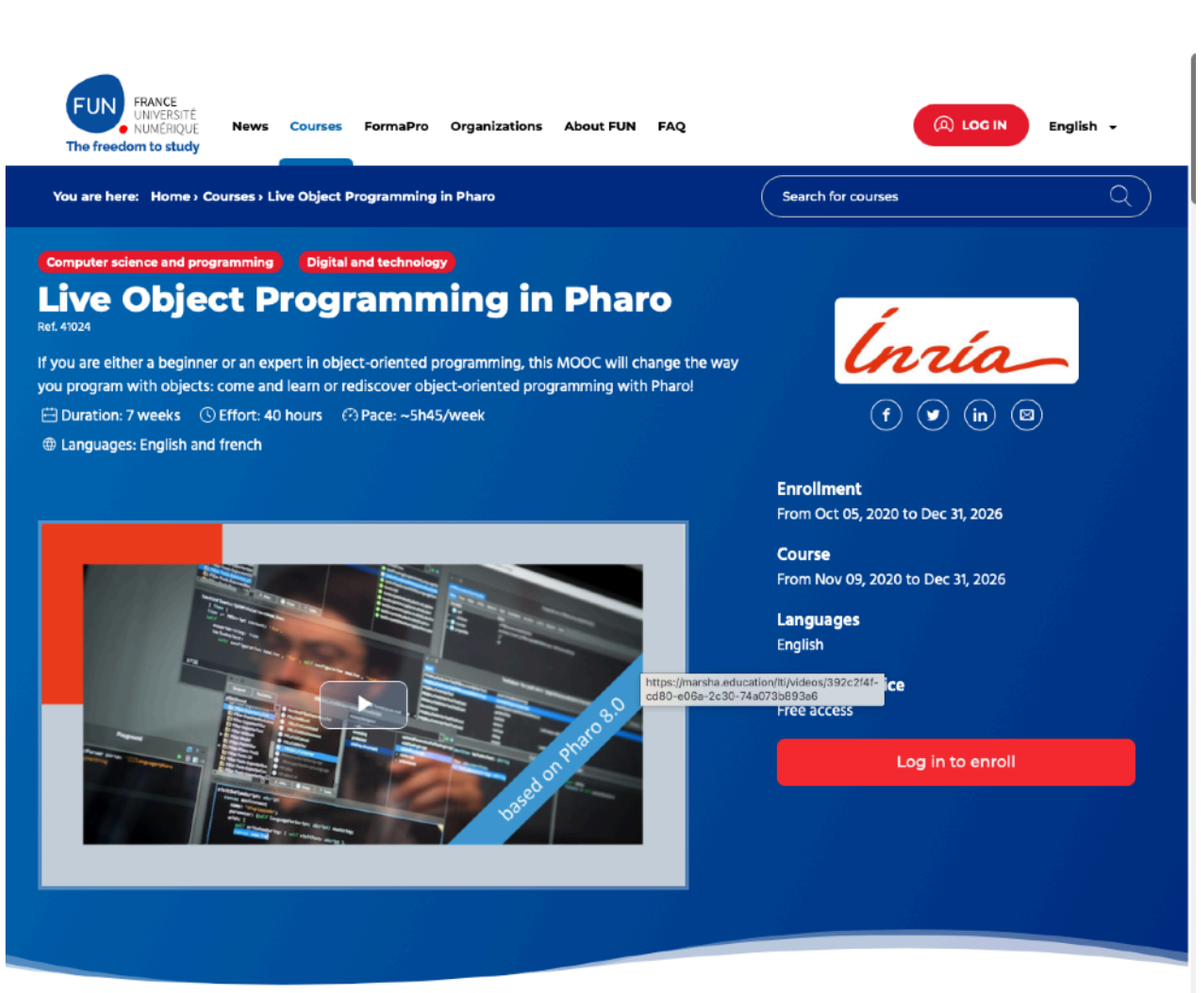


https://mooc.pharo.org

> “I have spent the last 20 years or so in software development and, following this Mooc, I realized I hadn’t really grasped the essence of object oriented design. – Anonymous”

https://www.fun-mooc.fr/en/courses/live-object-programming-pharo/

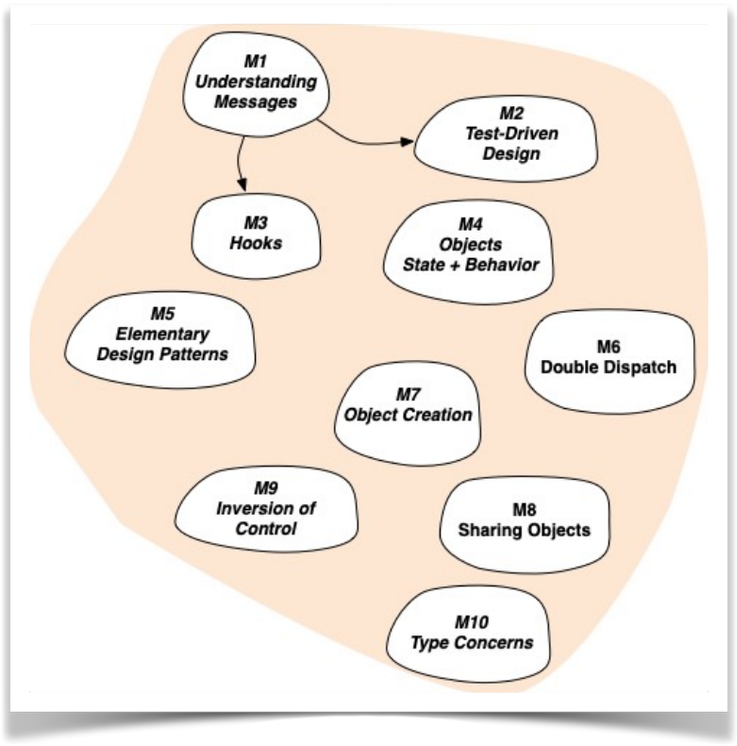
68 Lectures slides/videos
7 Challenges
2 Tutorials
1 Project



https://advanced-design-mooc.pharo.org/

> “Modules 1–4 should be taught to and understood by anyone programming with an OOP language. I appreciate that these pieces of invaluable knowledge keep being repeated in the other modules.”

60 Lectures slides/videos
One free book



Another and fresh look at Objects

- OOP at the root
- Turtles all the way down

Not, or (|), and friends

- An example of available material
- First discover
- Second repeat
- Third generalize

Exercise 1: Implement Not

Propose an implementation of Not in a world where:

- You have: true, false objects
- You only have objects and messages

How would you implement the message not?

```
false not

–> true


true not

–> false
```

Hint 1: No conditionals

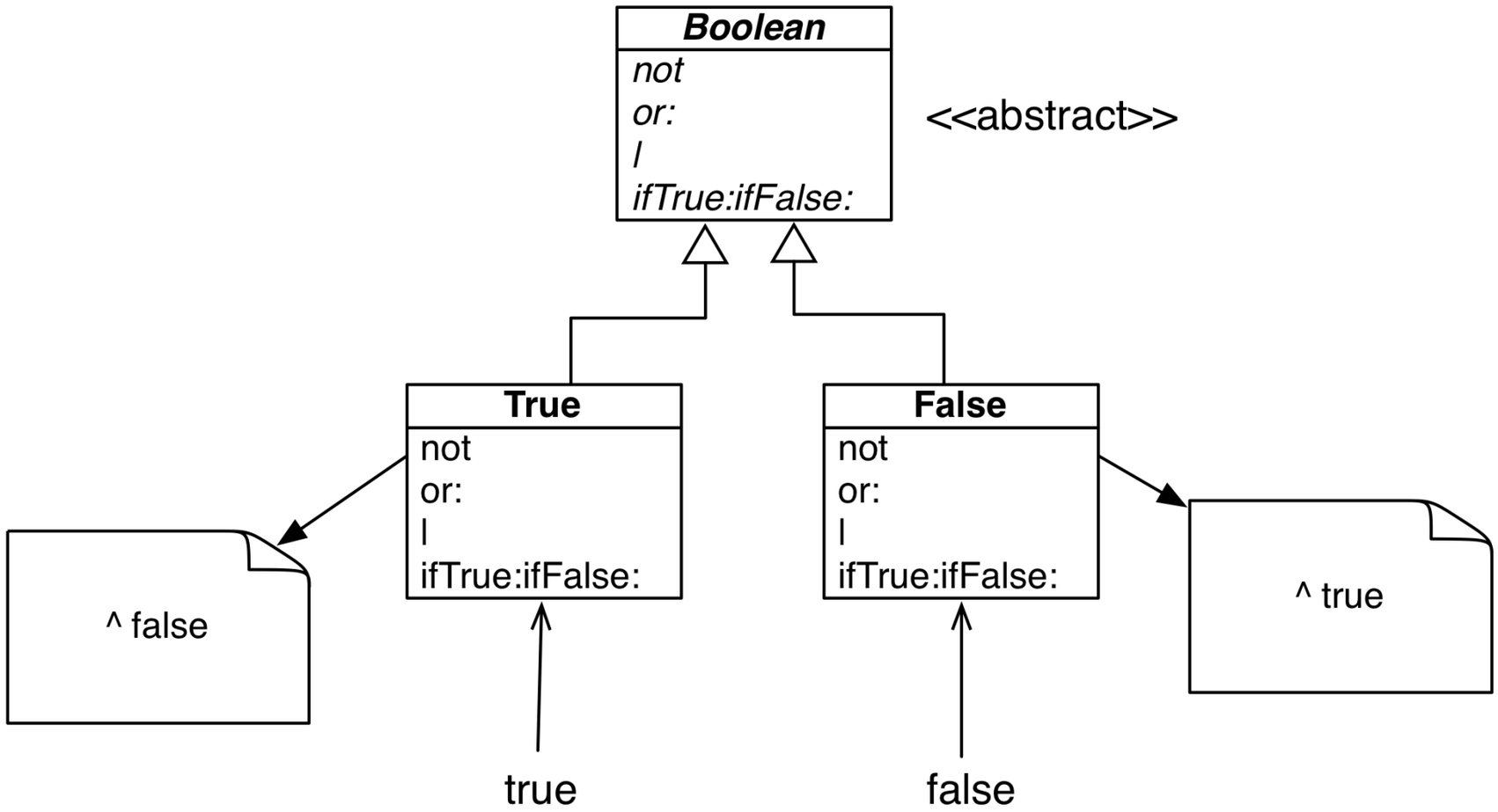
The solution does not use explicit conditionals (i.e., no `if`)

Hint 2: An example of choice in OOP

```
x open
```

- x can be a file, a window, a tool,...
- The method is **selected** based on x's class

Implementation hierarchy

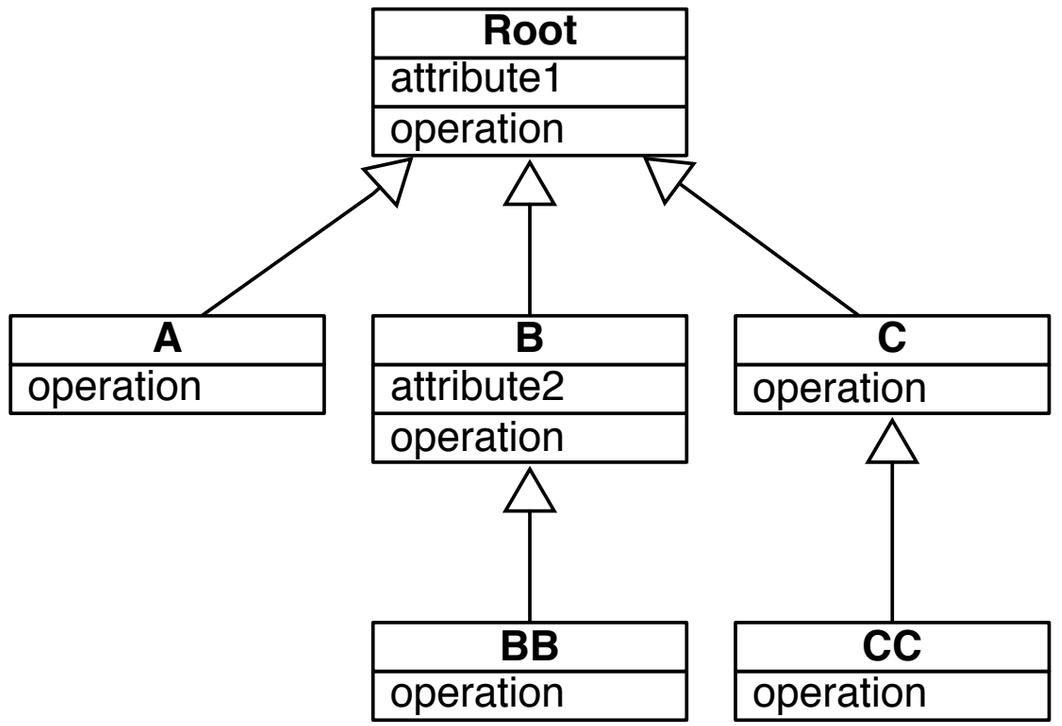


What is example is showing…

Sending a message is making a choice
Classes express choices
Modular design

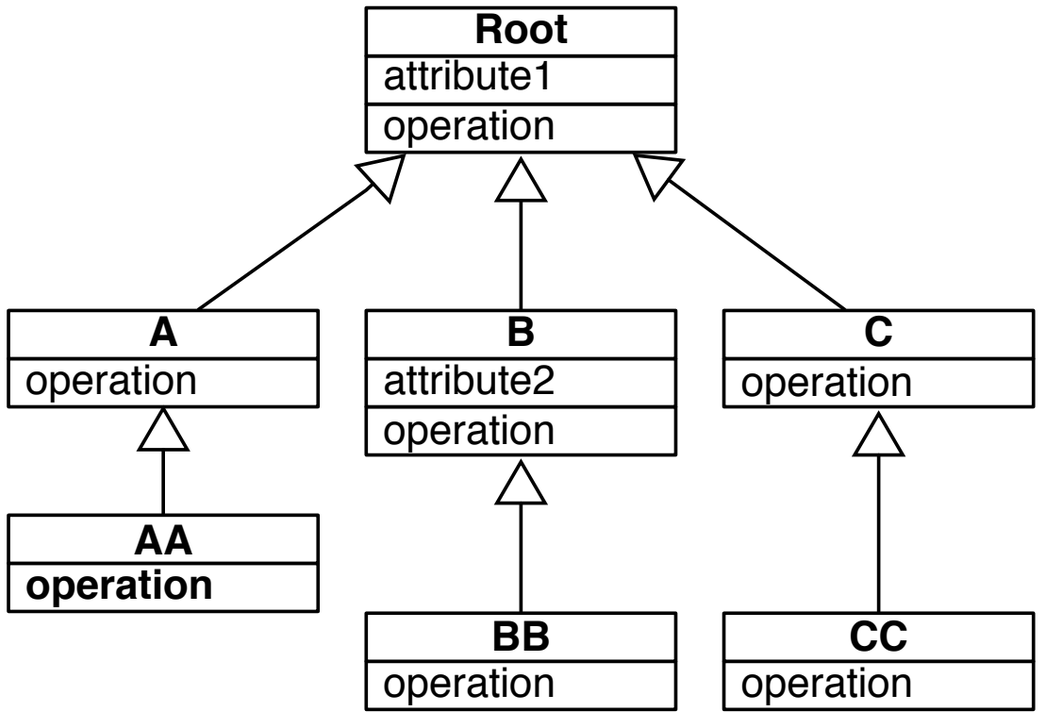
Select the right method

```
aCollection := {a . bb . c}.

...

aCollection do: [ :e |
    e operation]
```



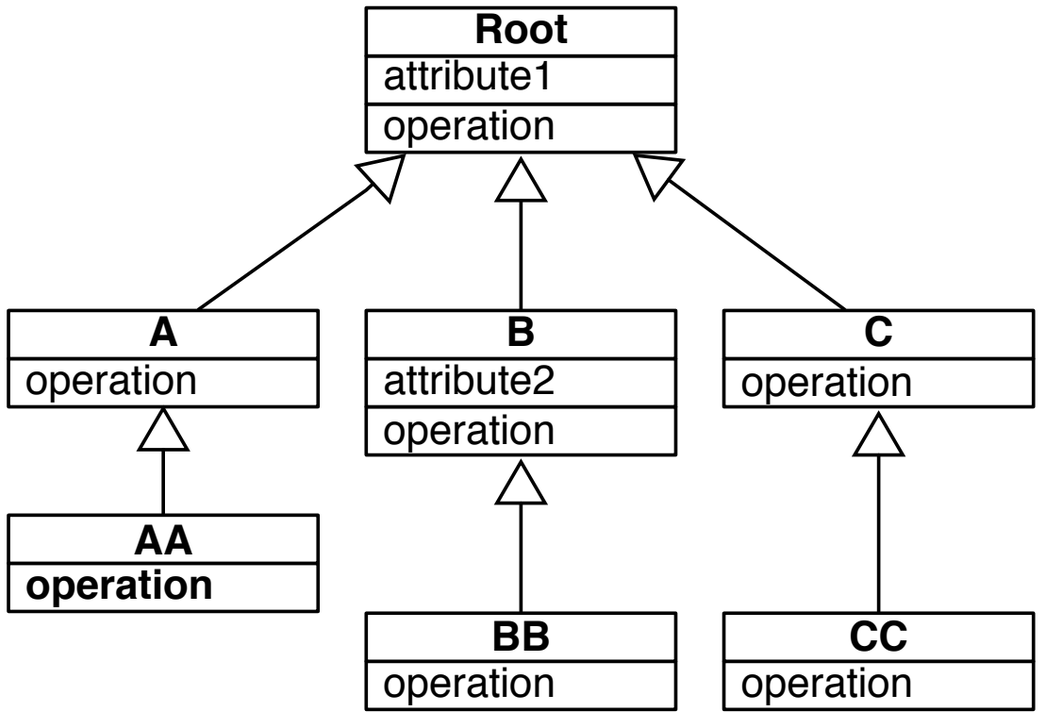
But dynamically: new objects can be chosen

```
aCollection := {a . bb . c . aa}.

...

aCollection do: [ :e |
   e operation]
```

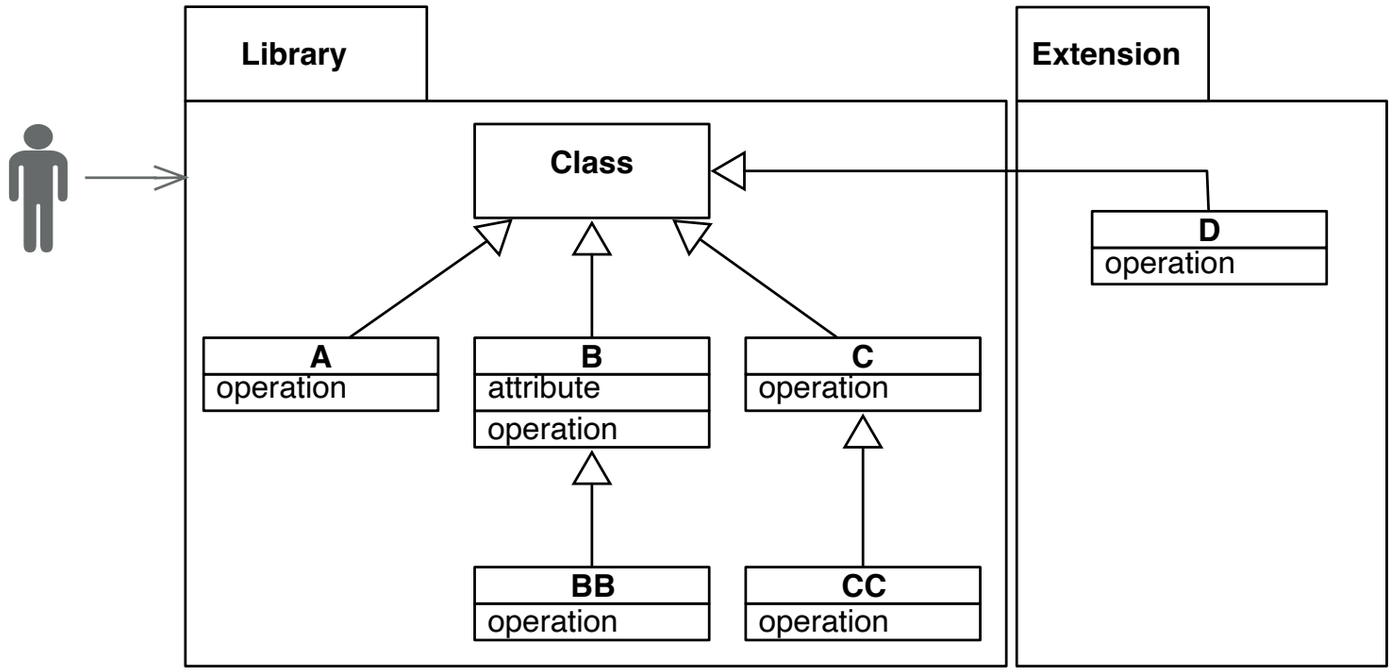


But dynamically: new objects can be chosen

```
aCollection := {a . bb . c . aa}.

...

aCollection do: [ :e |
   e operation]
```



Limit impact of changes



- If a client receives instances of D (in addition to classes of first package), its code does not have to change
- Method operation of D instances will be executed naturally

What about Research?

Yes we can build a real CV using Pharo!

- Check https://Ducasse.github.io/publications.html :)
- Friendly ecosystem
- Companies with challenges

Possible Research Venues

- Software testing
- Software engineering / evolution
- UX
- AI use/control
- Meta modeling
- Data visualization

Software engineering

- Assessing test quality
- Quality rules
- Enhancing debuggers
- Smarter code completion
- …

Infrastructure to collect user data

- Easy and nearly out of the box
- The community is ready to be a guinea pig

Open-source research platforms

Bienvenue sur le site de Cormas

CORMAS = COmmon-pool Resources and Multi- Agent Simulations

Cormas est une plateforme de modélisation multi-agent (ABM) dédiée à la modélisation des relations entre les sociétés et leur environnement.

Cormas est destinée à faciliter la conception d'ABM, ainsi que le suivi et l'analyse des simulations. La plateforme est basée sur l'environnement de programmation VisualWorks qui permet de développer des modèles en Smalltalk.

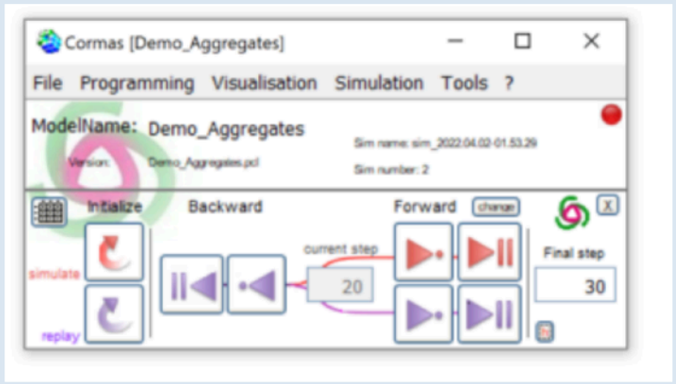
Cormas est un framework à partir duquel, par spécialisation et raffinement, les utilisateurs peuvent créer des entités spécifiques pour leur propre modèle.

cormas
COmmon pool Resources and Multi-Agent Simulations

Le logiciel Cormas, libre de droits (licence MIT), peut être téléchargé.

Découvrez la dernière version sortie en 2024.

Vous pouvez aussi consulter des exemples de modèles et des éléments de bibliographie

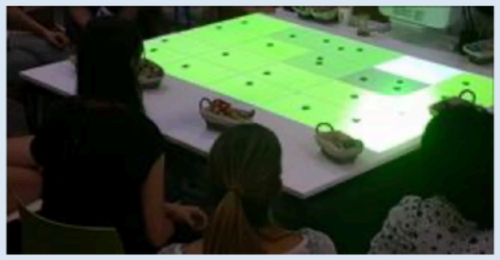


Interface principale de Cormas

- Pharo

Une nouvelle version de Cormas sous Pharo est en développement. Visiter le site

Nous proposons la démarche ComMod (Companion Modelling)



Session de ReHab à l'IUCN 2021 avec Cormas.

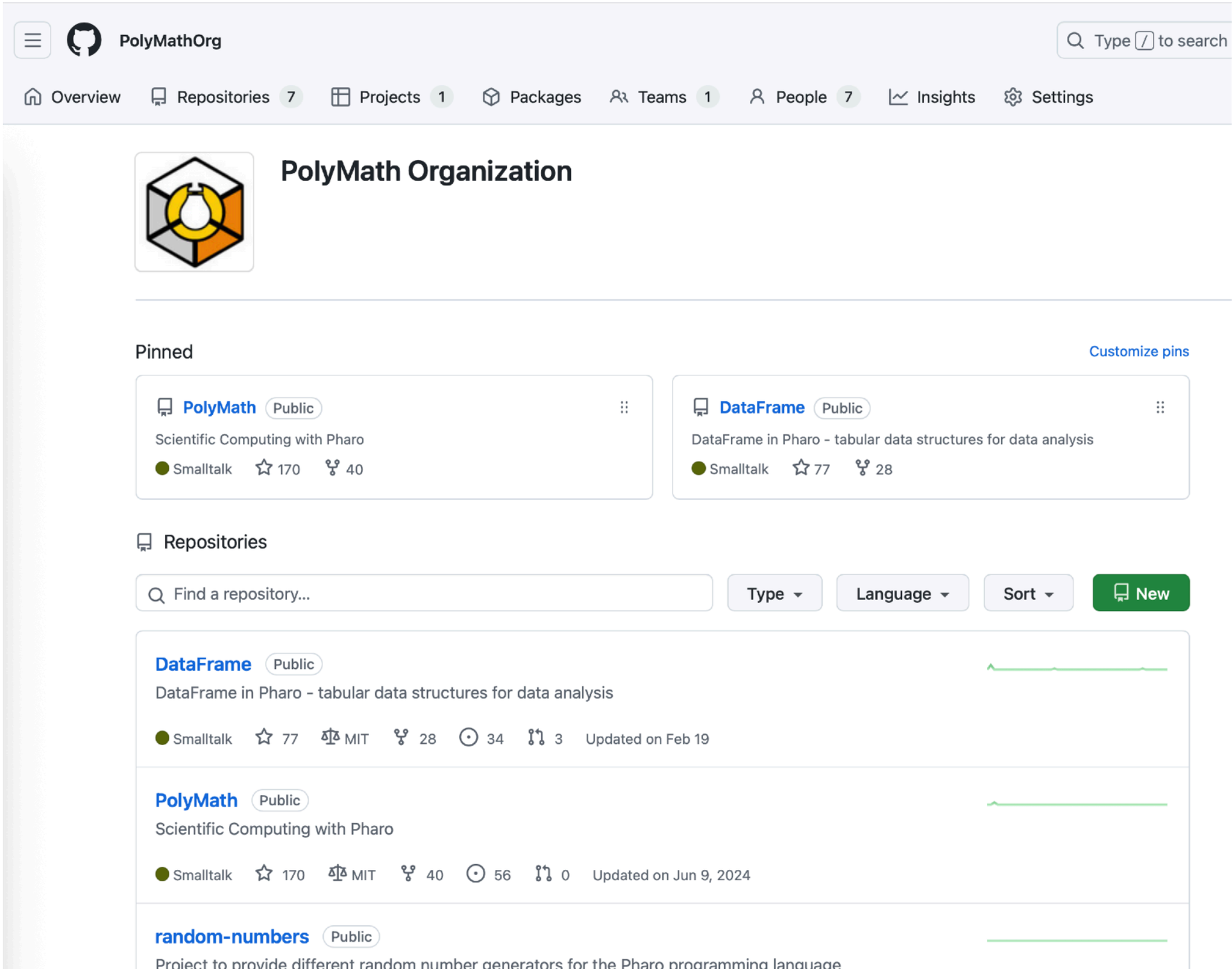
Visiter le site www.commod.org

Modular Moose: Software analysis

Moose is a platform for software analysis.

It allows to represent software system in a model, to query, manipulate, transform, visualize this model.

Polymath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ČVUT ČESKÉ VYSOKÉ UČENÍ TECHNICKÉ V PRAZE

FAKULTA INFORMAČNÍCH TECHNOLOGIÍ ČVUT V PRAZE

CENTRE FOR CONCEPTUAL MODELLING AND IMPLEMENTATION

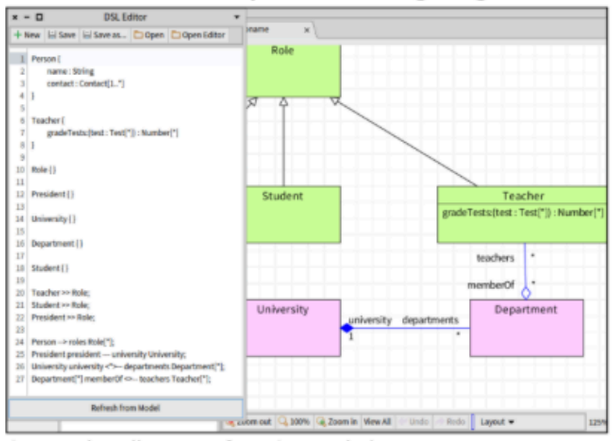
Pharo

OpenPonk

Modelling Platform

Conceptual modeling platform
for teaching, research, industry

DSL Support
(domain specific language)

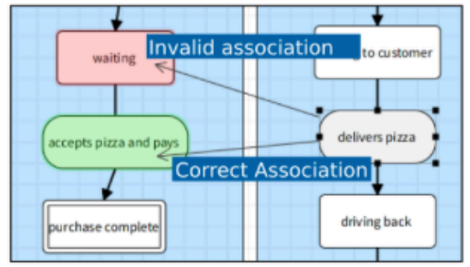


Generating diagrams for DSLs and vice versa
Some DSLs are part of OpenPonk,
but you can create your own.

OpenPonk goals

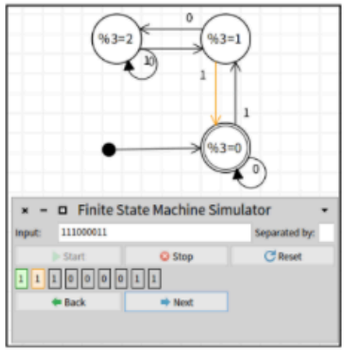
- Intuitive creation and interaction with models
- Support for existing and custom notations
- Extensibility for custom scenarios
- Open platform (MIT license)

Model validations



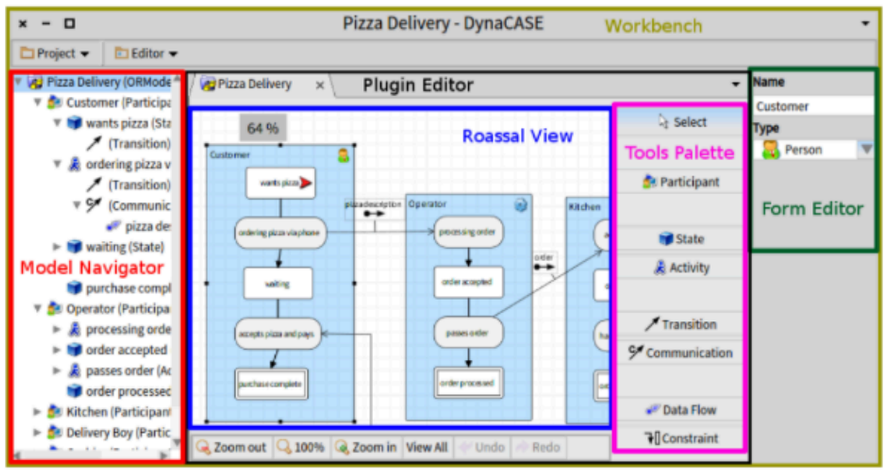
OpenPonk offers basic rule validation
during the creation process.
Complex validations are available for some notations.

Simulation and Visualization

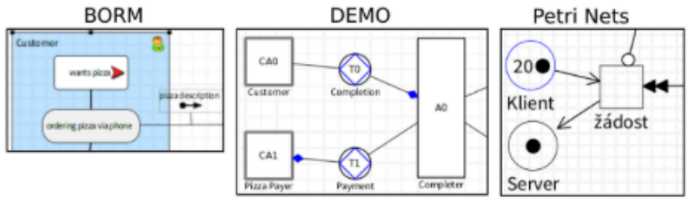


Step-by-step model simulation
and progress visualization.

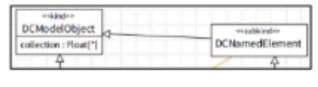
Petri nets and State Automata
simulation is available.



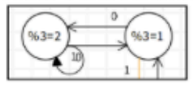
Support for common and custom notations



UML class diagrams and
structural OntoUML



Finite State Automata



Several models and notations are available.
You can use your own model and implement just the notation
according to your needs.

openponk.org

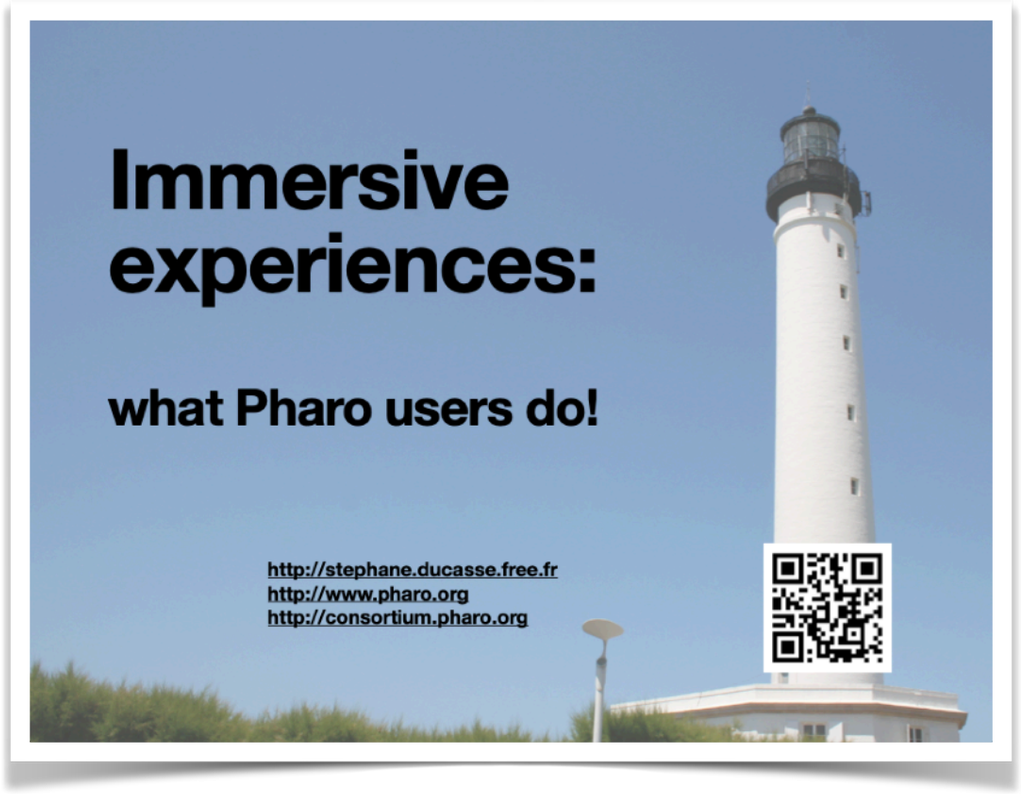
If you are interested in research or industrial cooperation, please contact robert.pergl@fit.cvut.cz

BTW: what is the goal of Pharo

- Better world, better people
- Open source
- Let people make money
- Have fun

What companies are doing with Pharo

- https://archive.esug.org/ESUG2025/day4/401-duca-what-devs-are-doing-with-pharo.pdf
- https://archive.esug.org/ESUG2025/day4/408-duca-what-people-are-doing-with-pharo.pdf



A powerful engine
to invent (y)our future

An ecosystem where innovation/business bloom

Some Pharo's Teachers

• Uni. of Fianarantsoa • Uni. of Novi Sad • Uni. of Maroua • Uni. of Brussels • Ecole des Mines de Douai • Uni. of Savoie • Ivan Franko Nat. Uni. of Lviv • Czech Technical Uni. • CULS Prague • Uni. of Quilmes • Uni. of La Plata • Northern Michigan Uni. • Uni. Technologica Nacional • Uni. of Santiago • Uni. Policnica de Catalunya • Uni. de Bretagne Occidentale • INFTI • Uni. of Fernhagen • IT Un. of Copenhagen • Uni. Lille • Uni. San Martin

Research Groups

Lafhis (AR)
CAR (FR)
Evref (FR)
Cirad (IRD)
Reveal (CH)
Lysic (FR)
Berger-Levrault (FR)
Uqbar (AR)
OC (FR)
CCMI-FIT (CZ)
Pleiad (CL)
Jet Propulsion Lab (US)

https://consortium.pharo.org

- Simple
- Powerful
- Fun
- Empowering you
- Get in touch with us

Pharo

Get in touch with us

Berger Levrault

Inria
INVENTEURS DU MONDE NUMÉRIQUE

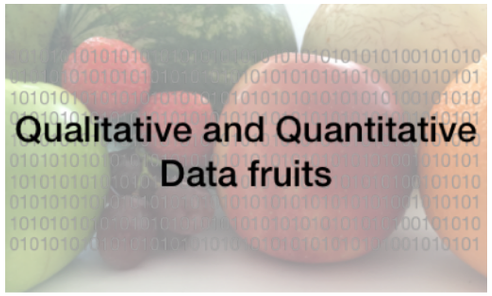
Lifeware

ENSTA Bretagne

THALES

ZWEIDENKER

Yesplan
Let's make it happen



esug
EUROPEAN SMALLTALK USER GROUP

GemTalk SYSTEMS

SPESENFUCHS

projector software

feenk

netstyle.ch

osoco
osoco software company

inspired!

SRA GROUP

telna

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FINWORKS
INFORMATION TECHNOLOGY SOLUTIONS